Thermia Heat Pumps

Duo

Electrical installation instructions

Economical heating solutions from Thermia

Art. nr. 9680–55353001Rev. 2

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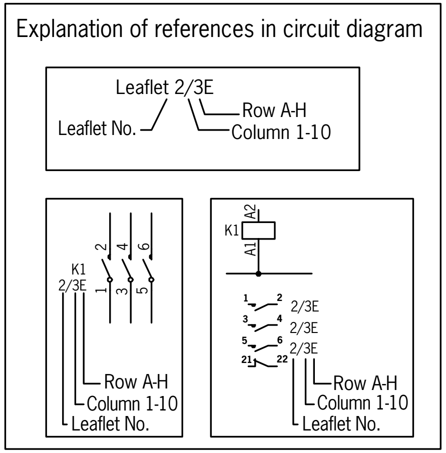

Explanation of references in circuit diagram

Electrical installation

Thermia Duo heat pumps are factory-wired internally. The leads that are to be connected are the mains supply , outdoor temperature sensor and room sensors (if any).

All electrical wiring must be carried out by a certified electrician in compliance with local and national codes and regulations. When in doubt, contact the local electricity distributor.

The mains lead is passed through the top of the unit.

Remove the front panel to get access to the terminal blocks. Connect the mains lead to the terminal block according to the accompanying schematic diagram.

The connection of the heat pump to the mains supply must be hard-wired and a multi-pole switch (safetyswitch) must be installed in the power supply line to the unit.

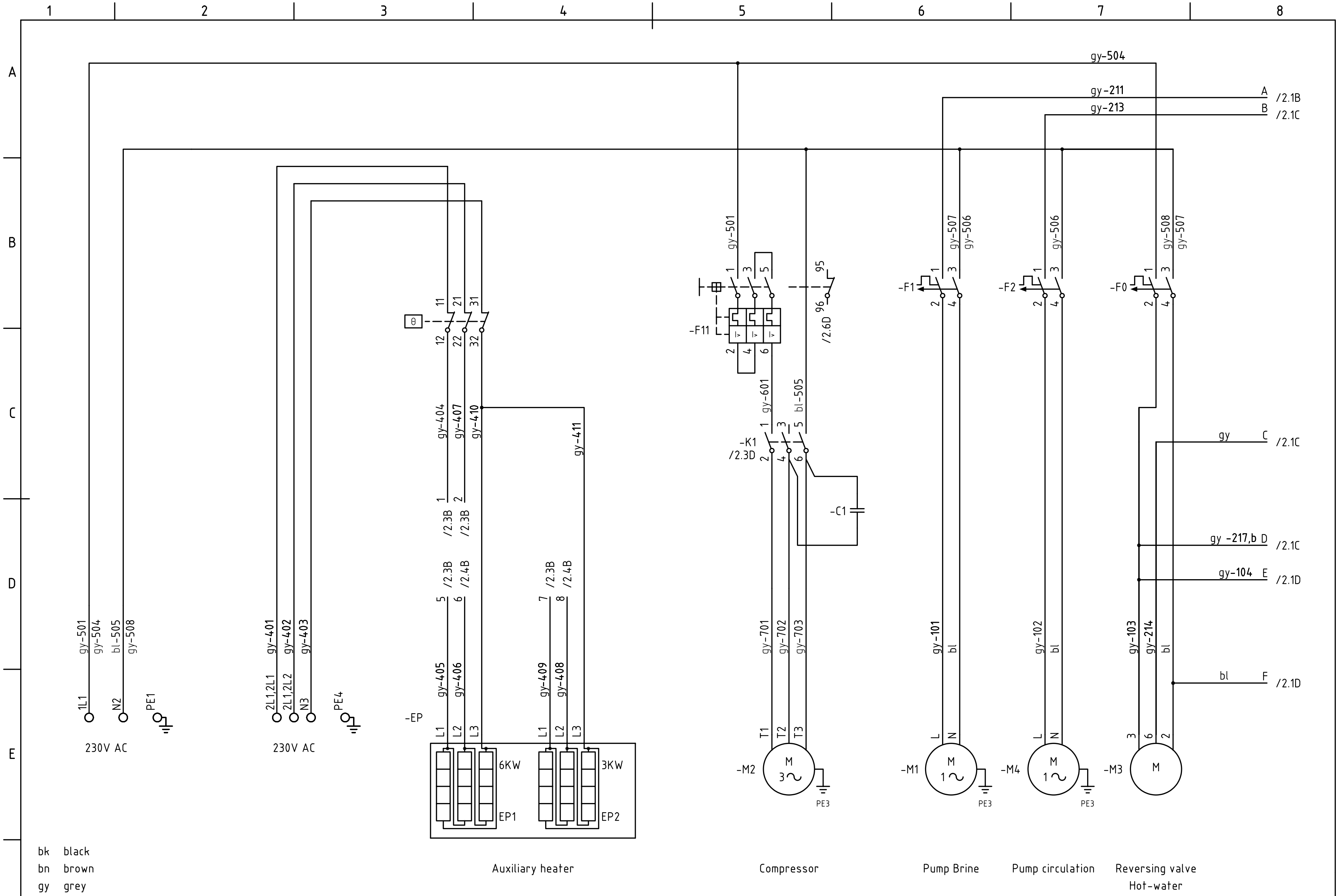
WARNING! To disconnect the power supply to the heat pump, the safety switch must be switched off.

NOTE!! Cables for outdoor and indoor sensors carry low voltage. Interference from surrounding cables may affect these and cause incorrect information to be sent to the control unit. It is therefore vitally important that careful consideration be given to positioning of cables. Low-voltage cables must be kept well separated from high-voltage cables. It is important that low-voltage cables are run parallel to high-voltage cables and separated by a distance of at least 25 cm for cable lengths up to 100 m. For longer cable lengths, the distance between cables should be increased.

Important information

Before switching on the power supply to the heat pump, you must ensure that the heating system, brine water line and hot water heater are filled and vented.

Otherwise, the circulation pumps can be severely damaged.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gy-508 gy-507
-F11
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/2.6D
-F1
-F2
-F0
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12 22 32
gy-404 gy-407 gy-410
gy-411
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
1 /2.3B
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5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-501 gy-504 bl-505 gy-508
gy-401 gy-402 gy-403
gy-405 gy-406
gy-409 gy-408
gy-701 gy-702 gy-703
gy-101 bl
gy-102 bl
gy-103 gy-214 bl
1L1 N2 PE1
2L1,2L1 2L1,2L2 N3 PE4
-EP
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L1 L2 L3
T1 T2 T3
L N
L N
3 6 2
230V AC
230V AC
6KW
EP1
3KW
EP2
-M2
M
3~
PE3
-M1
M
1~
PE3
-M4
M
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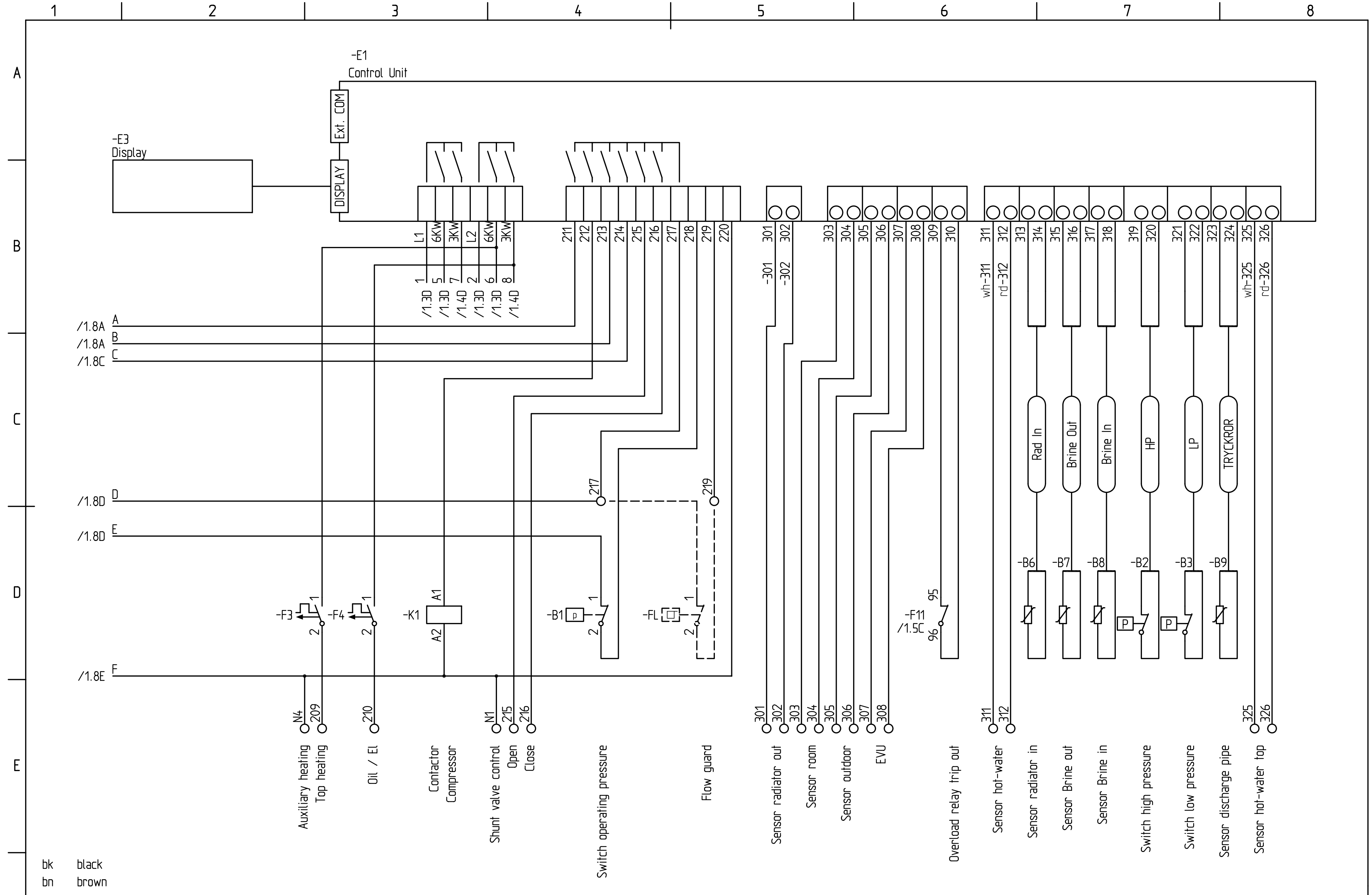
PE3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
		Duo 4ES 230V 1N Circuit diagram		DRAWING NR. 57283	CONTINUED 2 SHEET 1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-E1
Control Unit
-E3
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DISPLAY
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rd-312
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wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
217
219
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
1
2
-F11
/1.5C
95
96
N4
209
210
N1
215
216
301
302
303
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306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
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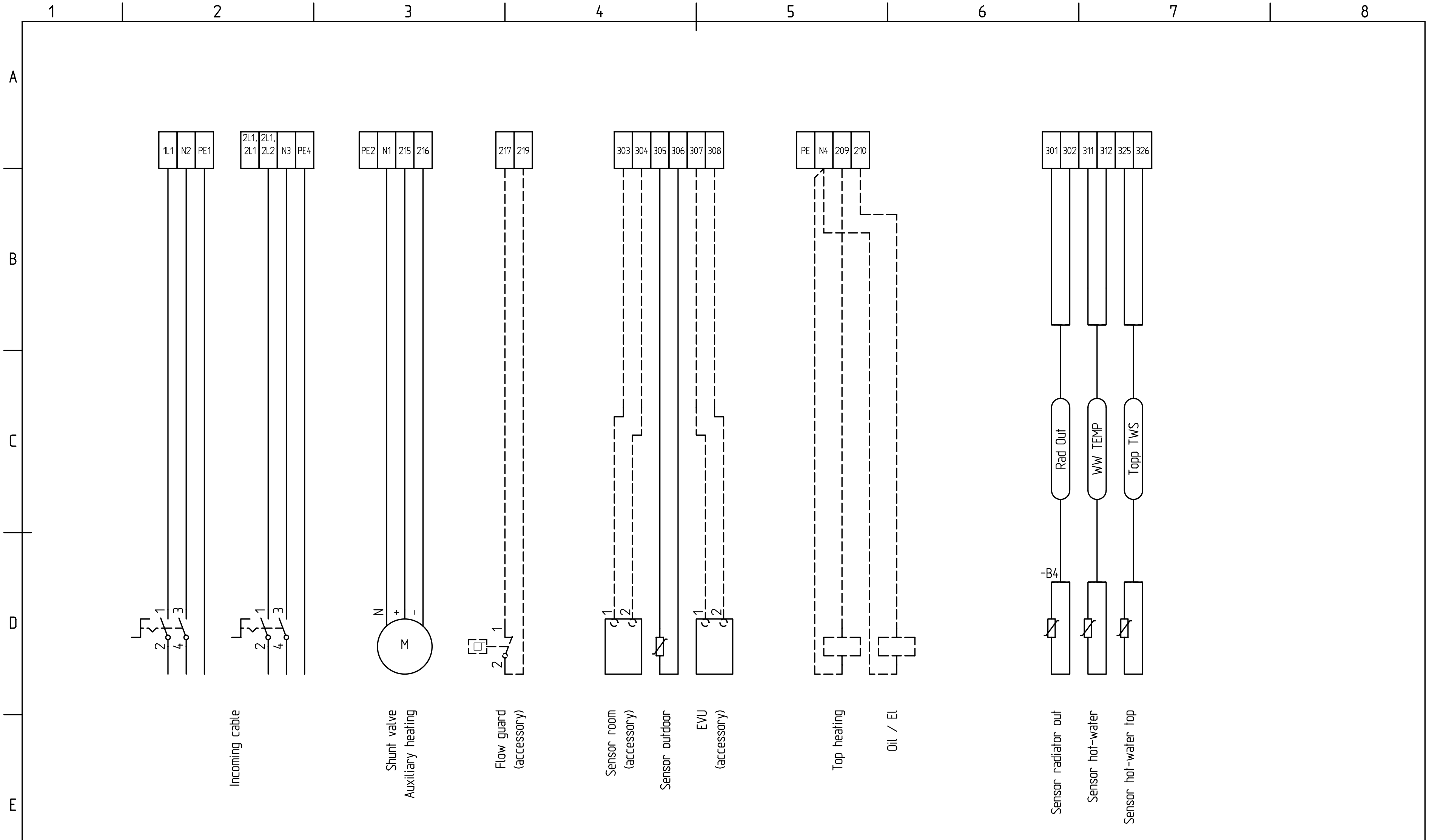
				DRAWING NR. 57283	SHEET 2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PE2 N1 215 216
217 219
303 304 305 306 307 308
PE N4 209 210
301 302 311 312 325 326
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1 3 2 4
N + -
M
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1 2
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 4ES 230V 1N External connections				CONTINUED 4
			DRAWING NR. 57283	SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	35 micF	9588-55265001	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	9-14A	9588-53794001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865001	/1.6E
-M2	1	Compressor	Copeland	230 V	9686-54808001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

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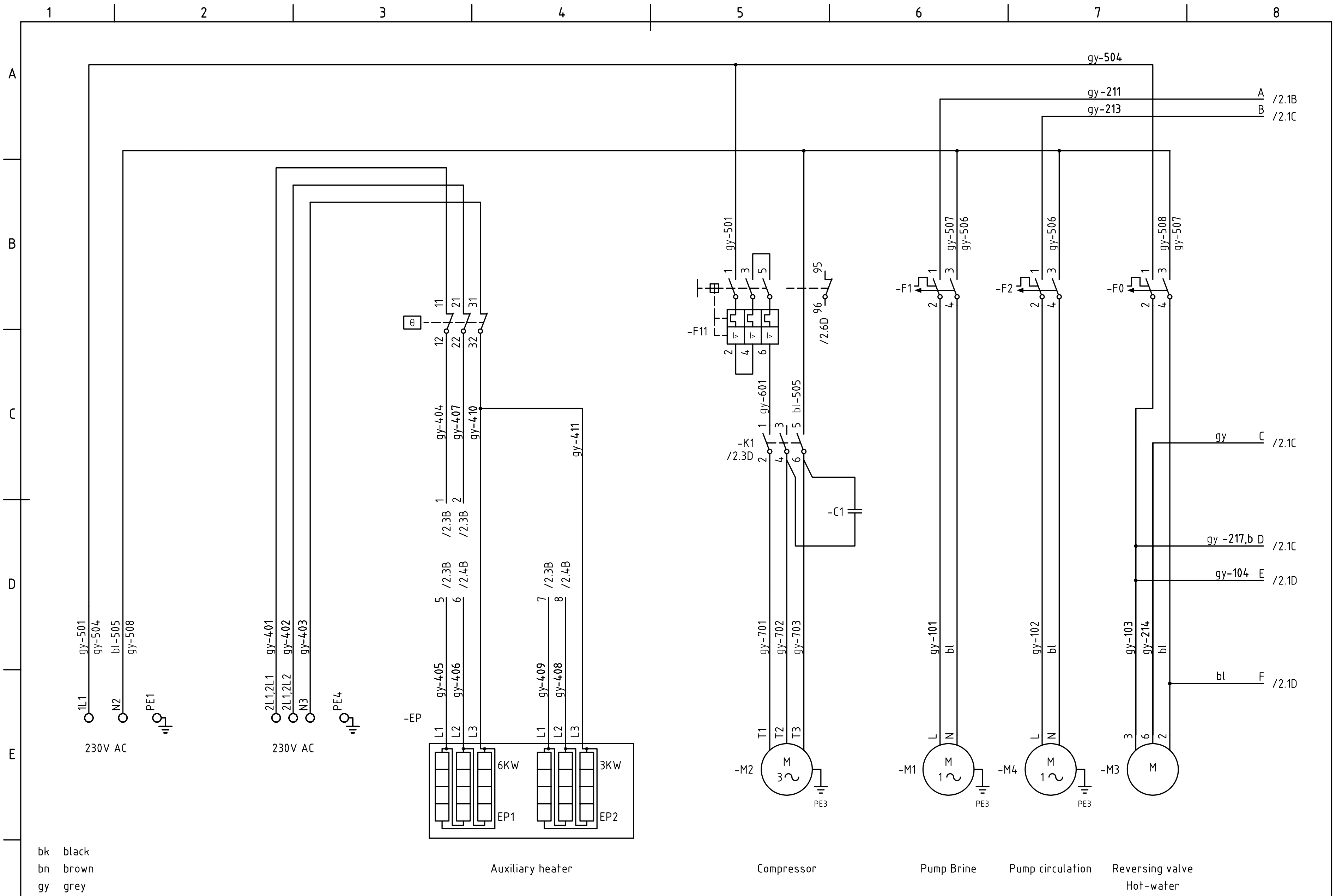
Duo 4ES 230V 1N Part list				CONTINUED -
			DRAWING NR. 57283	SHEET 4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bl-505
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-C1
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gy -217,b D /2.1C
gy-104 E /2.1D
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gy-103
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230V AC
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EP2
-M2
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PE3
-M4
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PE3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
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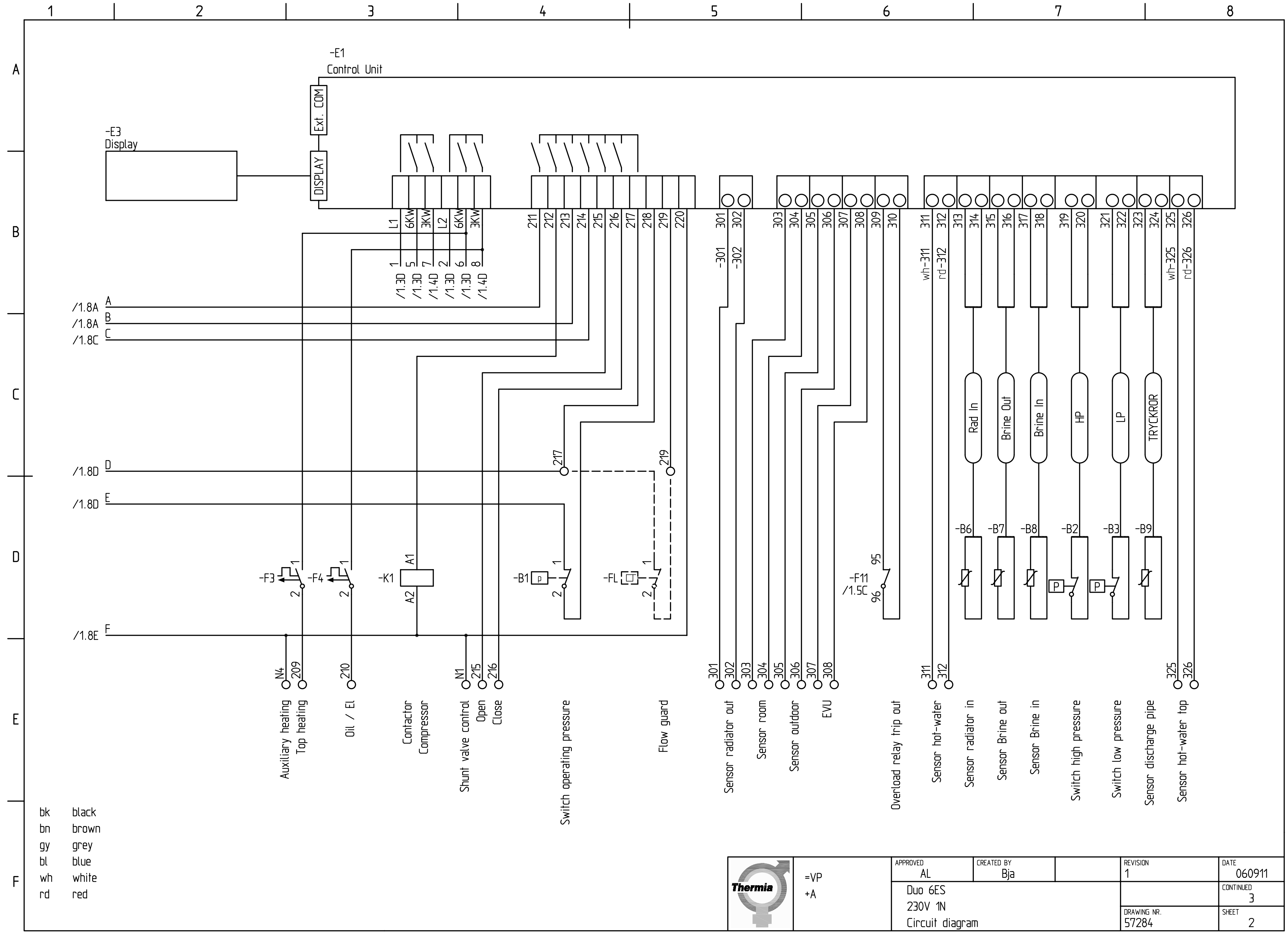
Duo 6ES 230V 1N Circuit diagram			DRAWING NR. 57284	CONTINUED 2 SHEET 1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Control Unit
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DISPLAY
-E3
Display
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3KW
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rd-312
wh-325
rd-326
/1.8A A
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/1.8C C
/1.8D D
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/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
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-FL
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-F11
/1.5C
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N4
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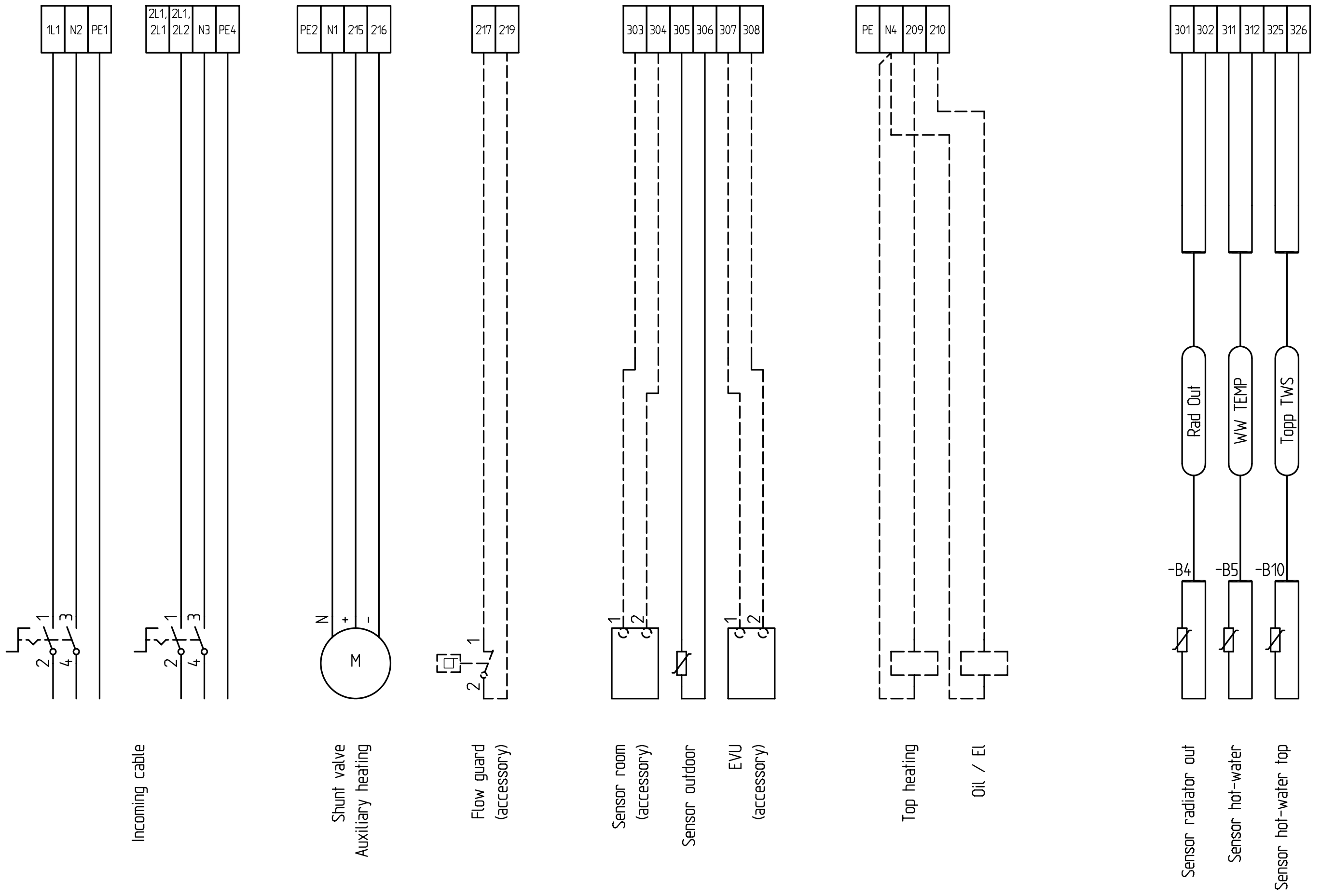
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
	Duo 6ES 230V 1N Circuit diagram		DRAWING NR. 57284	CONTINUED 3 SHEET 2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PE
N4
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312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
M
N
+
-
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
		Duo 6ES 230V 1N			CONTINUED 4
		External connections		DRAWING NR. 57284	SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	40 micF	9588-55265002	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	13-18A	9588-54715001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53310002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

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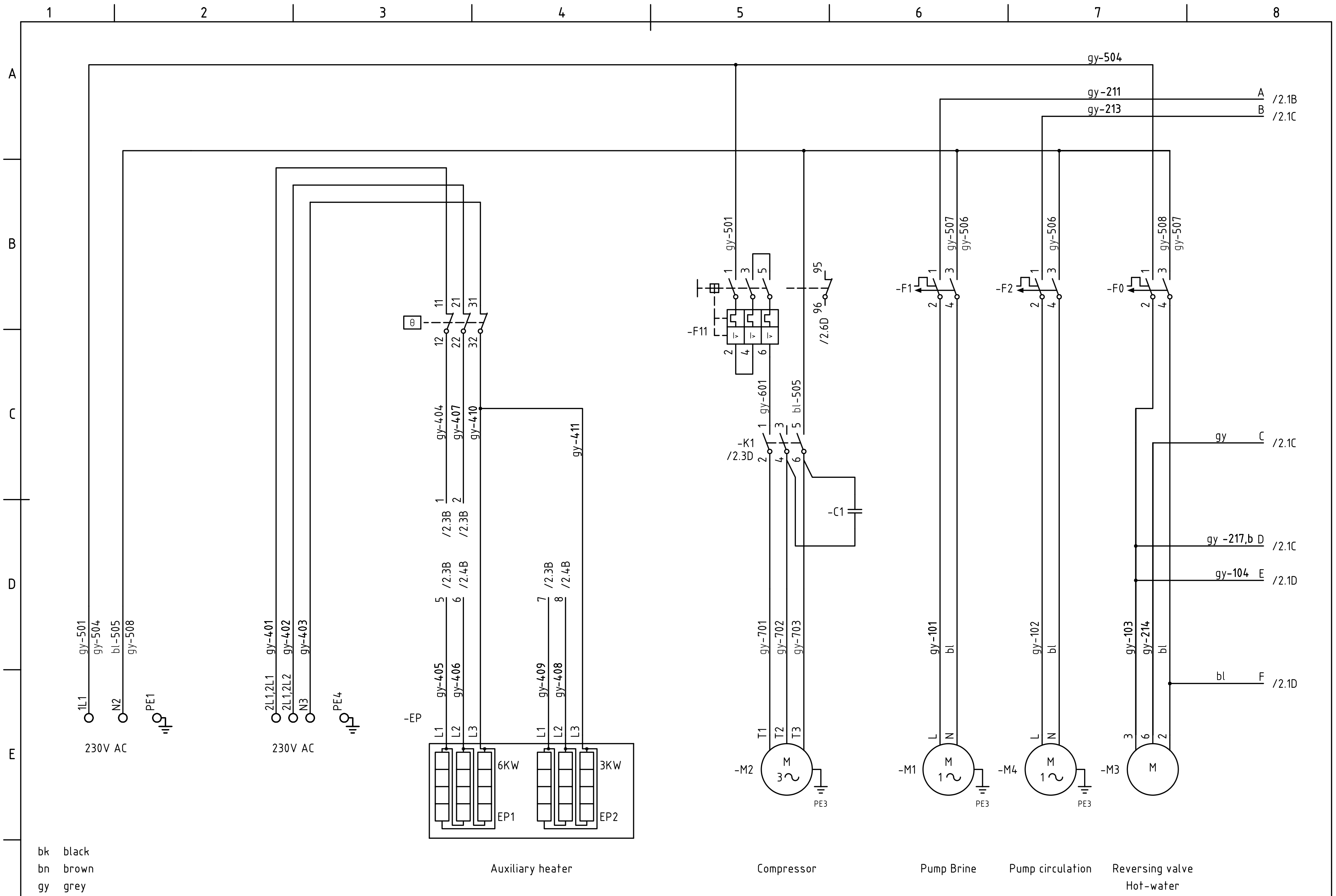
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gy-211
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gy-501
gy-507
gy-506
gy-508
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-F11
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/2.6D
-F1
-F2
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gy-404
gy-407
gy-410
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gy-601
bl-505
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-C1
gy C /2.1C
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6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-504
bl-505
gy-508
gy-401
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gy-403
gy-405
gy-406
gy-409
gy-408
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
gy-103
gy-214
bl
1L1
N2
PE1
2L1,2L1
2L1,2L2
N3
PE4
-EP
L1
L2
L3
T1
T2
T3
L
N
3
6
2
230V AC
230V AC
6KW
EP1
3KW
EP2
-M2
M
3~
PE3
-M1
M
1~
PE3
-M4
M
1~
PE3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

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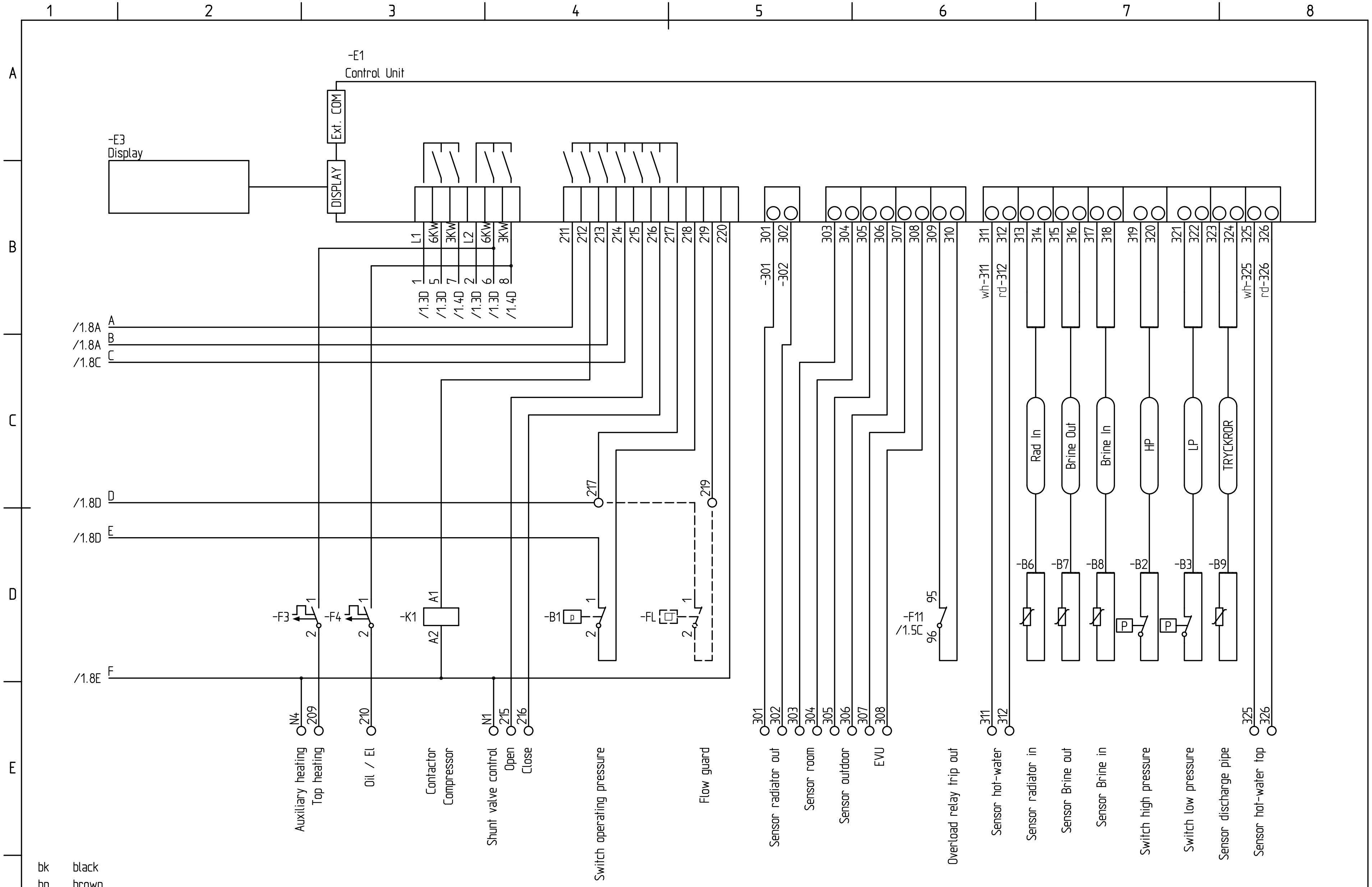
Duo 8ES 230V 1N Circuit diagram			DRAWING NR. 57285	CONTINUED 2 SHEET 1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A
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C
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F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
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/1.3D
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211
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rd-312
wh-325
rd-326
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/1.8A B
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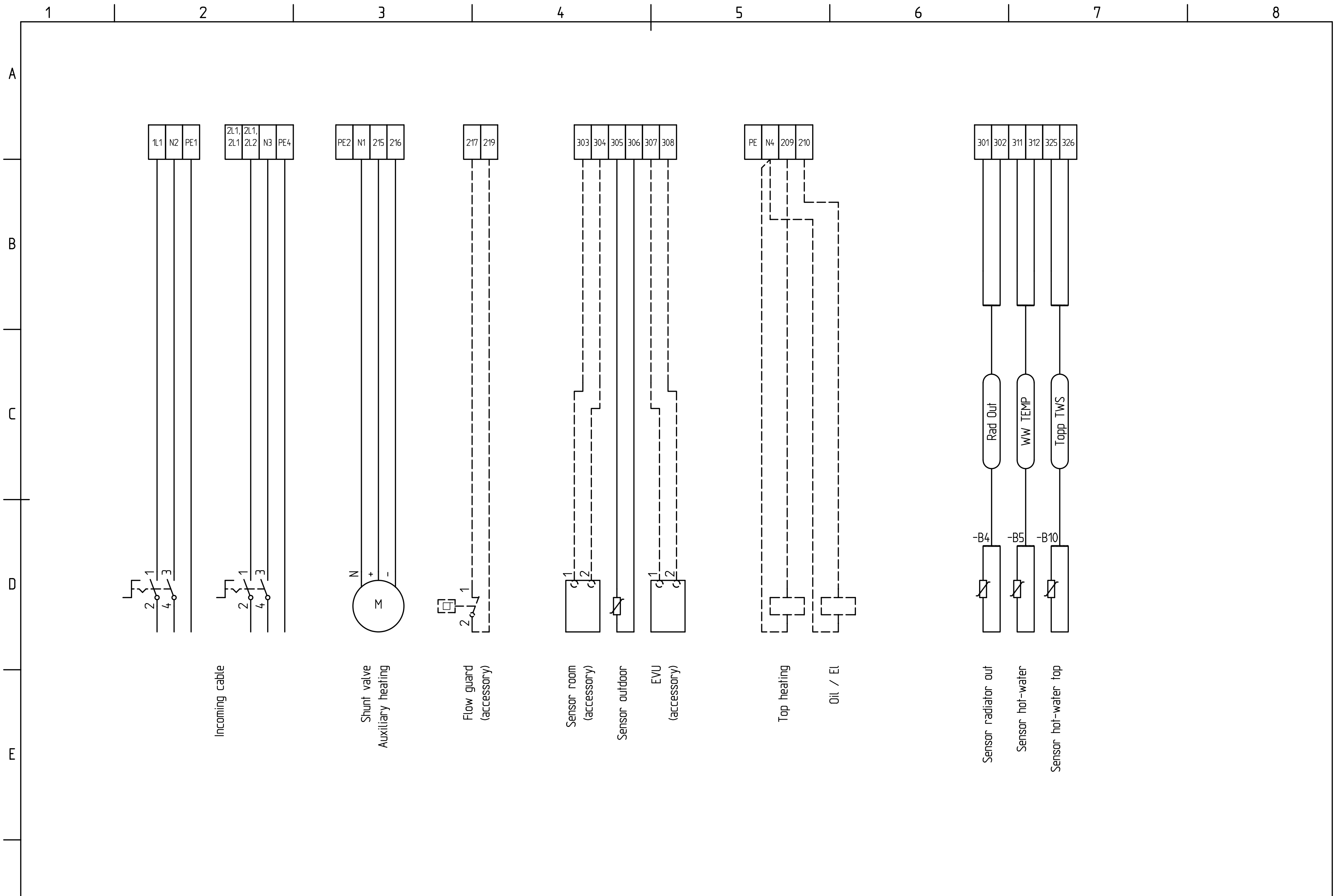
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
217
219
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 8ES 230V 1N Circuit diagram			DRAWING NR. 57285	CONTINUED 3 SHEET 2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A
B
C
D
E
F
1L1
N2
PE1
2L1, 2L1
2L1, 2L2
N3
PE4
PE2
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
N
+
-
M
Incoming cable
Shunt valve
Auxiliary heating
Flow guard (accessory)
Sensor room (accessory)
Sensor outdoor
EVU (accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

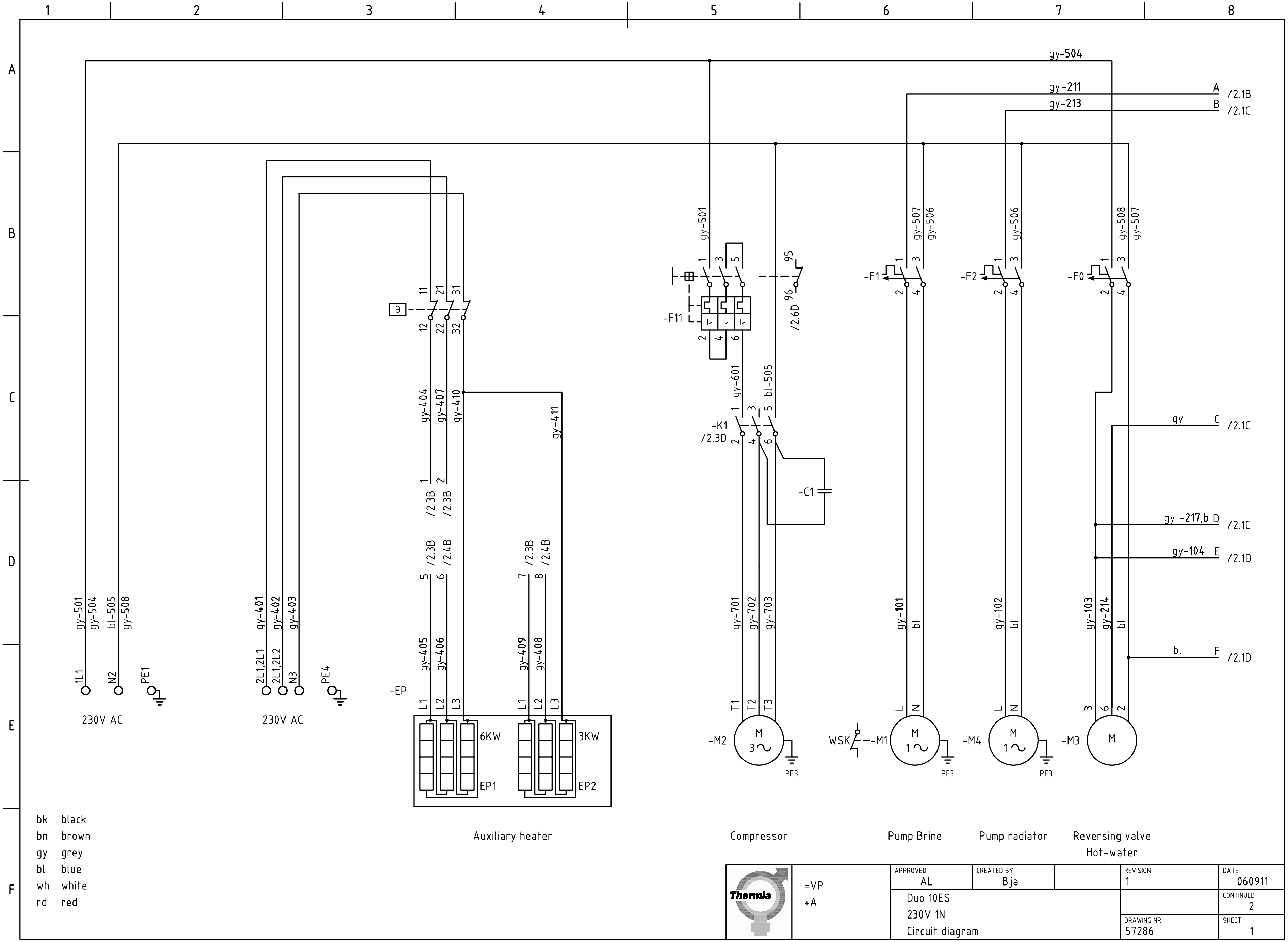
Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
		Duo 8ES 230V 1N External connections		DRAWING NR. 57285	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	50 micF	9588-55265004	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	17-23A	9588-55363001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53312002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 8ES 230V 1N Part list			CONTINUED	-
			DRAWING NR. 57285	SHEET 4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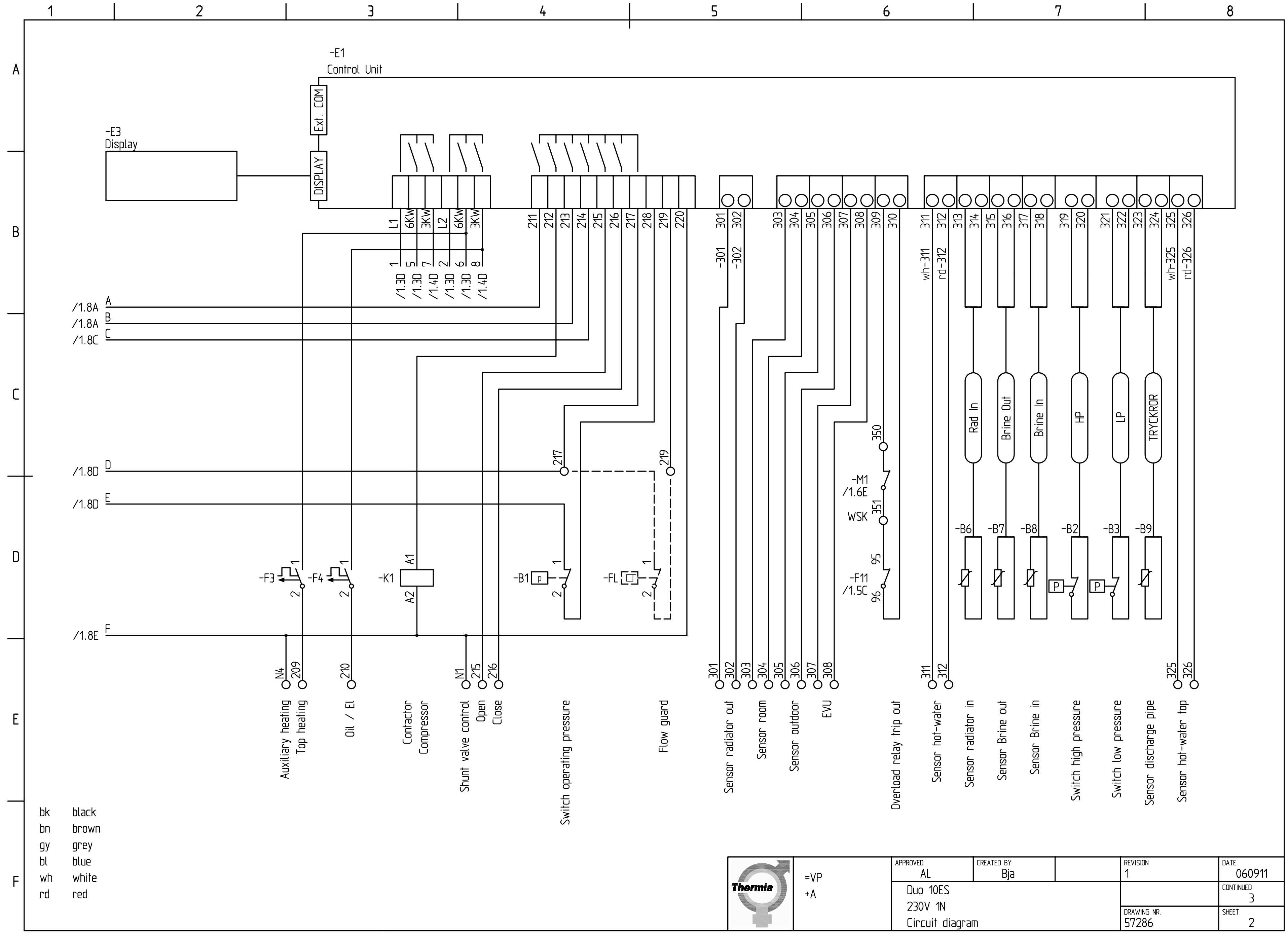
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
gy-507
gy-506
gy-506
gy-508
gy-507
-F11
95
96
/2.6D
-F1
-F2
-F0
θ
11 21 31
12 22 32
gy-404
gy-407
gy-410
gy-411
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-501
gy-504
bl-505
gy-508
gy-401
gy-402
gy-403
gy-405
gy-406
gy-409
gy-408
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
gy-103
gy-214
bl
1L1
N2
PE1
2L1,2L1
2L1,2L2
N3
PE4
-EP
L1 L2 L3
L1 L2 L3
T1 T2 T3
L N
L N
3 6 2
230V AC
230V AC
6KW
EP1
3KW
EP2
-M2
M
3~
PE3
WSK
-M1
M
1~
PE3
-M4
M
1~
PE3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump radiator
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 10ES				CONTINUED 2
230V 1N			DRAWING NR. 57286	SHEET 1
Circuit diagram				


1
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A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1 /1.3D
5 /1.3D
7 /1.4D
2 /1.3D
6 /1.3D
8 /1.4D
211
212
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311
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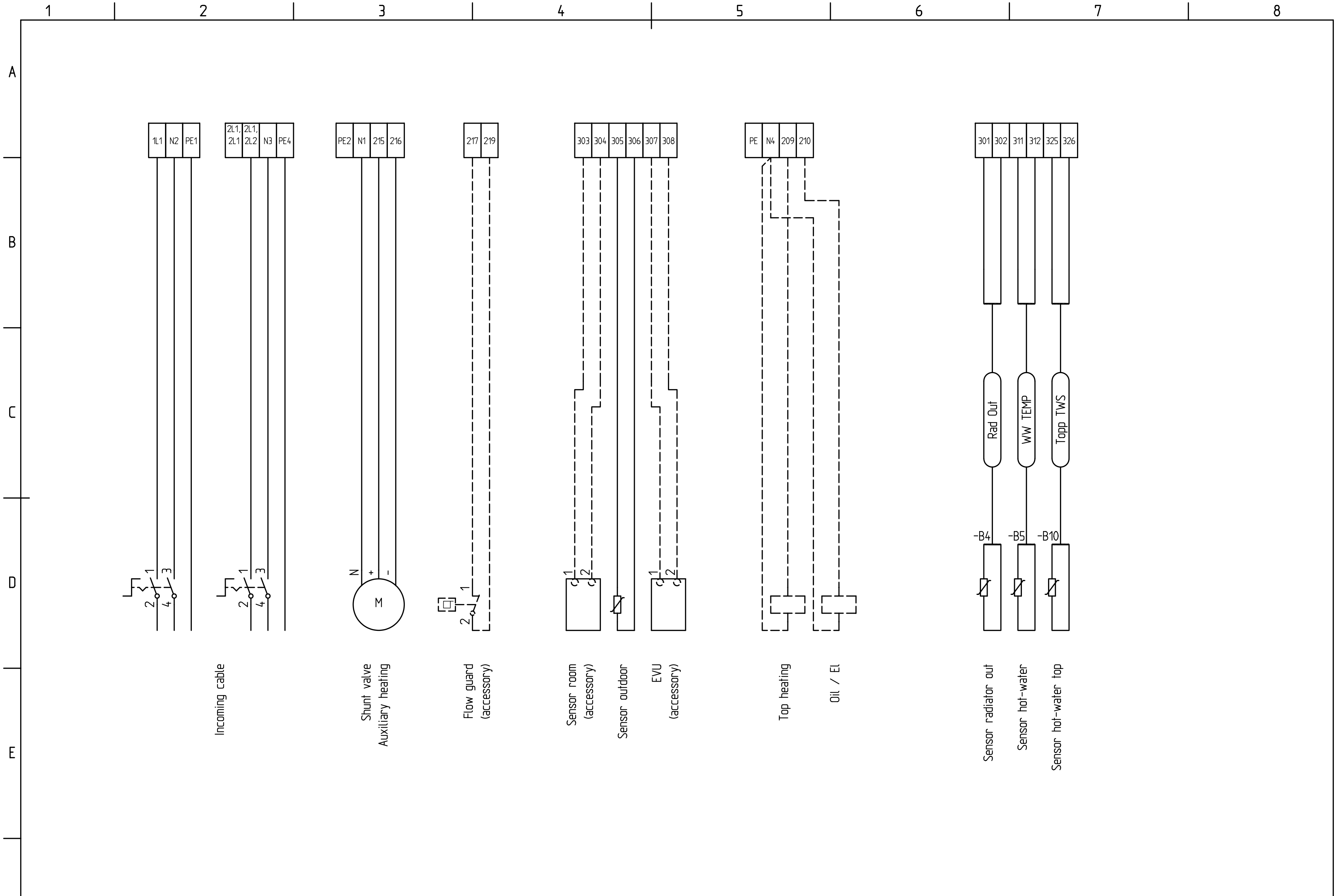
wh-311
rd-312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
350
-M1
/1.6E
351
WSK
95
-F11
/1.5C
96
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
217
219
N4
209
210
N1
215
216
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED
AL
CREATED BY
Bja
REVISION
1
DATE
060911
CONTINUED
3
Duo 10ES
230V 1N
Circuit diagram
DRAWING NR.
57286
SHEET
2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1L1
N2
PE1
2L1, 2L1
2L1, 2L2
N3
PE4
PE2
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
N
+
-
M
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911

Duo 10ES
230V 1N
External connections

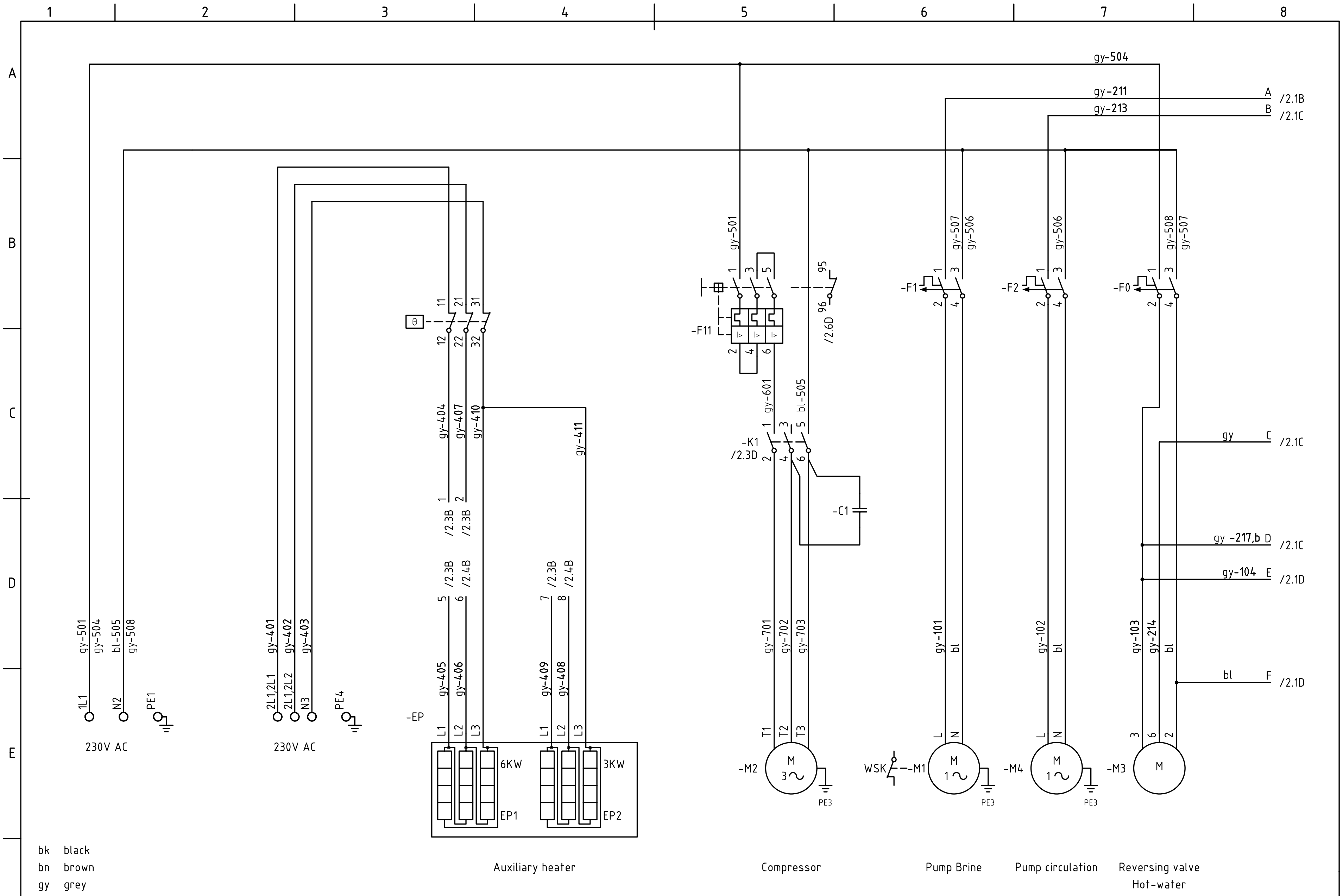
CONTINUED 4
DRAWING NR. 57286
SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	60 micF	9588-55265005	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	20-25A	9588-55546001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53313002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	
-X3	1	Terminal	Phoenix	WSK	9583-56967001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 10ES 230V 1N Part list			DRAWING NR. 57286	CONTINUED - SHEET 4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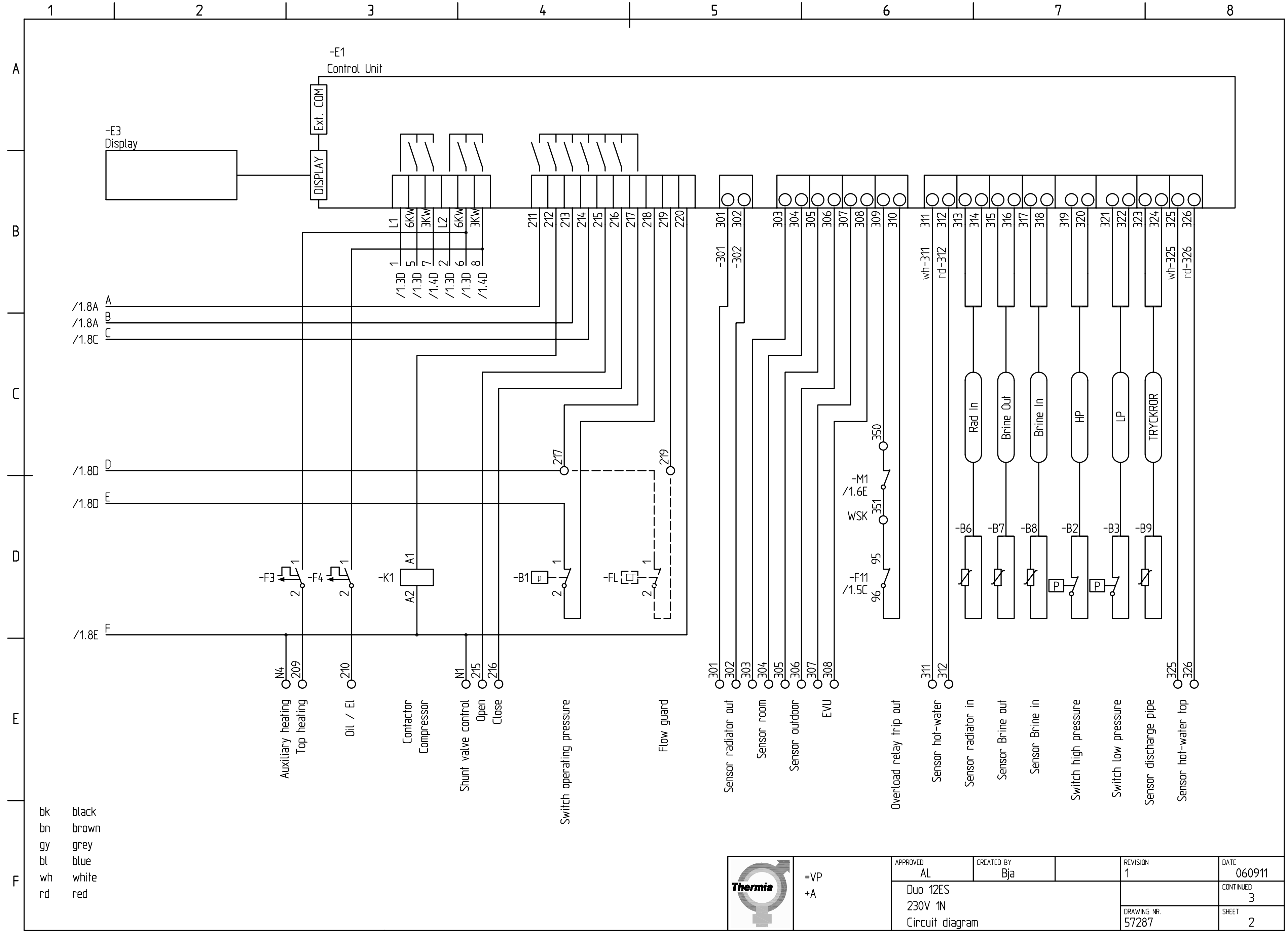
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
gy-507
gy-506
gy-508
gy-507
-F11
95
96
/2.6D
-F1
-F2
-F0
θ
11 21 31
12 22 32
gy-404
gy-407
gy-410
gy-411
1 2
/2.3B /2.3B
5 6 /2.3B /2.4B
7 8 /2.3B /2.4B
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501 gy-504 bl-505 gy-508
1L1 N2 PE1
230V AC
gy-401 gy-402 gy-403
2L1,2L1 2L1,2L2 N3 PE4
230V AC
gy-405 gy-406
gy-409 gy-408
-EP
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M
3~
PE3
gy-101 bl
L N
WSK
-M1
M
1~
PE3
gy-102 bl
L N
-M4
M
1~
PE3
gy-103 gy-214 bl
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
		Duo 12ES 230V 1N Circuit diagram			DRAWING NR. 57287	CONTINUED 2 SHEET 1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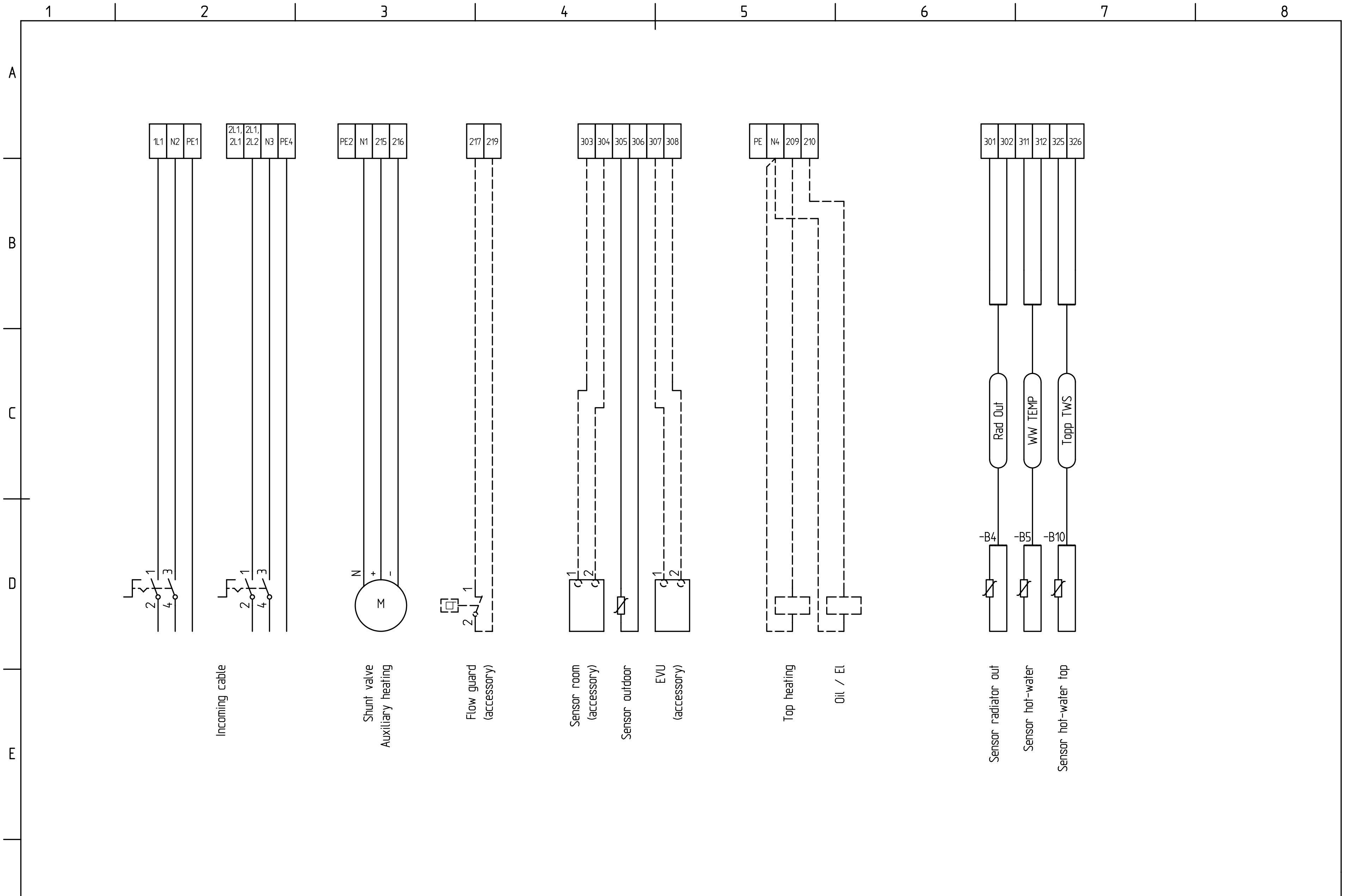
A B C D E F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1 6KW 3KW L2 6KW 3KW
1 5 7 2 6 8
/1.3D /1.3D /1.4D /1.3D /1.3D /1.4D
211 212 213 214 215 216 217 218 219 220
301 302 303 304 305 306 307 308 309 310
311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326
-301 -302
wh-311 rd-312
wh-325 rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
217 219
350
-M1
/1.6E
351
WSK
95
-F11
/1.5C
96
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6 -B7 -B8 -B2 -B3 -B9
P P
-F3 -F4 -K1 -B1 p -FL
1 2 A1 A2
N4 209 210 N1 215 216
301 302 303 304 305 306 307 308 311 312 325 326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
Duo 12ES
230V 1N
Circuit diagram
CONTINUED 3
DRAWING NR. 57287
SHEET 2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PE1
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2L1
2L1,
2L2
N3
PE4
PE2
N1
215
216
217
219
303
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306
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PE
N4
209
210
301
302
311
312
325
326
M
N
+
-
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
		Duo 12ES 230V 1N External connections			DRAWING NR. 57287	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	45 micF	9588-55265003	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	20-25A	9588-55546001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53314002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	
-X3	1	Terminal	Phoenix	WSK	9583-56967001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911

Duo 12ES
230V 1N
Part list

CONTINUED
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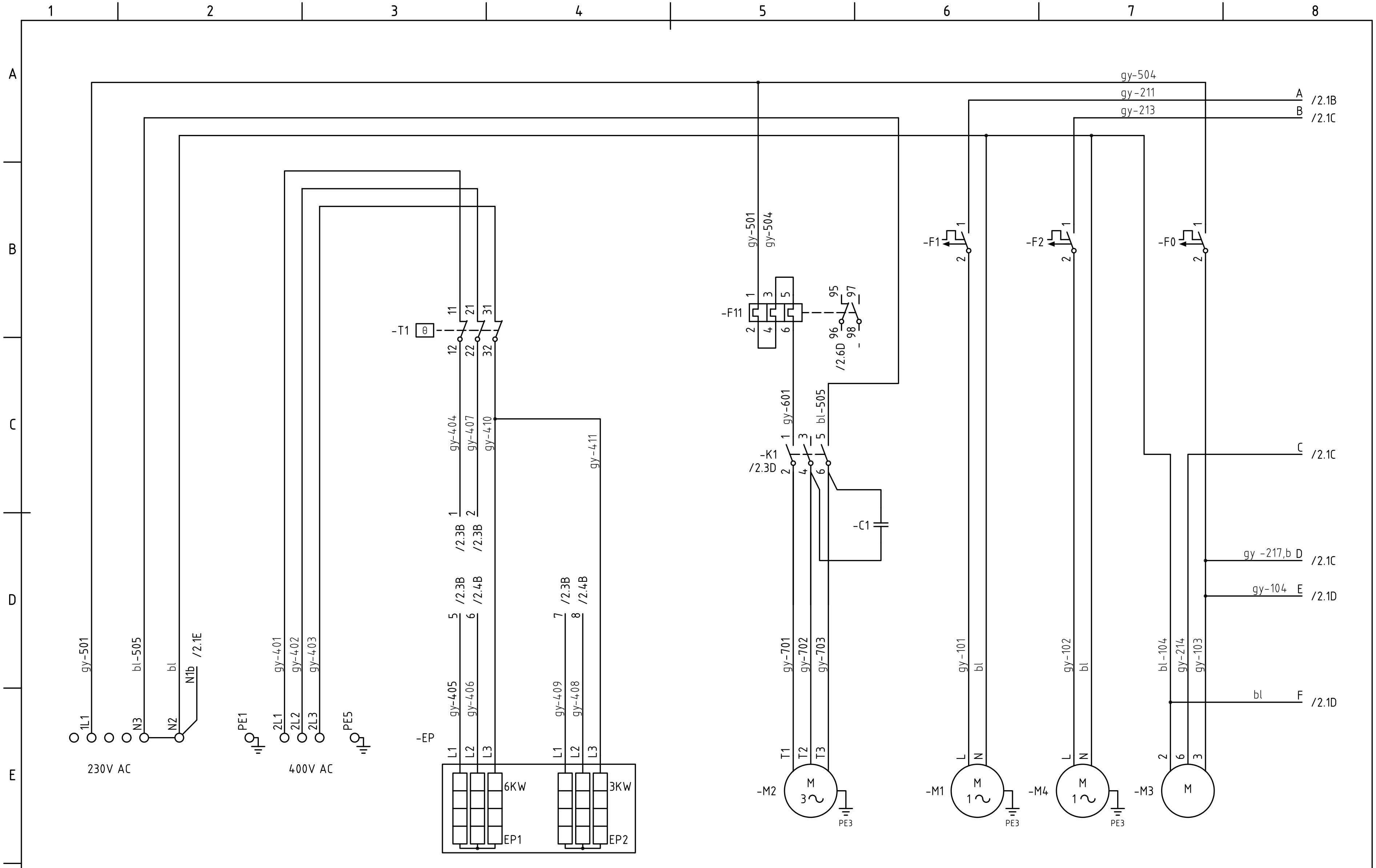
DRAWING NR.
57287

SHEET
4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gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
gy-501
gy-504
-F11
95
97
96
98
/2.6D
-T1 θ
11 21 31
12 22 32
gy-404
gy-407
gy-410
gy-411
gy-601
bl-505
-K1
/2.3D
-C1
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-501
bl-505
bl
N1b /2.1E
1L1
N3
N2
230V AC
PE1
2L1
2L2
2L3
PE5
gy-401
gy-402
gy-403
400V AC
-EP
gy-405
gy-406
gy-409
gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2 M 3~
PE3
gy-101
bl
L N
-M1 M 1~
gy-102
-M4 M 1~
bl-104
gy-214
gy-103
2 6 3
-M3 M

Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

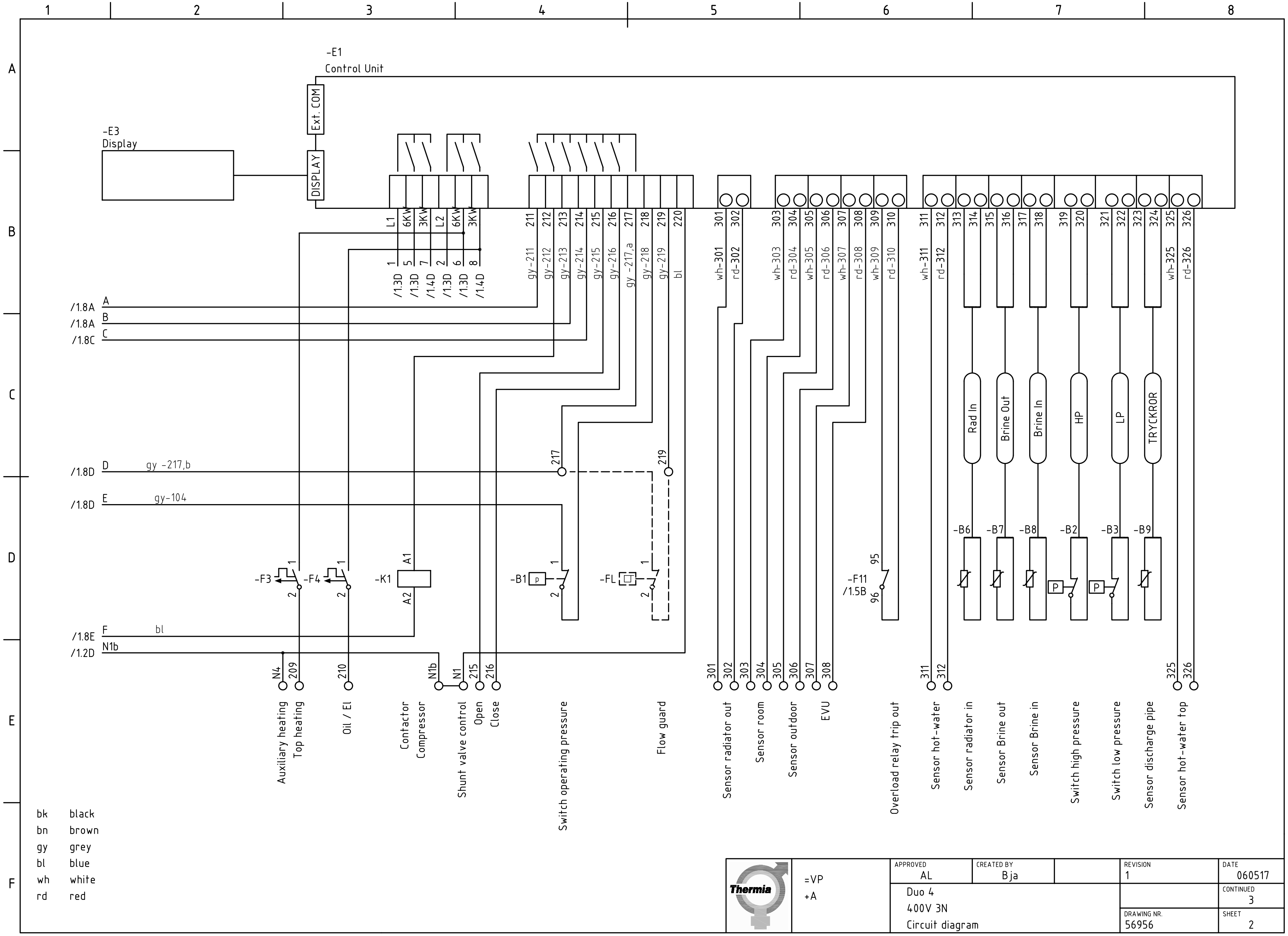
bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Note! Can be connected to 230V single phase
if auxiliary heater is disconnected.

Thermia

=VP
+A

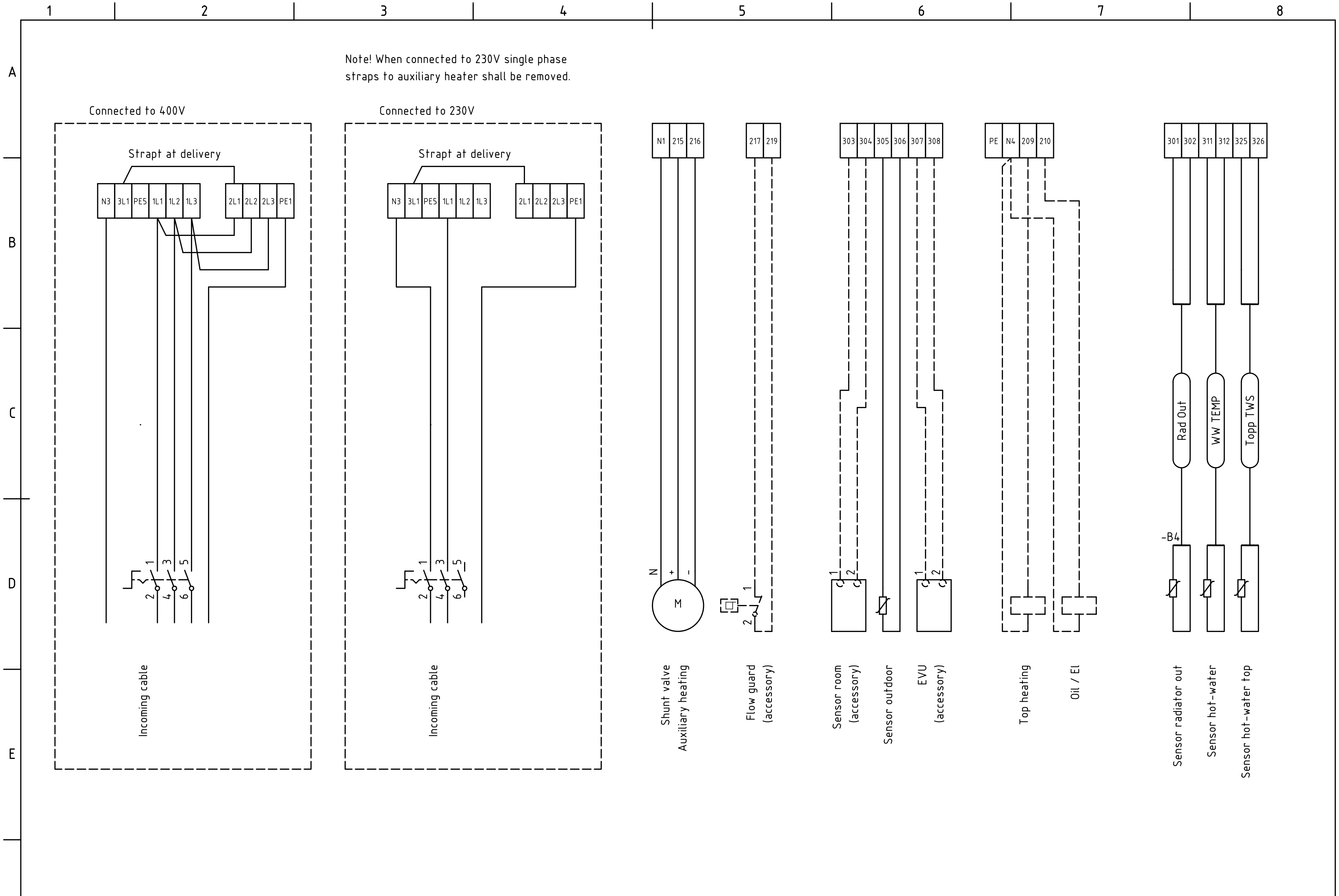
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060517
Duo 4 400V 3N Circuit diagram				CONTINUED 2
			DRAWING NR. 56956	SHEET 1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
gy-211
gy-212
gy-213
gy-214
gy-215
gy-216
gy -217,a
gy-218
gy-219
bl
301
302
wh-301
rd-302
303
304
305
306
307
308
309
310
wh-303
rd-304
wh-305
rd-306
wh-307
rd-308
wh-309
rd-310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D gy -217,b
/1.8D E gy-104
/1.8E F bl
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
217
219
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060517
Duo 4 400V 3N Circuit diagram			DRAWING NR. 56956	CONTINUED 3 SHEET 2



Note! When connected to 230V single phase straps to auxiliary heater shall be removed.

1
2
3
4
5
6
7
8
A
B
C
D
E
F
Connected to 400V
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
Incoming cable
Connected to 230V
Strapt at delivery
Incoming cable
N1
215
216
N
+
-
M
Shunt valve
Auxiliary heating
217
219
Flow guard
(accessory)
303
304
305
306
307
308
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
PE
N4
209
210
Top heating
Oil / El
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
Sensor radiator out
Sensor hot-water
Sensor hot-water top

=VP
+A

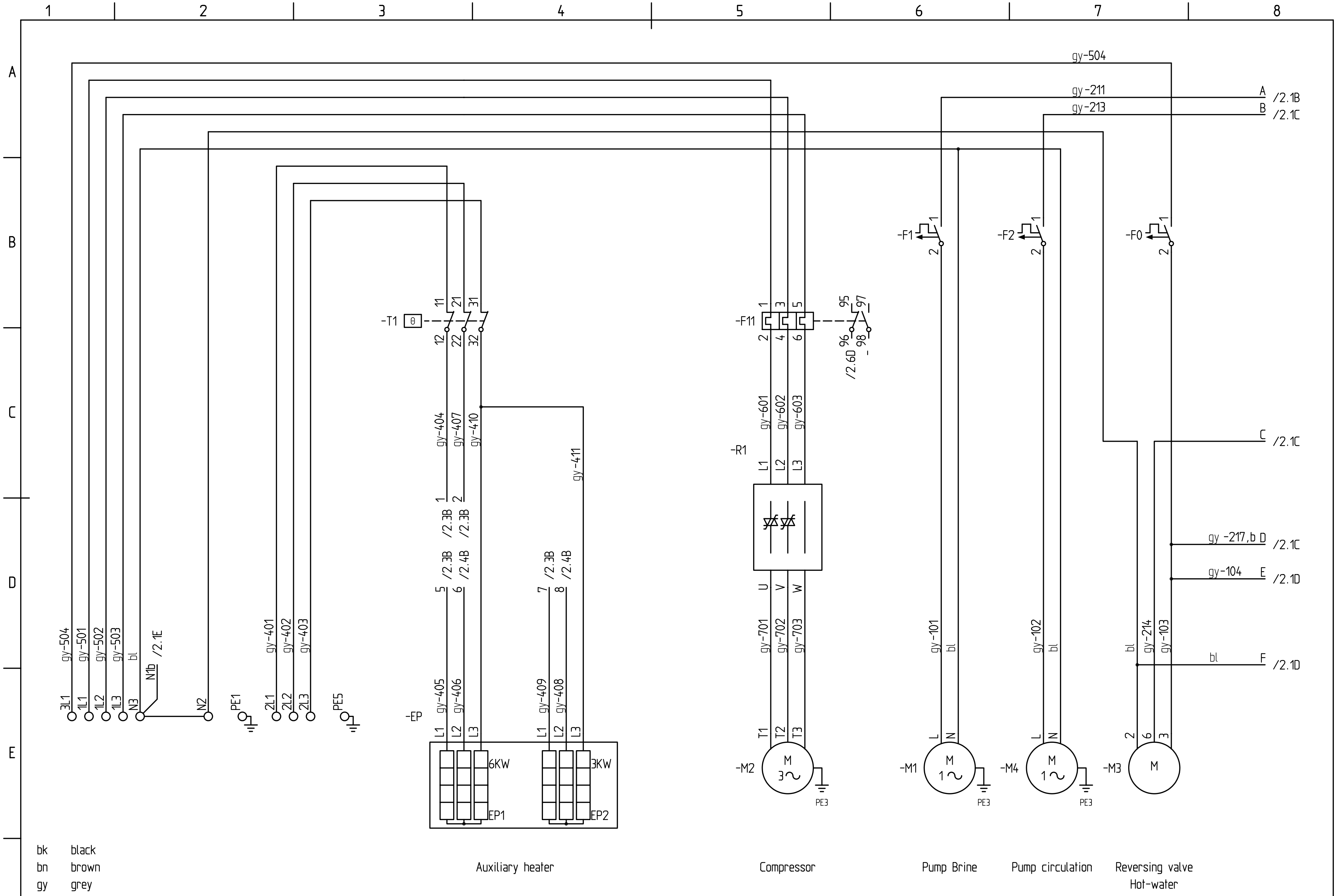
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060517
Duo 4 400V 3N External connections			DRAWING NR. 56956	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	35 micF	9588-55265001	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	9-13A	9588-38379012	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5B
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865001	/1.6E
-M2	1	Compressor	Copeland	230 V	9686-54808001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat	9583-55836001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060517
Duo 4 400V 3N Part list			DRAWING NR. 56956	CONTINUED - SHEET 4

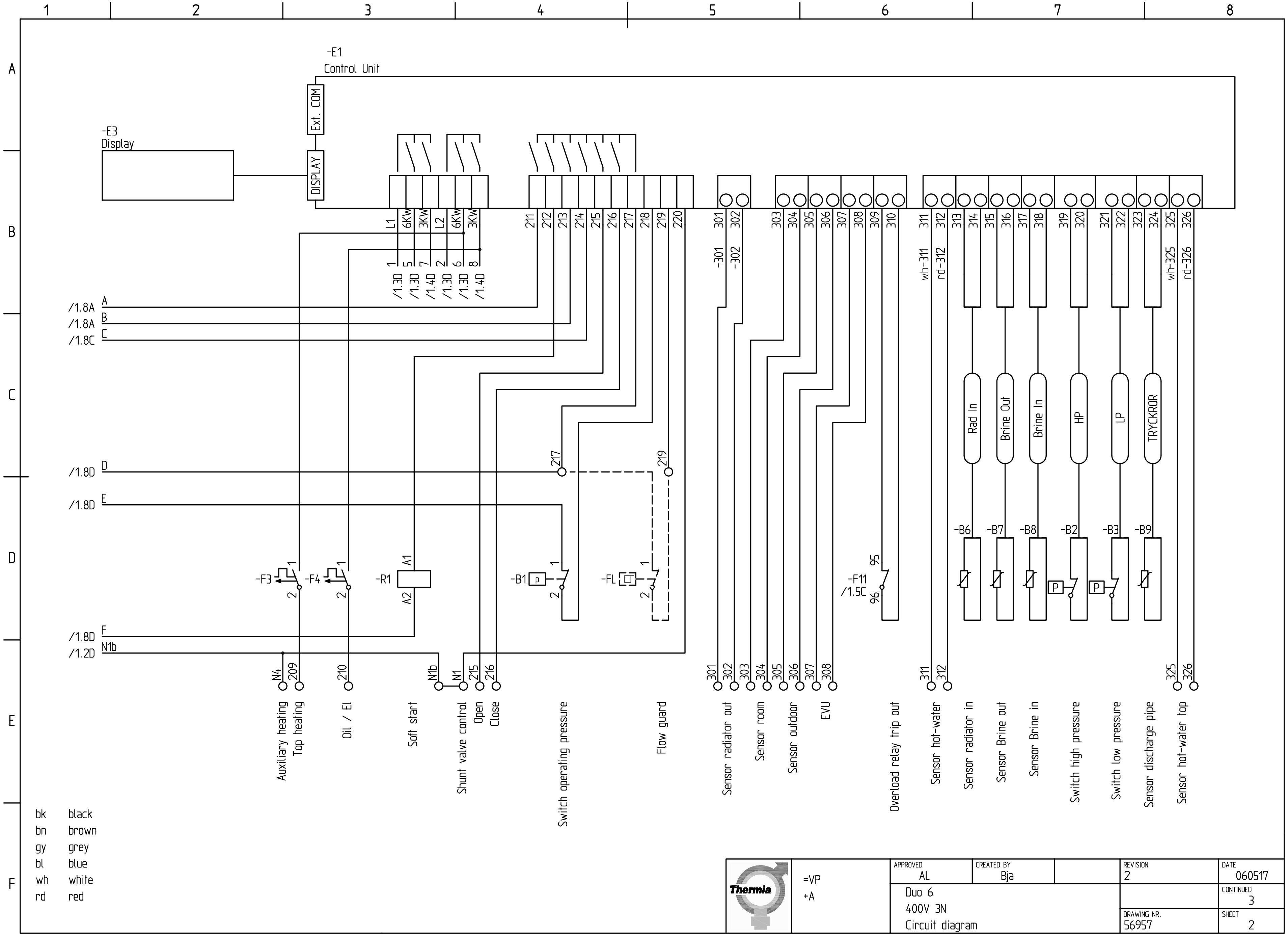

1
2
3
4
5
6
7
8
A
B
C
D
E
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1 L2 L3
U V W
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3 N2
PE1
gy-401
gy-402
gy-403
2L1 2L2 2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M 3~
PE3
gy-101
bl
L N
-M1
M 1~
PE3
gy-102
bl
L N
-M4
M 1~
PE3
bl
gy-214
gy-103
2 6 3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

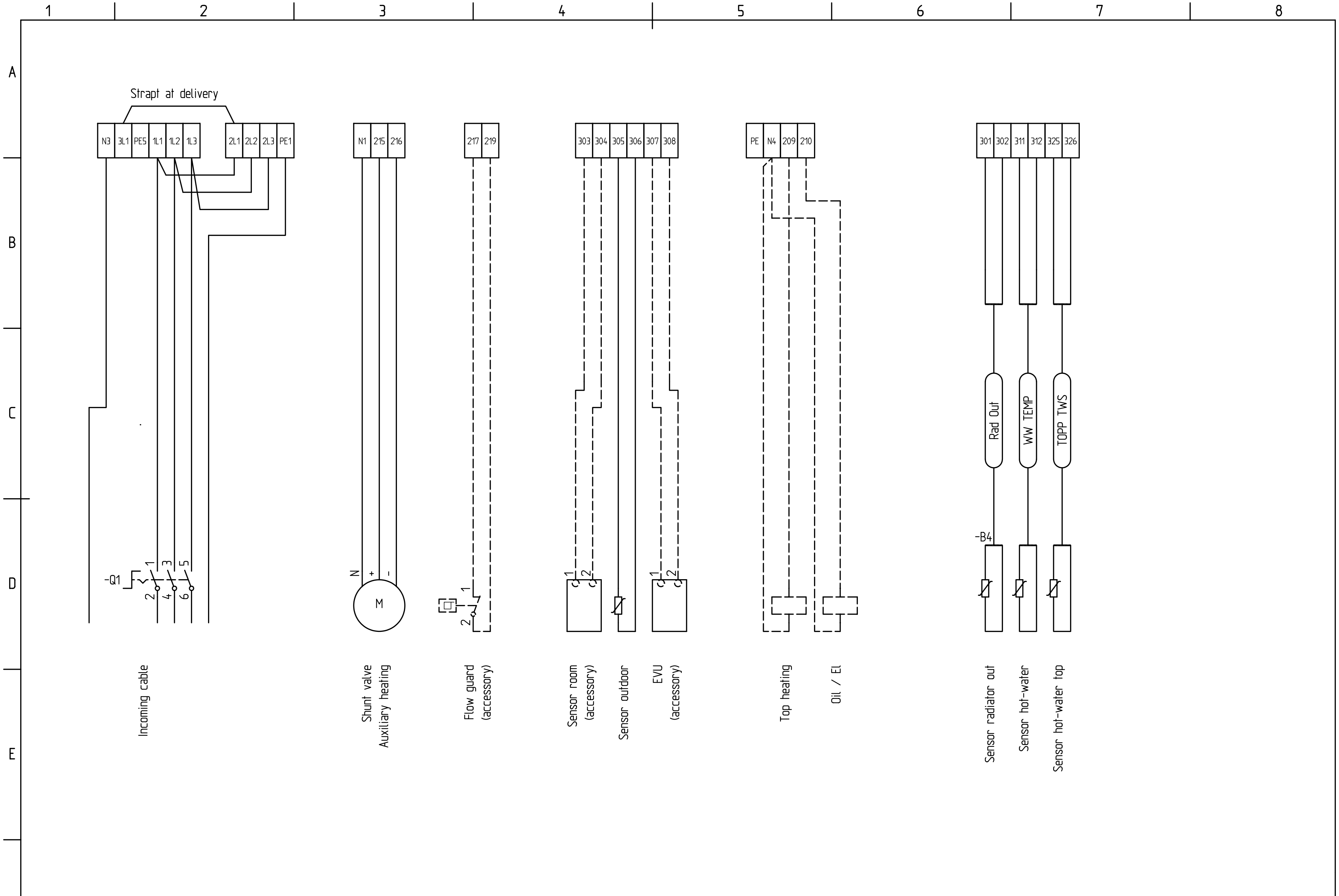
APPROVED AL	CREATED BY AL		REVISION 2	DATE 060821
Duo 6 400V 3N Circuit diagram			DRAWING NR. 56957	CONTINUED 2 SHEET 1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
217
219
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		2	060517
Duo 6				CONTINUED 3
400V 3N			DRAWING NR. 56957	SHEET 2
Circuit diagram				


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
TOPP TWS
-B4
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

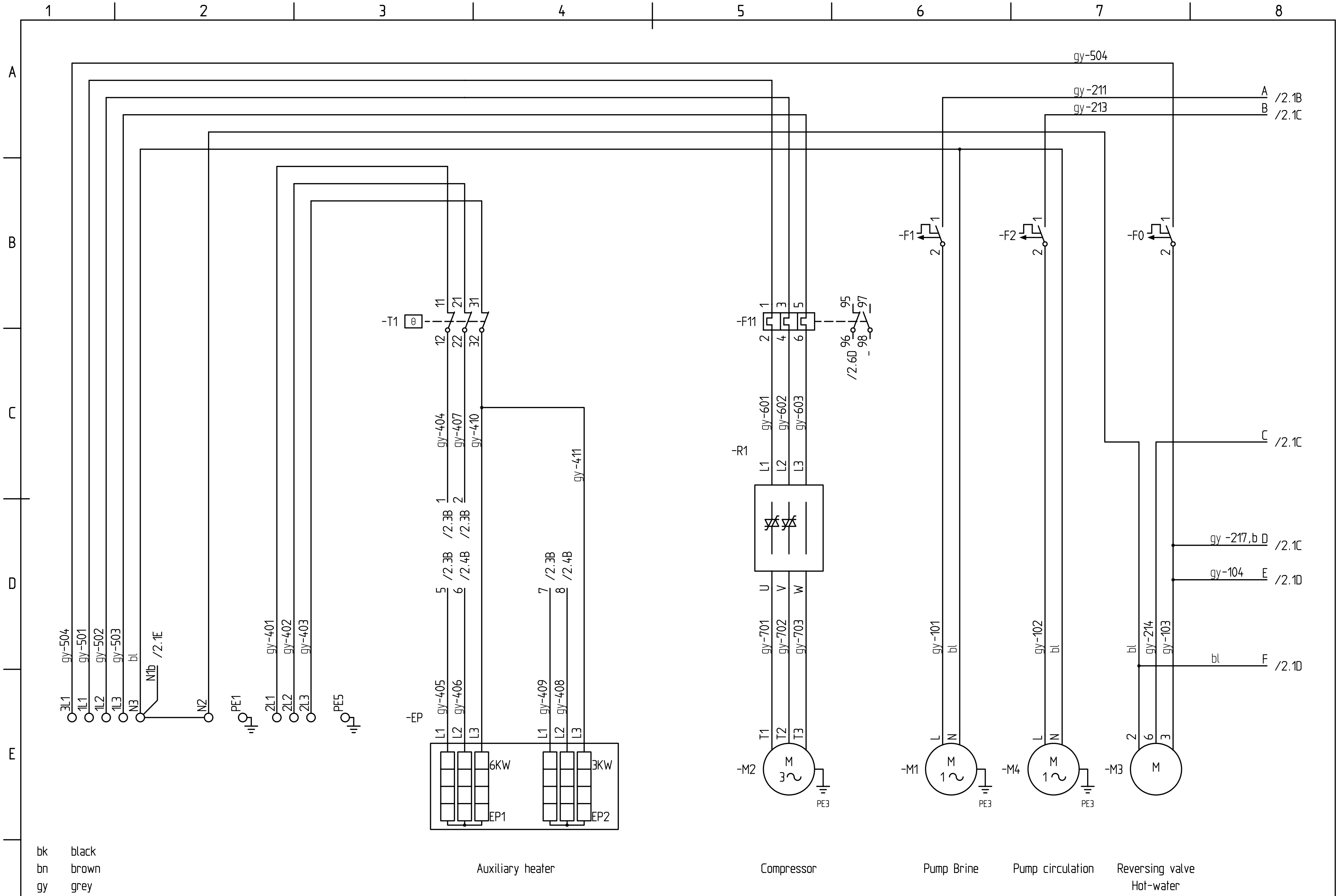
		APPROVED	CREATED BY		REVISION	DATE
Thermia	=VP +A	AL	Bja		2	060517
		Duo 6 400V 3N External connections			DRAWING NR. 56957	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	4-6A	9588-38379009	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-53865002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53310001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		2	061018
Duo 6 400V 3N Part list			DRAWING NR. 56957	CONTINUED - SHEET 4

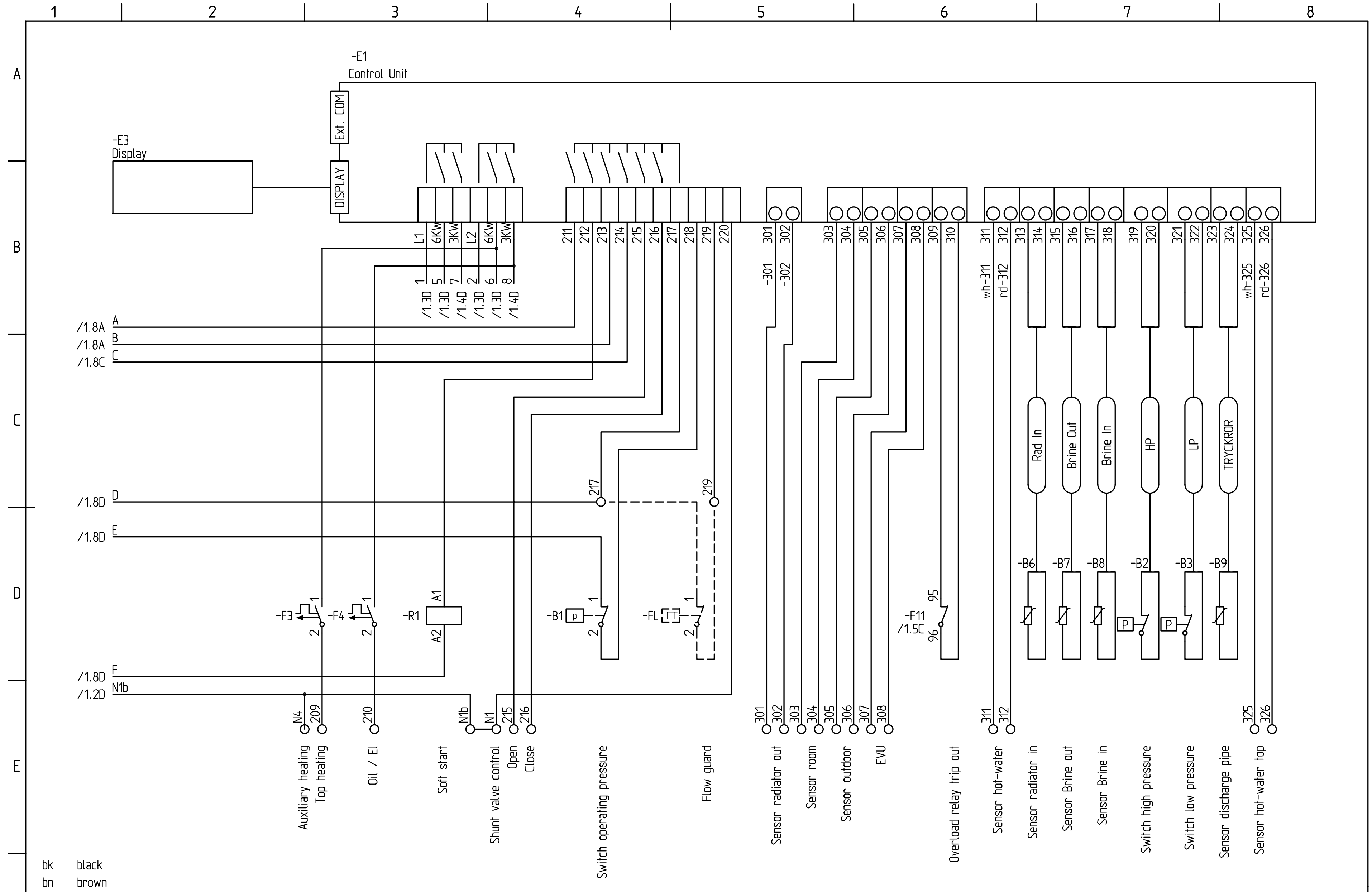

1
2
3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
1
2
-T1
θ
11
21
31
12
22
32
-F11
1
3
5
2
4
6
95
97
96
98
/2.6D
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1
L2
L3
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
U
V
W
gy-217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1
1L1
1L2
1L3
N3
N2
PE1
gy-401
gy-402
gy-403
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1
L2
L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1
T2
T3
-M2
M
3~
PE3
gy-101
bl
L
N
-M1
M
1~
PE3
gy-102
bl
L
N
-M4
M
1~
PE3
bl
gy-214
gy-103
2
6
3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

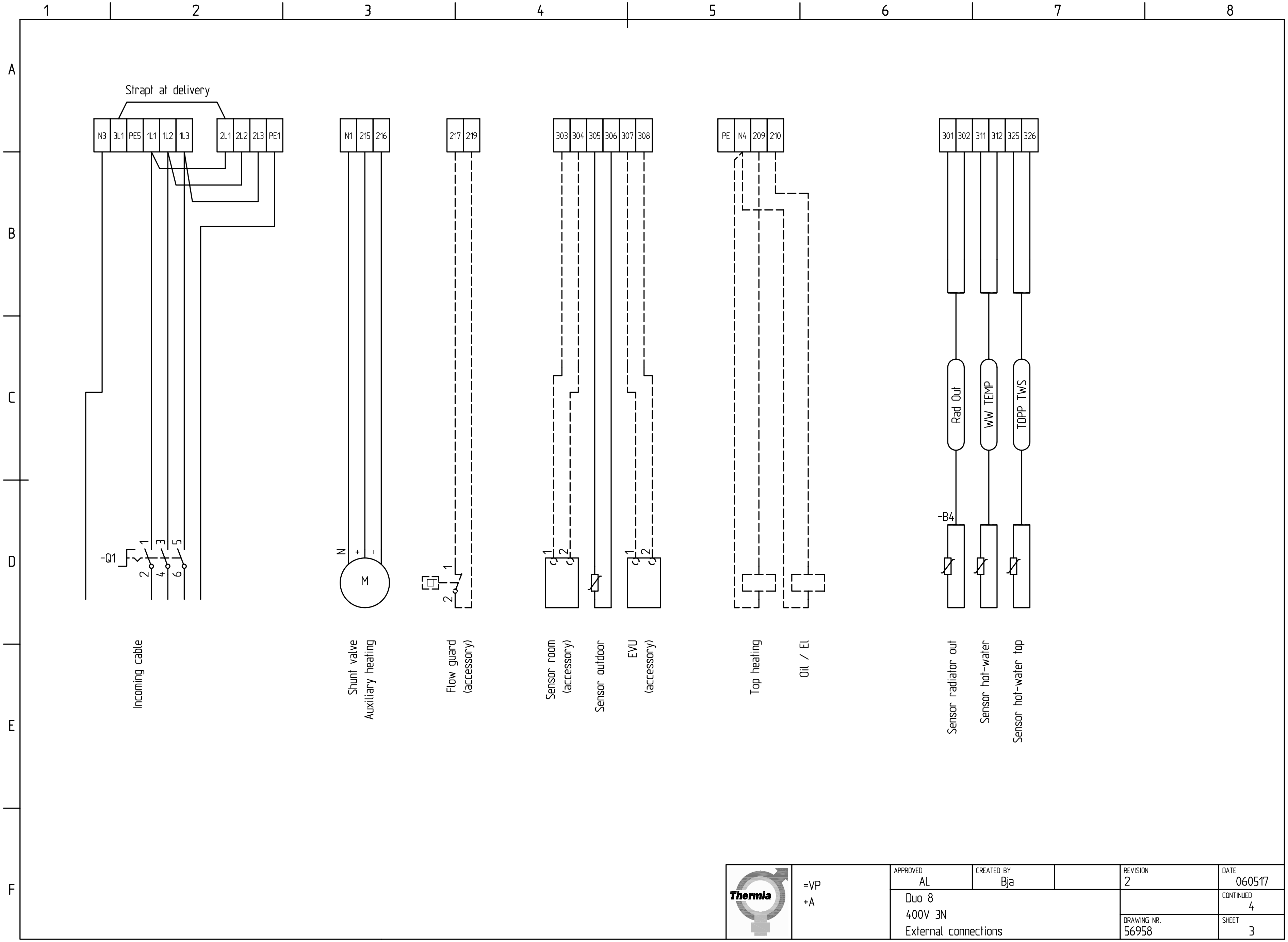
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		2	060821
DHP-L 8				CONTINUED 2
400V 3N			DRAWING NR.	SHEET
Circuit diagram			56958	1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1b
N1
215
216
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

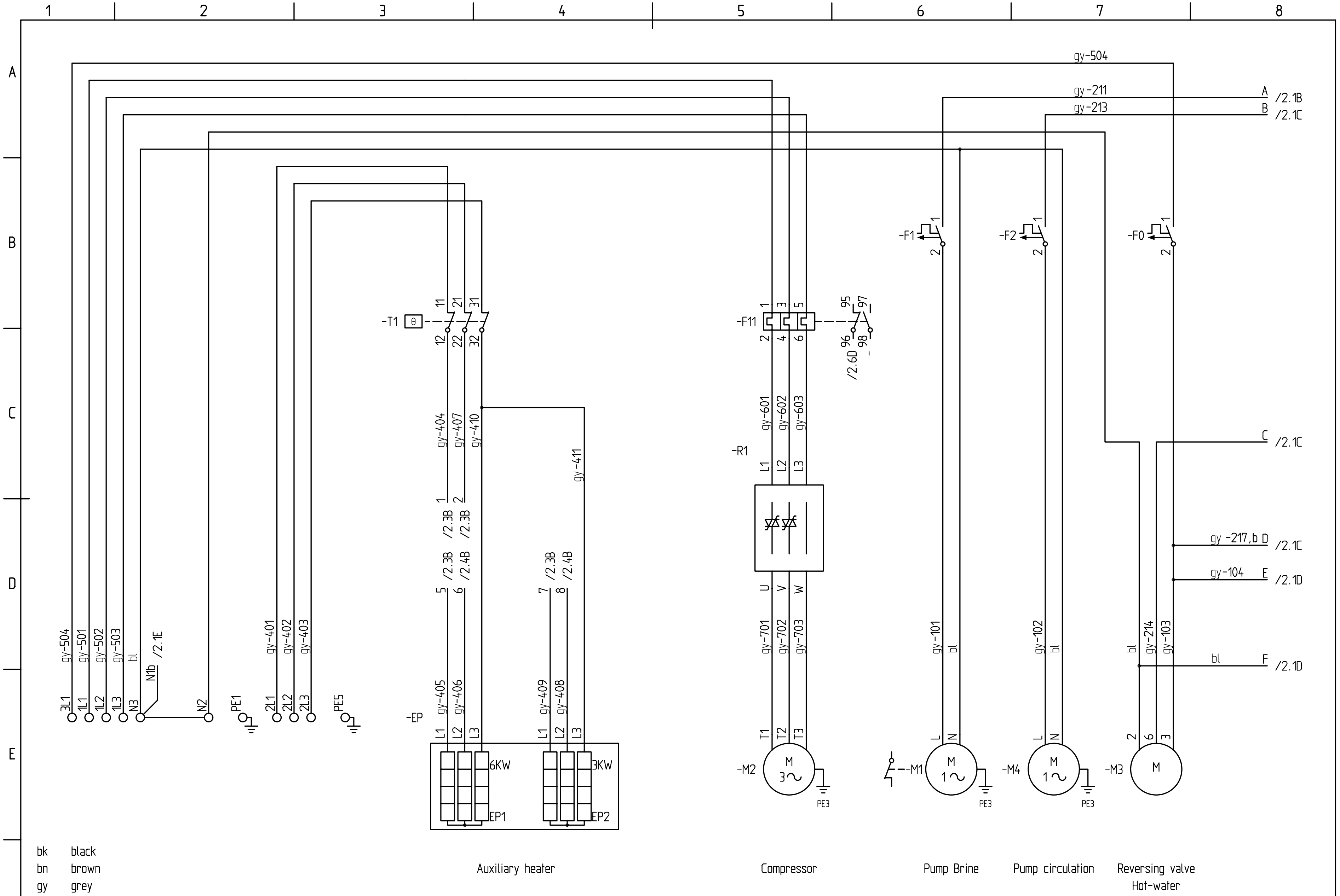
Thermia
=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		2	060517
Duo 8 400V 3N Circuit diagram				CONTINUED 3
			DRAWING NR. 56958	SHEET 2


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Rad Out
WW TEMP
TOPP TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia
=VP
+A
APPROVED
AL
CREATED BY
Bja
REVISION
2
DATE
060517
Duo 8
400V 3N
External connections
CONTINUED
4
DRAWING NR.
56958
SHEET
3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	5,5-8A	9588-38379010	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-53865002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53312001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 061018
		Duo 8 400V 3N Part list		DRAWING NR. 56958	CONTINUED - SHEET 4


1
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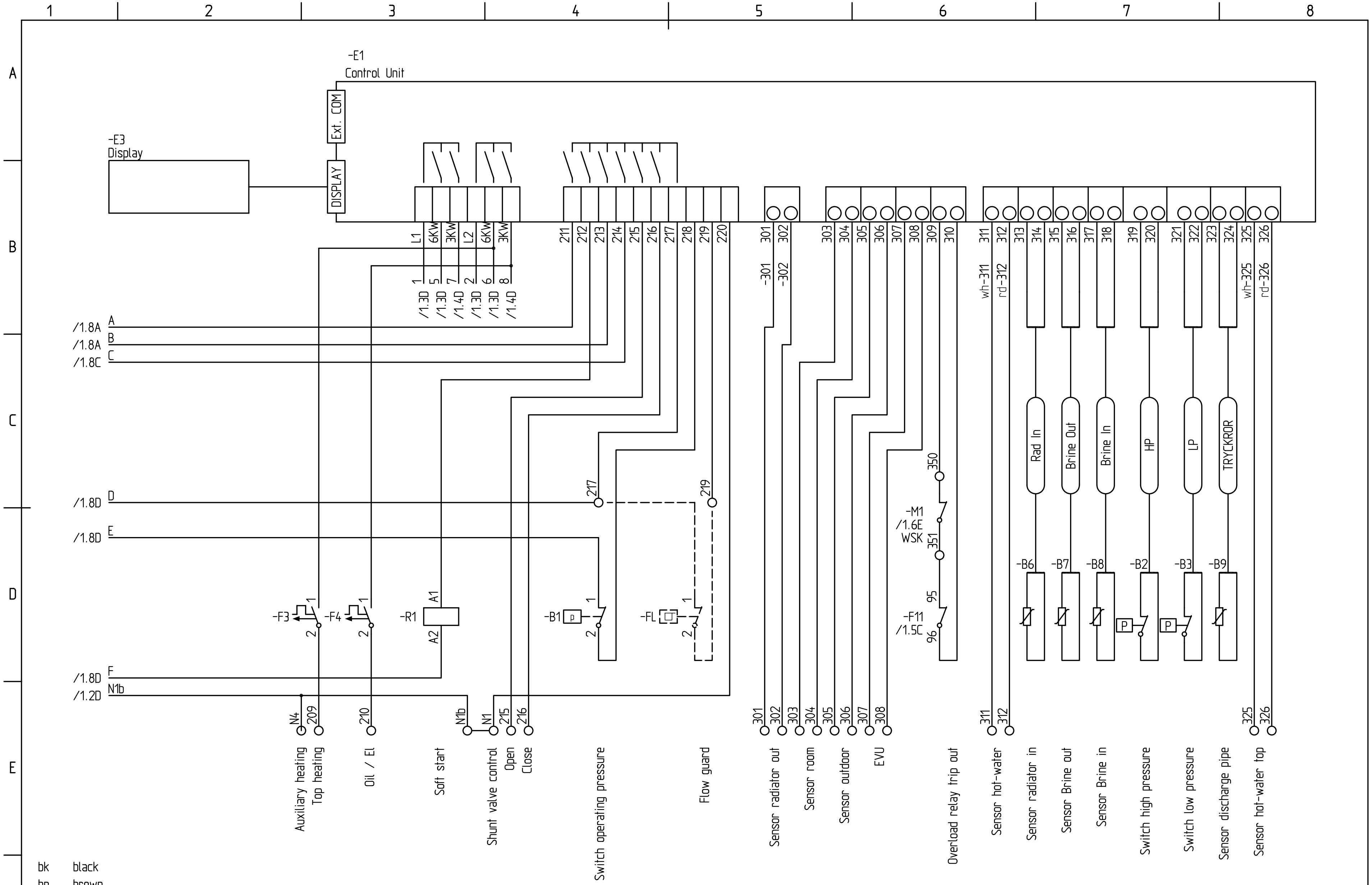
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
gy-601 gy-602 gy-603
-R1
L1 L2 L3
U V W
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M 3~
PE3
C /2.1C
gy-404 gy-407 gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504 gy-501 gy-502 gy-503 bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3 N2 PE1
gy-401 gy-402 gy-403
2L1 2L2 2L3 PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-101 bl
gy-102 bl
bl gy-214 gy-103
L N
2 6 3
-M1
M 1~
-M4
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 2
DATE 060821
CONTINUED 2
Duo 10
400V 3N
Circuit diagram
DRAWING NR. 56959
SHEET 1

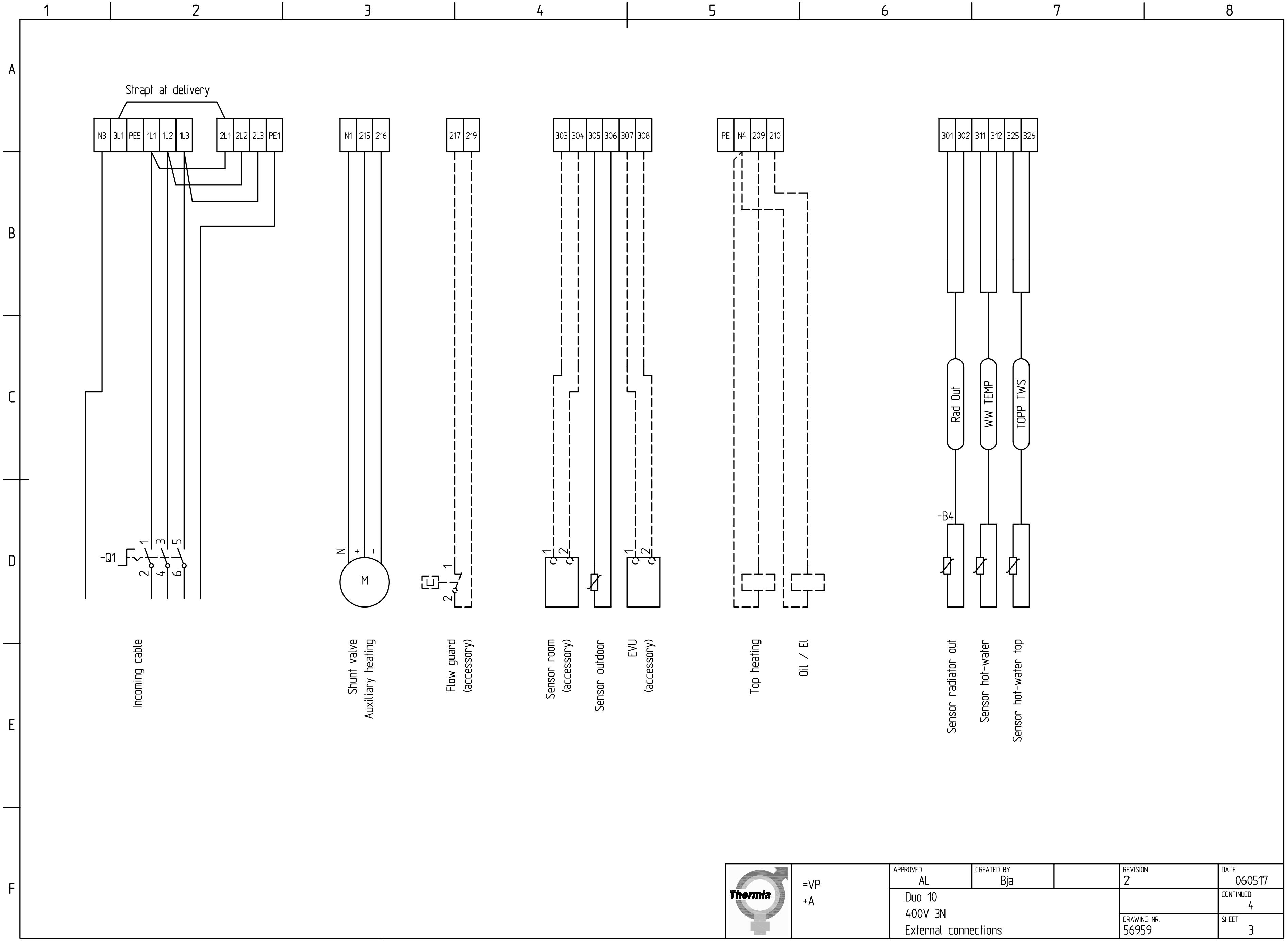

1
2
3
4
5
6
7
8
A
B
C
D
E
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
217
219
350
-M1
/1.6E
WSK
351
95
-F11
/1.5C
96
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
1
2
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

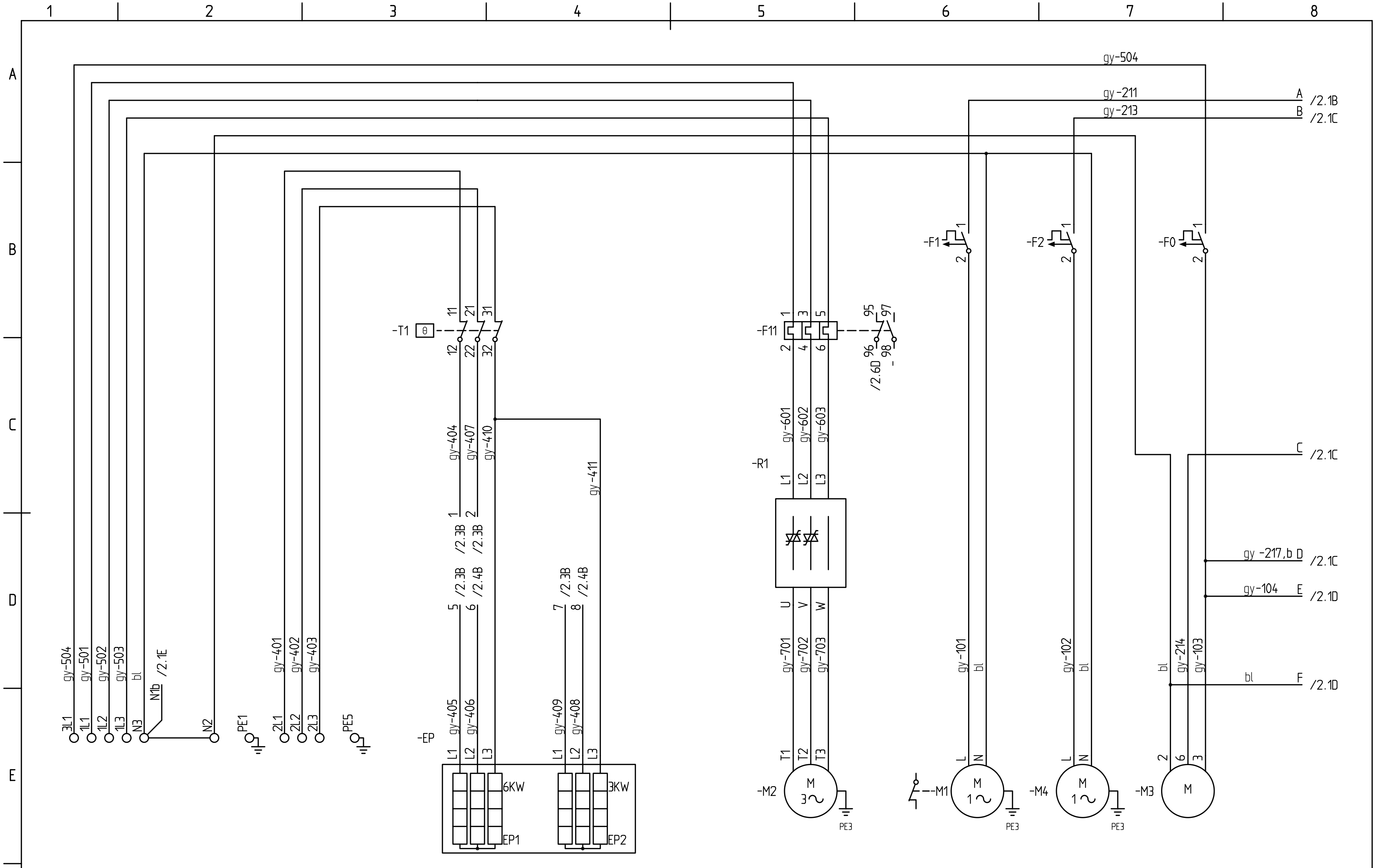
APPROVED AL	CREATED BY Bja		REVISION 2	DATE 060517
Duo 10 400V 3N Circuit diagram			DRAWING NR. 56959	CONTINUED 3 SHEET 2


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
TOPP TWS
-B4
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia

=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 060517
	Duo 10			CONTINUED 4
	400V 3N External connections		DRAWING NR. 56959	SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	9-13A	9588-38379012	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53313001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1’	Terminal block	Phoenix	Duo	9583-55837001	
-X3	1	Terminal	Phoenix	WSK	9583-56967001	

	APPROVED	CREATED BY		REVISION	DATE
=VP +A	AL	Bja		2	061018
	Duo 10 400V 3N Part list				CONTINUED -
				DRAWING NR. 56959	SHEET 4


1
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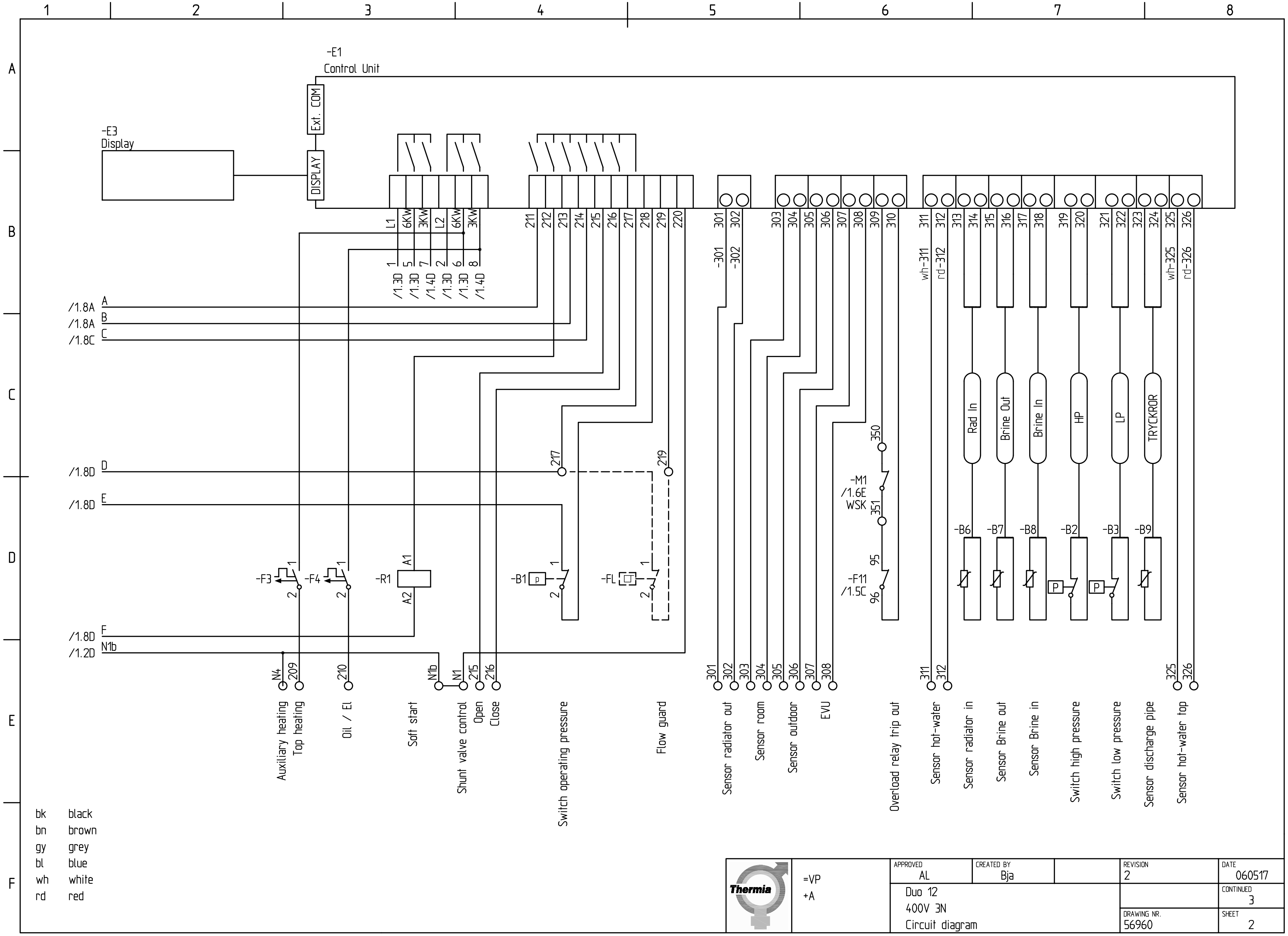
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1 L2 L3
U V W
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3
N2
PE1
gy-401
gy-402
gy-403
2L1 2L2 2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M 3~
PE3
gy-101
bl
-M1
M 1~
gy-102
-M4
bl
gy-214
gy-103
-M3
M
2 6 3
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 060821
		Duo 12 400V 3N Circuit diagram		DRAWING NR. 56960	CONTINUED 2 SHEET 1


1
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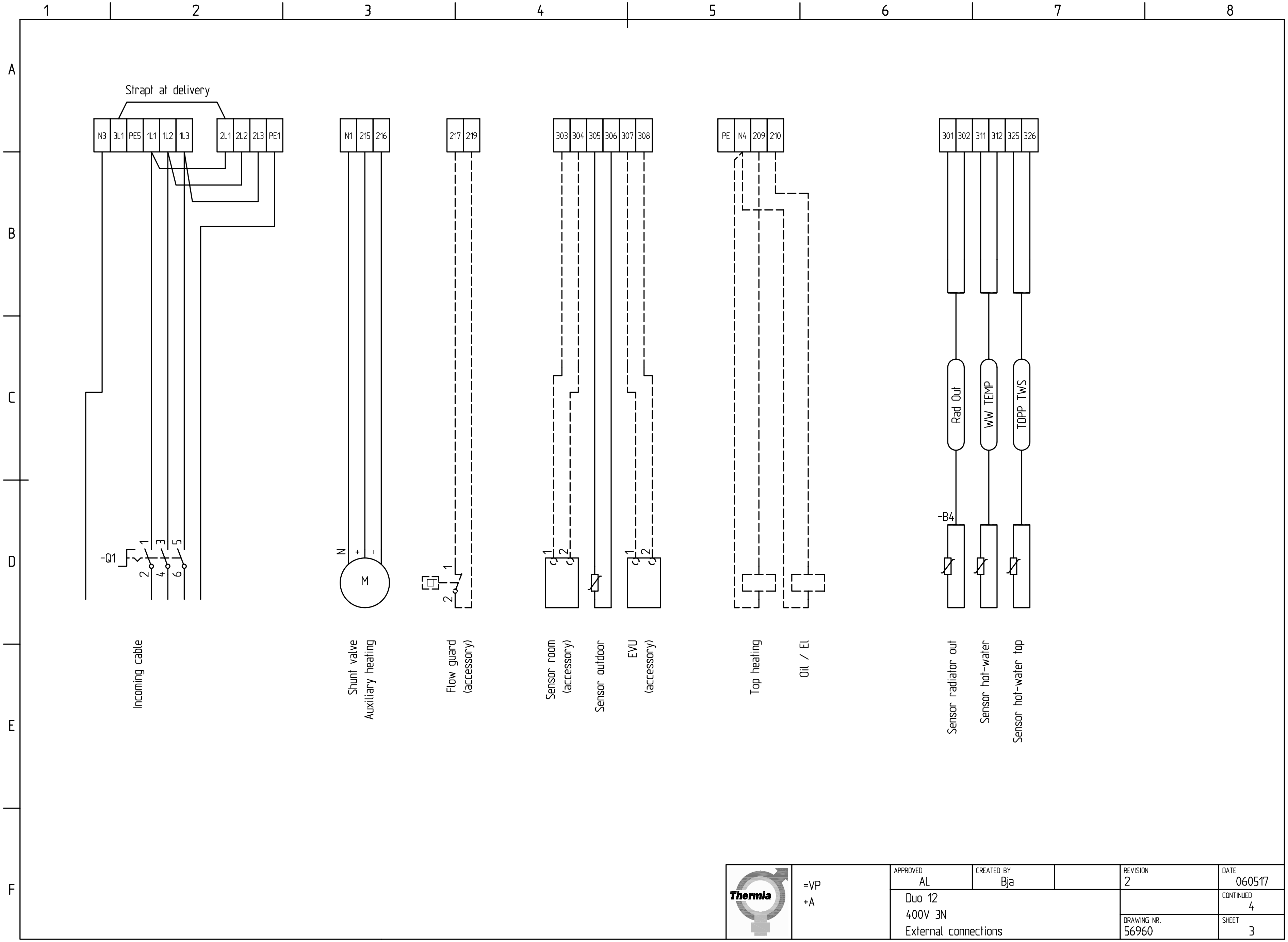
7
8
A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
wh-311
rd-312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
217
219
350
-M1
/1.6E
WSK
351
95
-F11
/1.5C
96
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
1
2
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED
AL
CREATED BY
Bja
REVISION
2
DATE
060517
Duo 12
400V 3N
Circuit diagram
CONTINUED
3
DRAWING NR.
56960
SHEET
2


1
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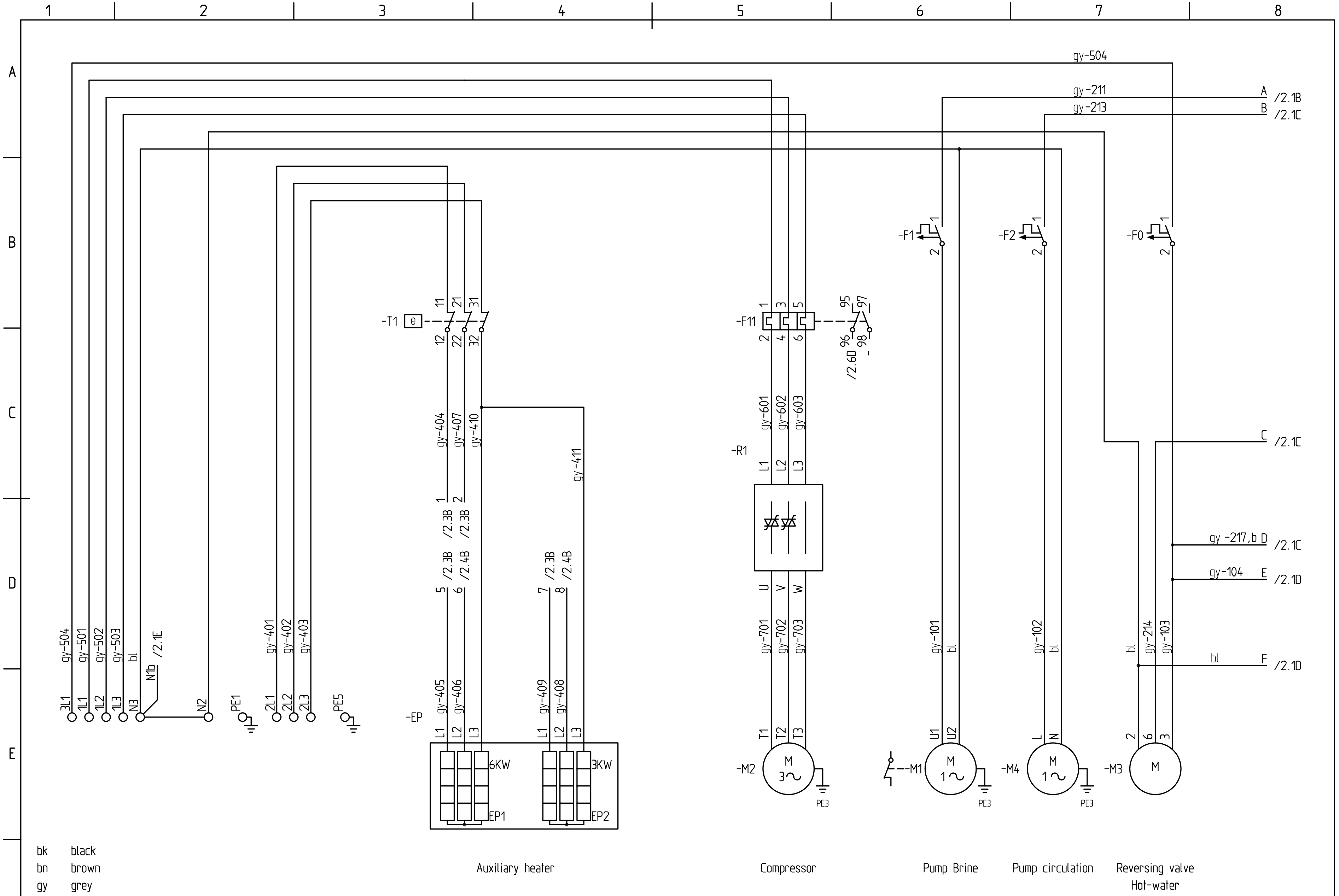
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Rad Out
WW TEMP
TOPP TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja		REVISION 2	DATE 060517
		Duo 12 400V 3N External connections			DRAWING NR. 56960	CONTINUED 4 SHEET 3

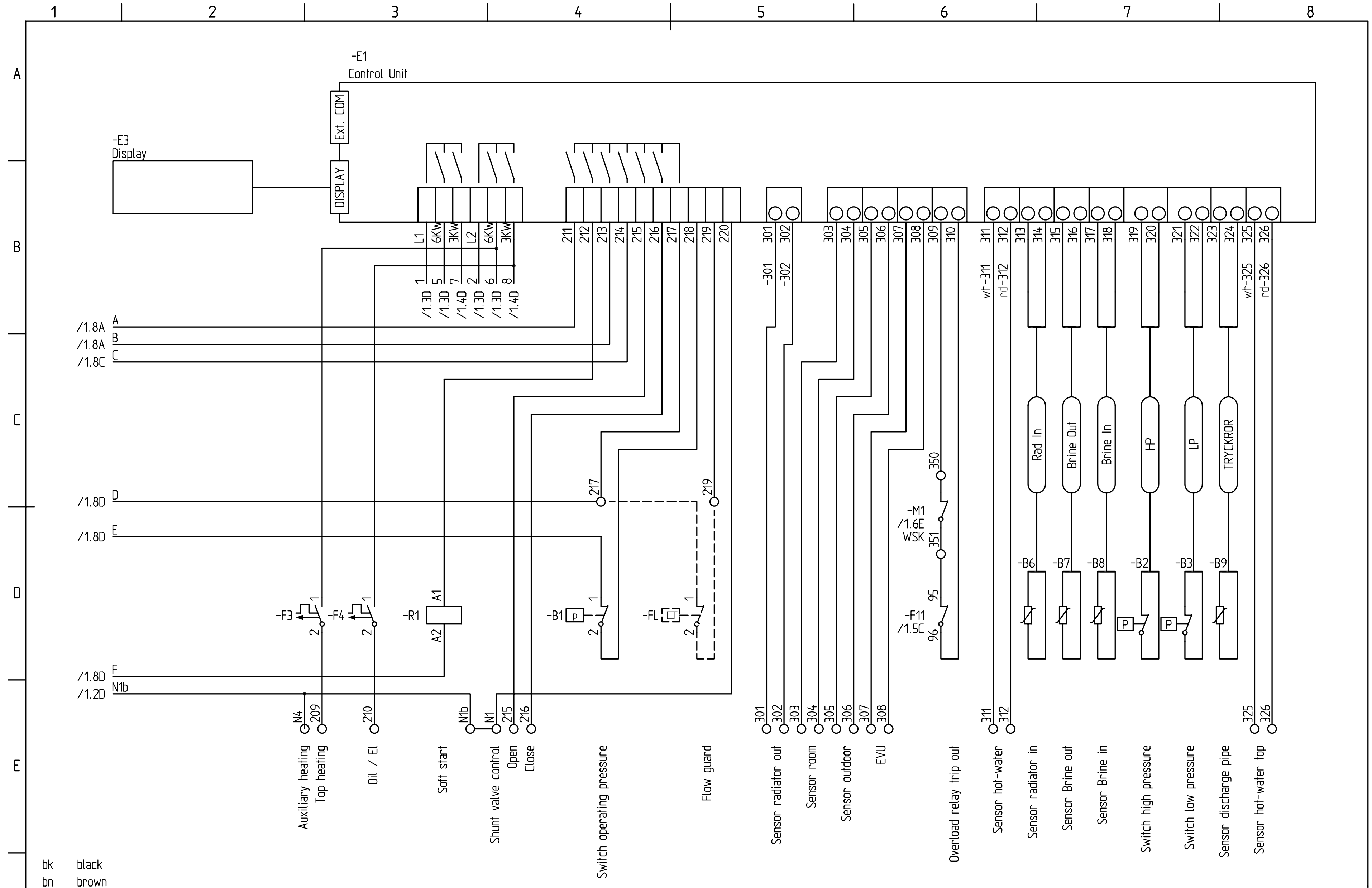
Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	9-13A	9588-38379012	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53314001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	
-X3	1	Terminal	Phoenix	WSK	9583-56967001	

Thermia	=VP +A	APPROVED AL	CREATED BY Bja		REVISION 2	DATE 061018
		Duo 12 400V 3N Part list				CONTINUED -
					DRAWING NR. 56960	SHEET 4


1
2
3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
1
2
-T1
θ
11
21
31
12
22
32
-F11
1
3
5
2
4
6
95
97
96
98
/2.6D
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1
L2
L3
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
U
V
W
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
gy-401
gy-402
gy-403
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
bl
gy-214
gy-103
bl F /2.1D
3L1
1L1
1L2
1L3
N3
N2
PE1
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1
L2
L3
6KW
EP1
3KW
EP2
T1
T2
T3
-M2
M
3~
PE3
U1
U2
-M1
M
1~
PE3
L
N
-M4
M
1~
PE3
2
6
3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 060821
		Duo 16 400V 3N Circuit diagram		DRAWING NR. 56961	CONTINUED 2 SHEET 1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control Unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
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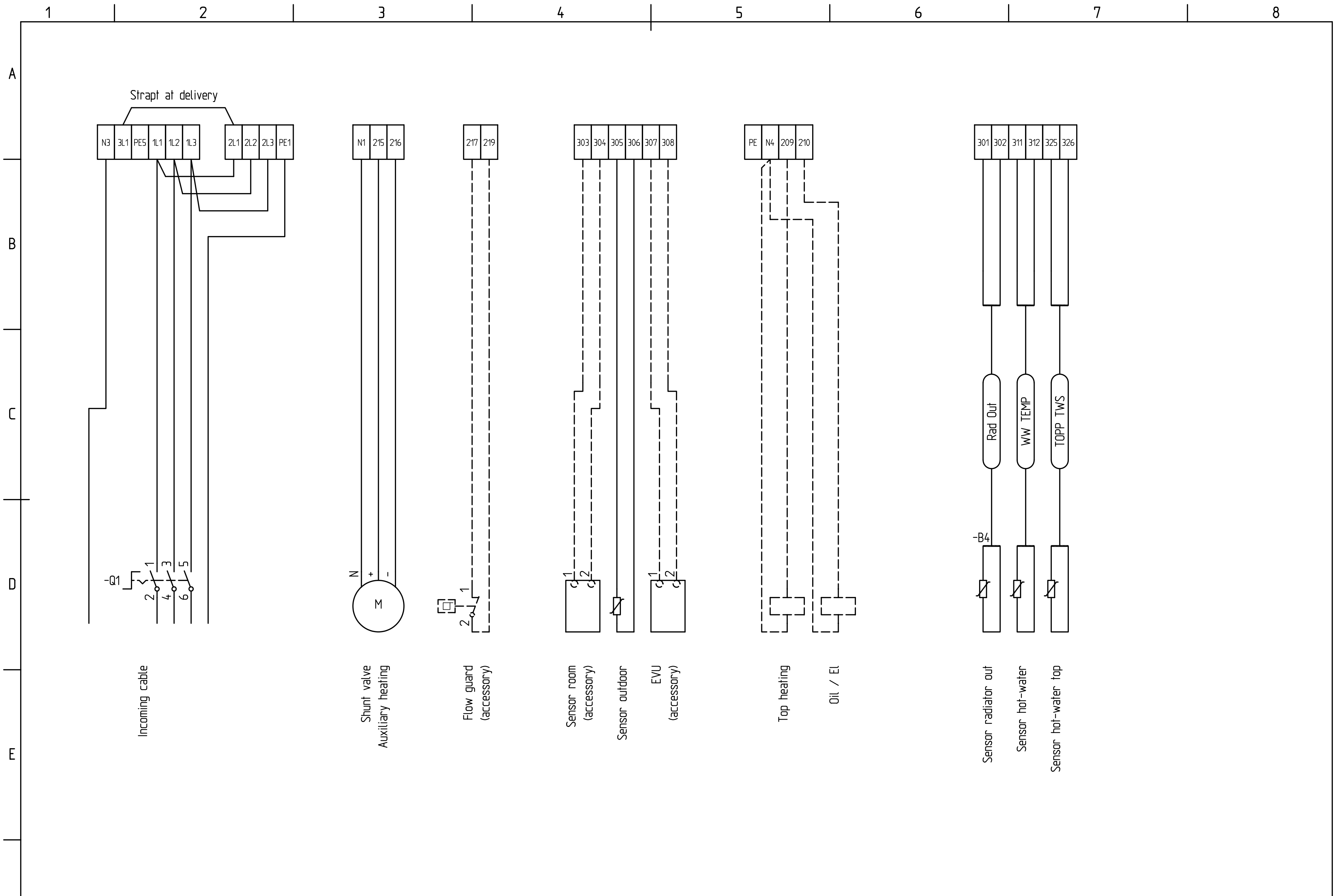
308
309
310
311
312
wh-311
rd-312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
217
219
350
-M1
/1.6E
WSK
351
95
-F11
/1.5C
96
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
1
2
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 060517
		Duo 16 400V 3N Circuit diagram			CONTINUED 3
				DRAWING NR. 56961	SHEET 2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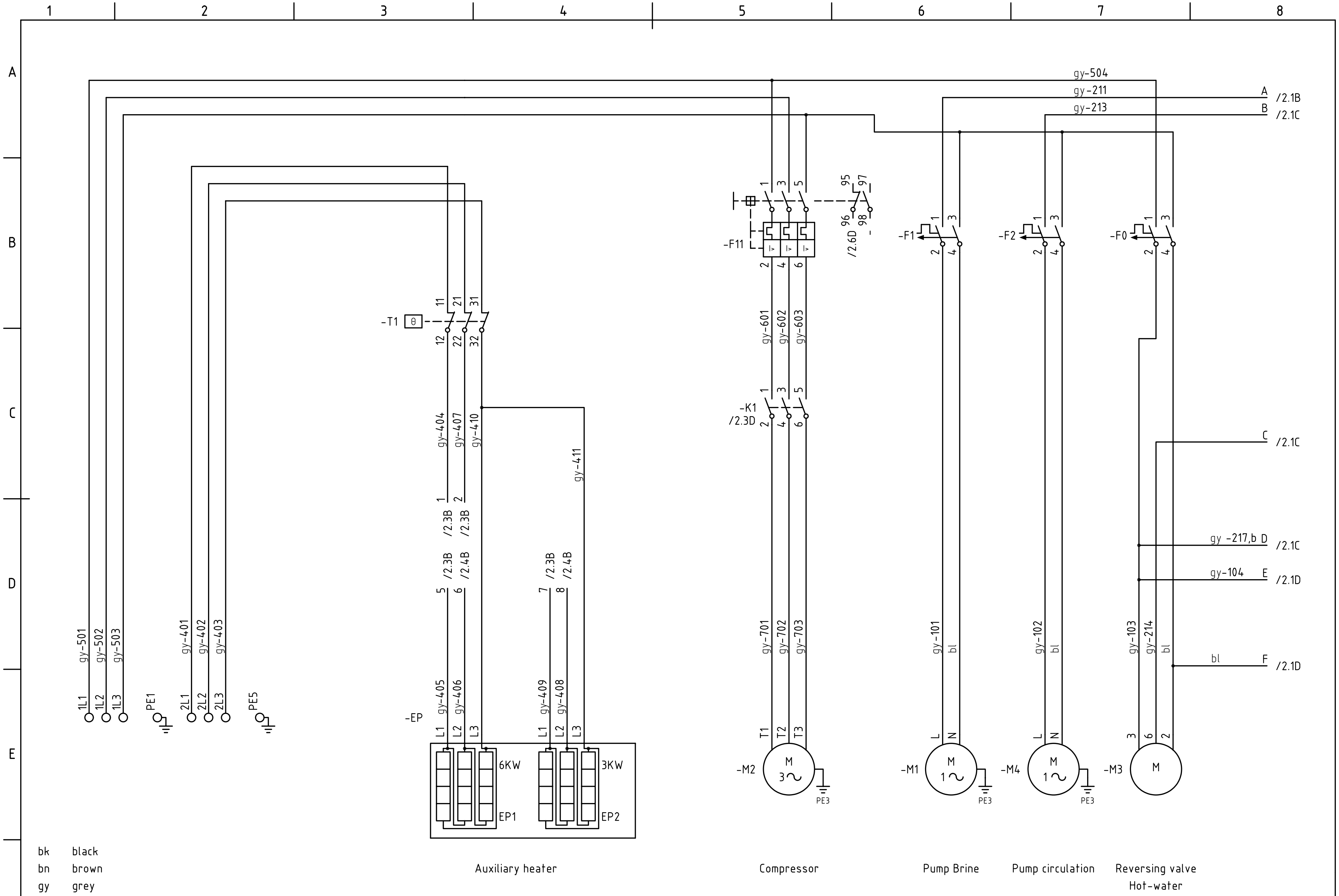
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Rad Out
WW TEMP
TOPP TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 060517
		Duo 16 400V 3N External connections			CONTINUED 4
				DRAWING NR. 56961	SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	12-18A	9588-38379013	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53316001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865003	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	
-X3	1	Terminal	Phoenix	WSK	9583-56967001	

=VP +A	APPROVED AL	CREATED BY Bja	REVISION 2	DATE 061018
	Duo 16 400V 3N Part list		DRAWING NR. 56961	CONTINUED - SHEET 4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7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
gy-601 gy-602 gy-603
-K1
/2.3D
gy-404 gy-407 gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501 gy-502 gy-503
gy-401 gy-402 gy-403
gy-701 gy-702 gy-703
gy-101 bl
gy-102 bl
gy-103 gy-214 bl
1L1 1L2 1L3
PE1
2L1 2L2 2L3
PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
T1 T2 T3
-M2
M 3~
PE3
L N
-M1
M 1~
-M4
M 1~
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED
AL

CREATED BY
Bja

REVISION
1

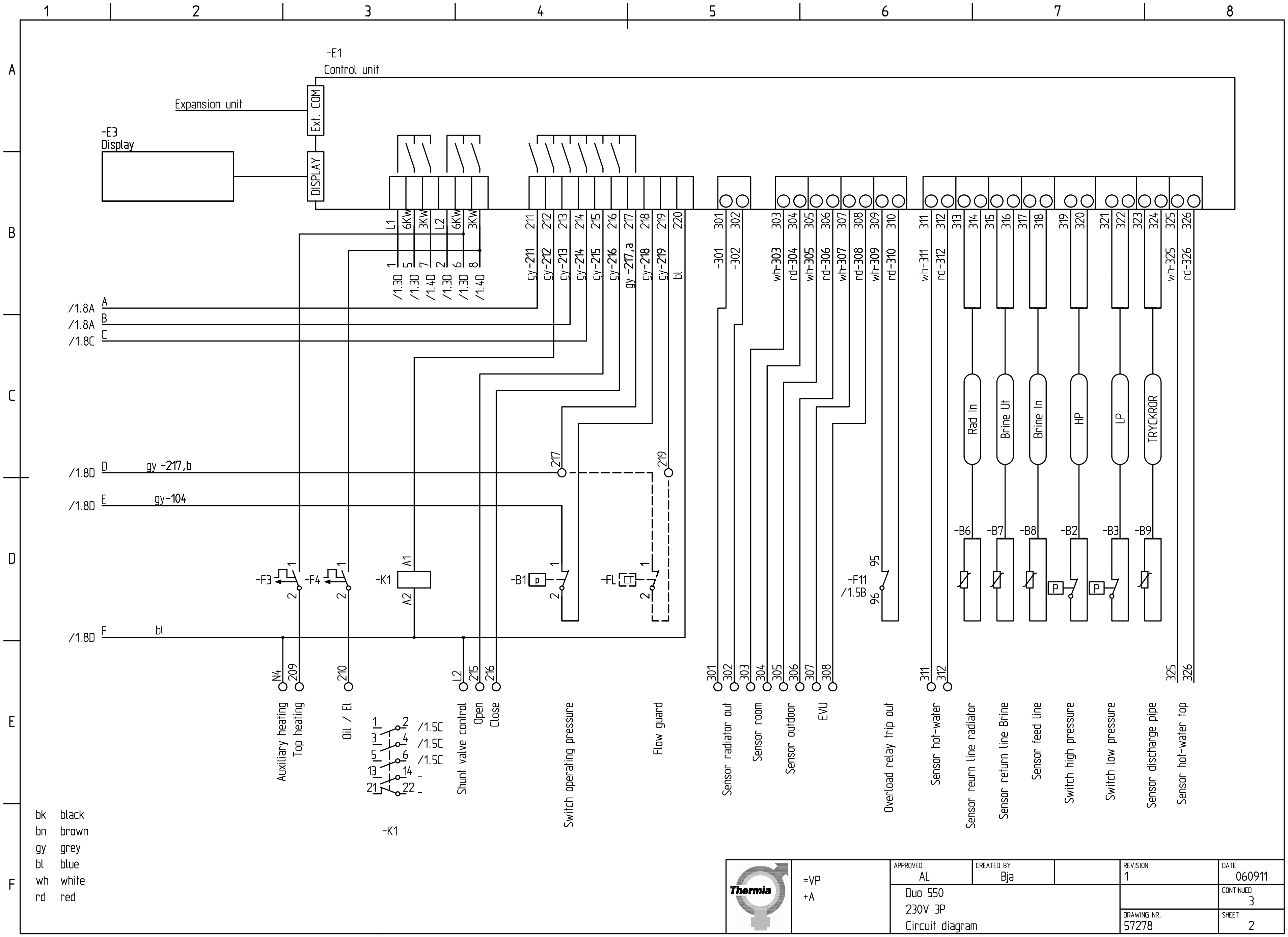
DATE
060911

Duo 550
230V 3P
Circuit diagram

CONTINUED
2

DRAWING NR.
57278

SHEET
1


1
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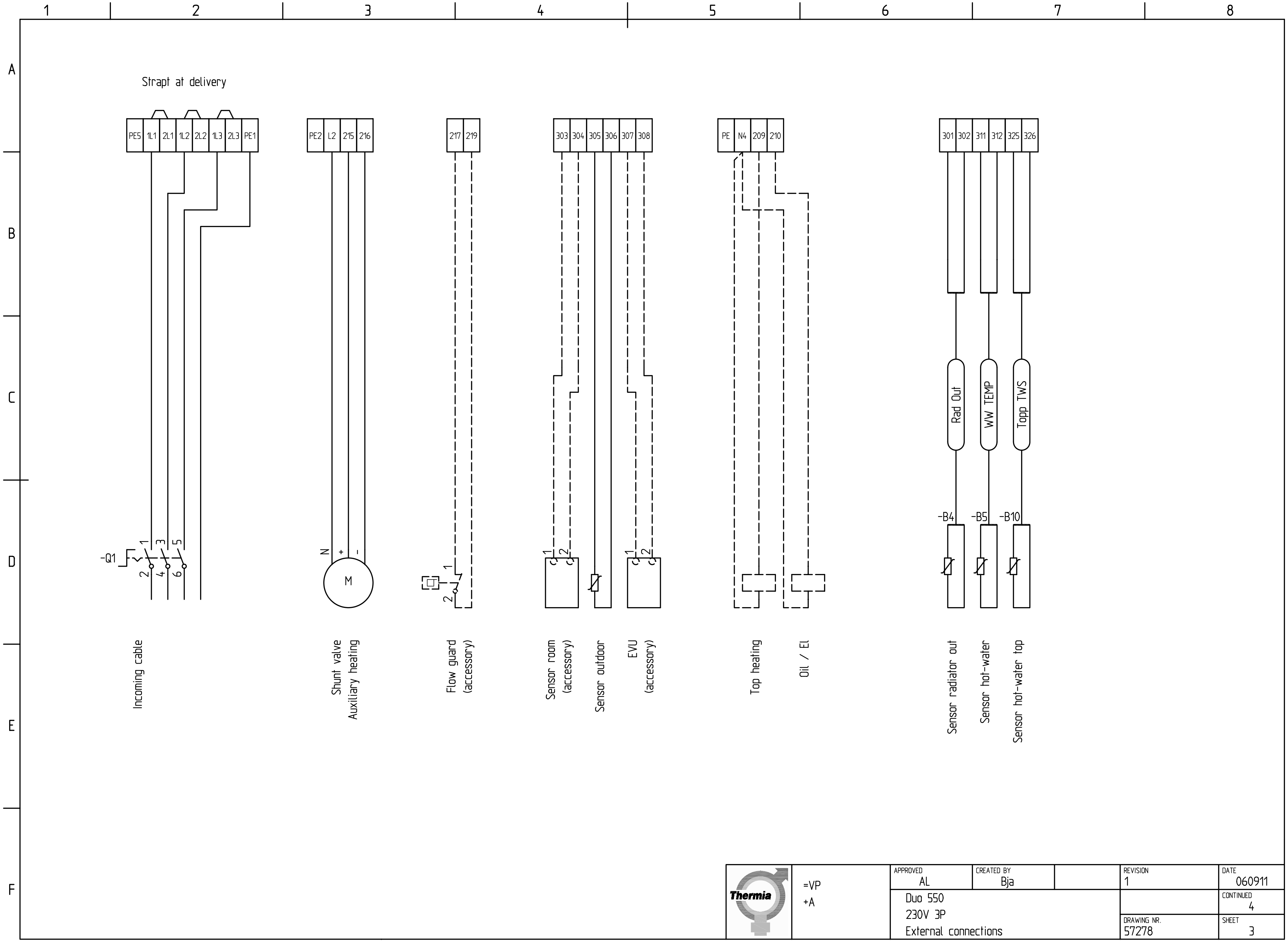
7
8
A
B
C
D
E
F
-E1
Control unit
Expansion unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
gy-211
gy-212
gy-213
gy-214
gy-215
gy-216
gy -217,a
gy-218
gy-219
bl
301
302
-301
-302
303
304
305
306
307
308
309
310
wh-303
rd-304
wh-305
rd-306
wh-307
rd-308
wh-309
rd-310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
gy -217,b
/1.8D E
gy-104
/1.8D F
bl
217
219
Rad In
Brine Ut
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
1
2
N4
209
210
L2
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
1
2
/1.5C
3
4
/1.5C
5
6
/1.5C
13
14
21
22
-K1
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor reurn line radiator
Sensor return line Brine
Sensor feed line
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
	Duo 550 230V 3P Circuit diagram			DRAWING NR. 57278	CONTINUED 3 SHEET 2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A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
PE2
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia

		APPROVED	CREATED BY		REVISION	DATE
Thermia	=VP +A	AL	Bja		1	060911
		Duo 550 230V 3P External connections			DRAWING NR. 57278	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	1,5 bar	9578-37142001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	9-14A	9588-53794001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	12A 5,5KW 230V	9588-38378002	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865001	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51332002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

Thermia

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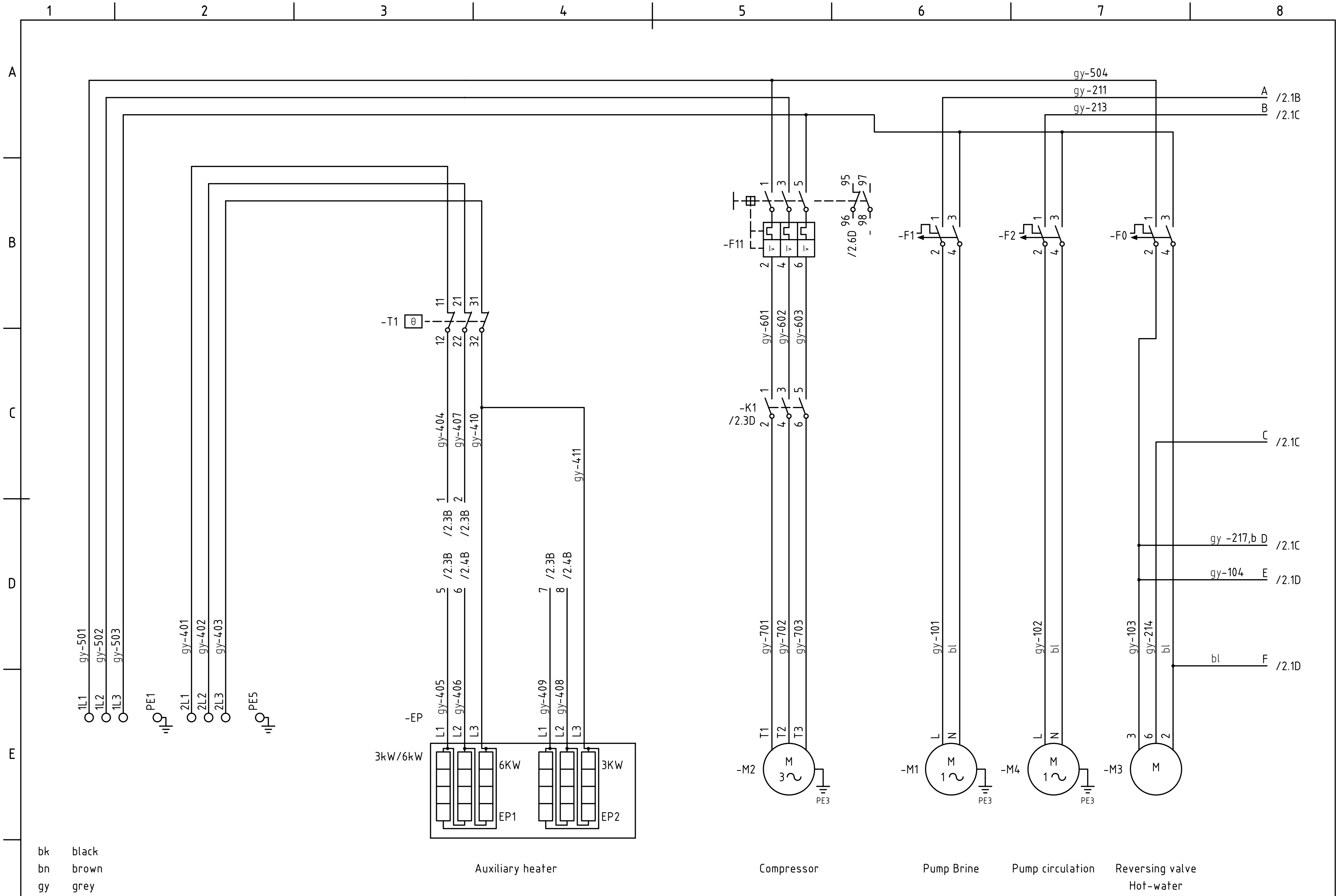
APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
Duo 550 230V 3P Part list				CONTINUED -
			DRAWING NR. 57278	SHEET 4


1
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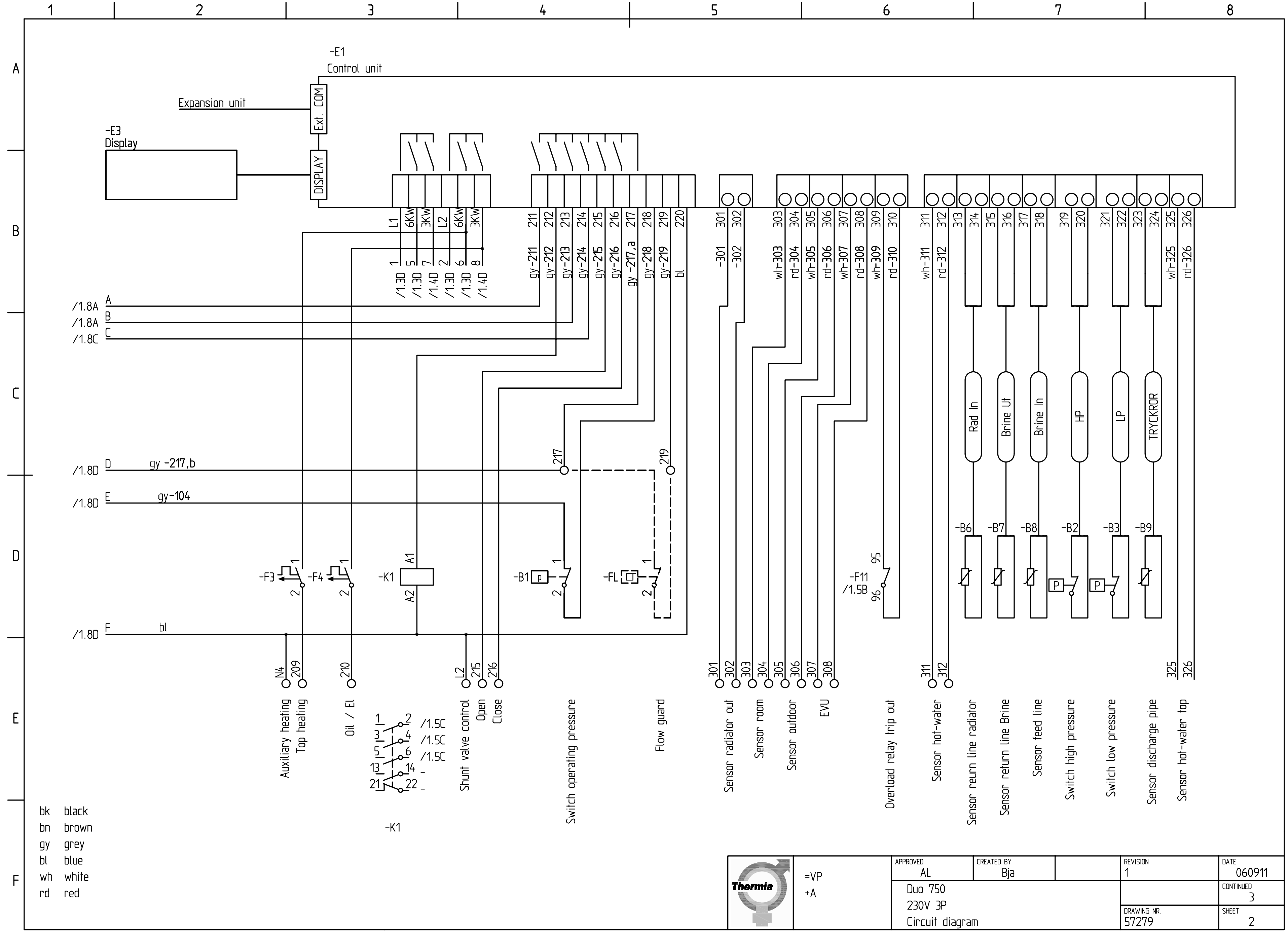
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
gy-601
gy-602
gy-603
-K1
/2.3D
gy-404
gy-407
gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-502
gy-503
gy-401
gy-402
gy-403
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
gy-103
gy-214
bl
1L1
1L2
1L3
PE1
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1 L2 L3
L1 L2 L3
3kW/6kW
6KW
EP1
3KW
EP2
T1 T2 T3
-M2
M
3~
PE3
L N
-M1
M
1~
PE3
L N
-M4
M
1~
PE3
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

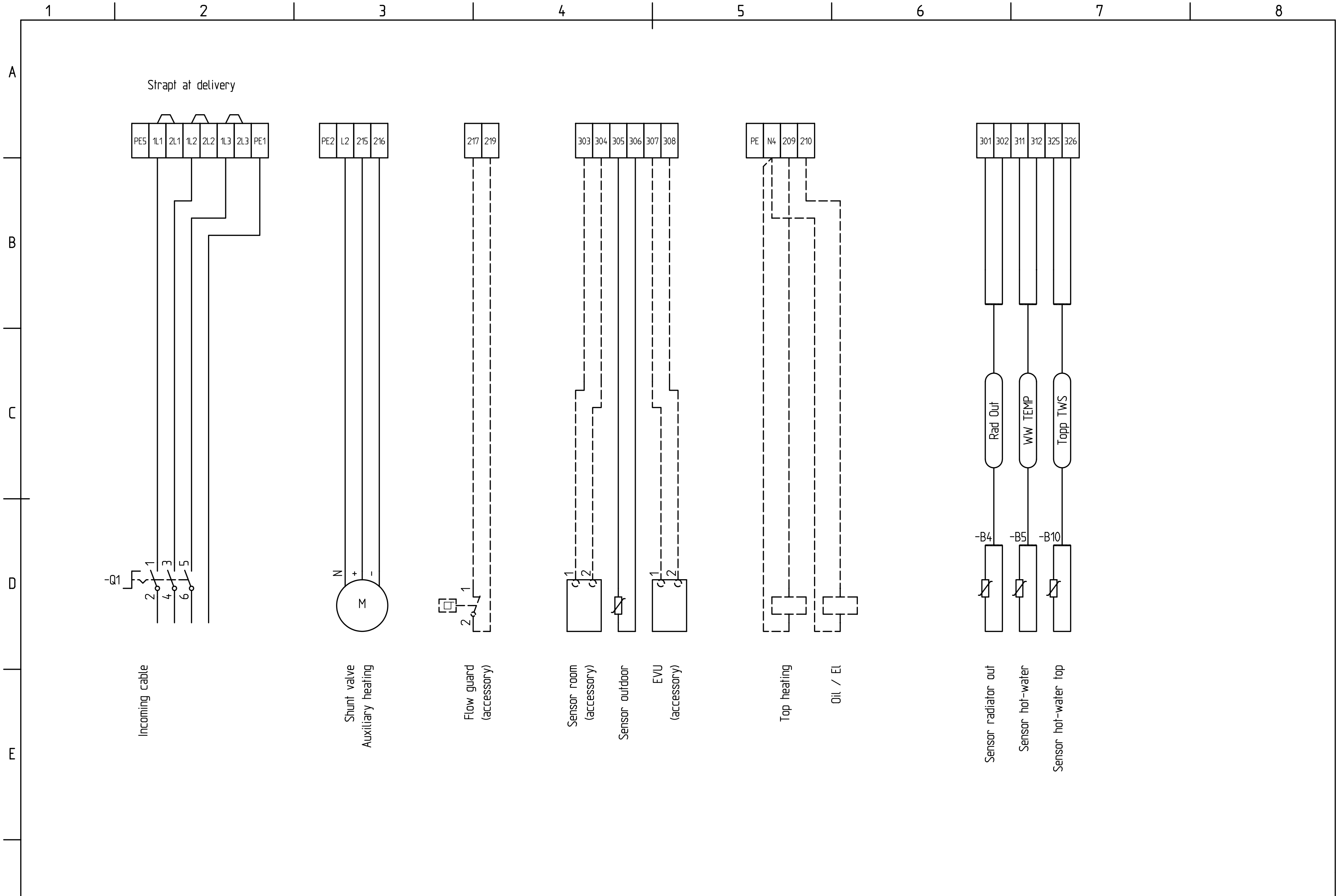
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 750				CONTINUED 2
230V 3P			DRAWING NR. 57279	SHEET 1
Circuit diagram				


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Expansion unit
Ext. COM
-E3
Display
DISPLAY
L1 6KW 3KW L2 6KW 3KW
211 212 213 214 215 216 217 218 219 220
301 302 303 304 305 306 307 308 309 310
311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326
1 5 7 2 6 8
/1.3D /1.3D /1.4D /1.3D /1.3D /1.4D
gy-211 gy-212 gy-213 gy-214 gy-215 gy-216 gy-217,a gy-218 gy-219 bl
-301 -302
wh-303 rd-304 wh-305 rd-306 wh-307 rd-308 wh-309 rd-310
wh-311 rd-312
wh-325 rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D gy -217,b
/1.8D E gy-104
/1.8D F bl
Rad In
Brine Ut
Brine In
HP
LP
TRYCKROR
-B6 -B7 -B8 -B2 -B3 -B9
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4 209 210 L2 215 216
301 302 303 304 305 306 307 308
311 312
325 326
Auxiliary heating
Top heating
Oil / El
1 2 /1.5C
3 4 /1.5C
5 6 /1.5C
13 14 -
21 22 -
-K1
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor reurn line radiator
Sensor return line Brine
Sensor feed line
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
Duo 750				CONTINUED 3
230V 3P Circuit diagram			DRAWING NR. 57279	SHEET 2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A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
PE2
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

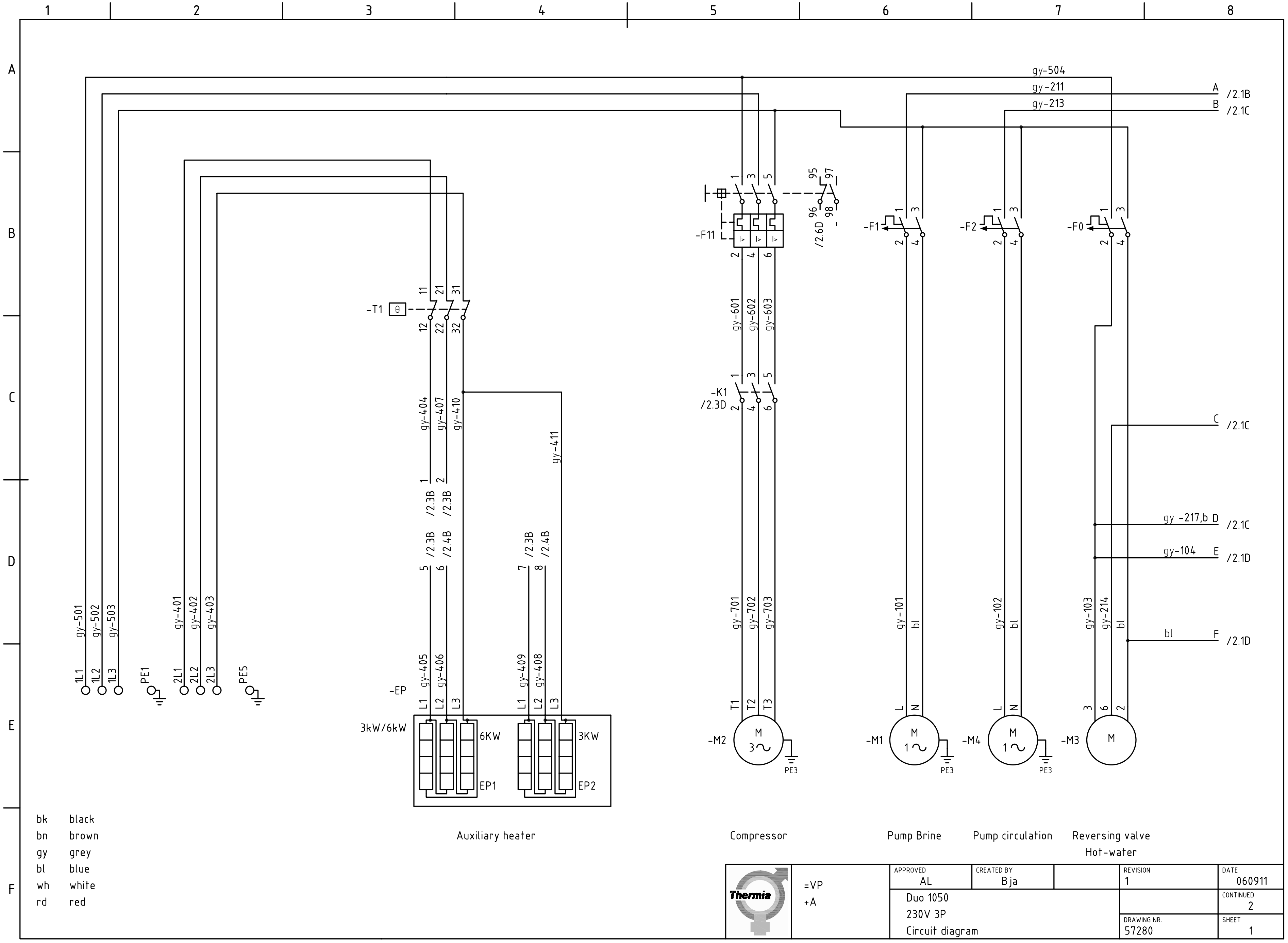
Thermia

=VP
+A

APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
Duo 750 230V 3P External connections			DRAWING NR. 57279	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	1,5 bar	9578-37142001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	9-14A	9588-53794001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51333002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865001	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 060911
	Duo 750 230V 3P Part list				CONTINUED -
				DRAWING NR. 57279	SHEET 4


1
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7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
gy-601 gy-602 gy-603
-K1
/2.3D
gy-404 gy-407 gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501 gy-502 gy-503
gy-401 gy-402 gy-403
gy-701 gy-702 gy-703
gy-101 bl
gy-102 bl
gy-103 gy-214 bl
1L1 1L2 1L3 PE1
2L1 2L2 2L3 PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
3kW/6kW
6KW
3KW
EP1
EP2
T1 T2 T3
L N
3 6 2
-M2
M 3~
-M1
M 1~
-M4
M 1~
-M3
M
PE3
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

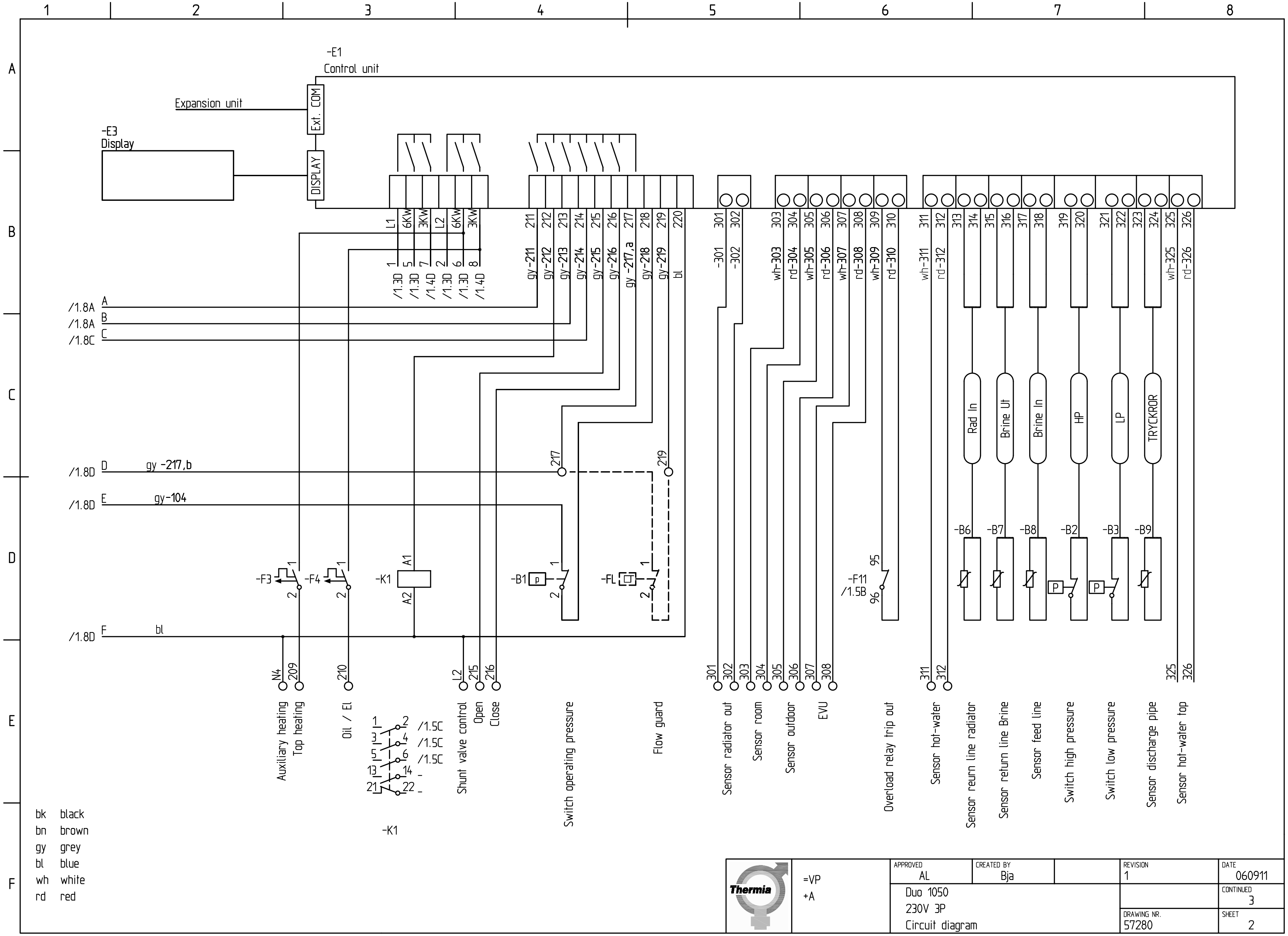
Thermia

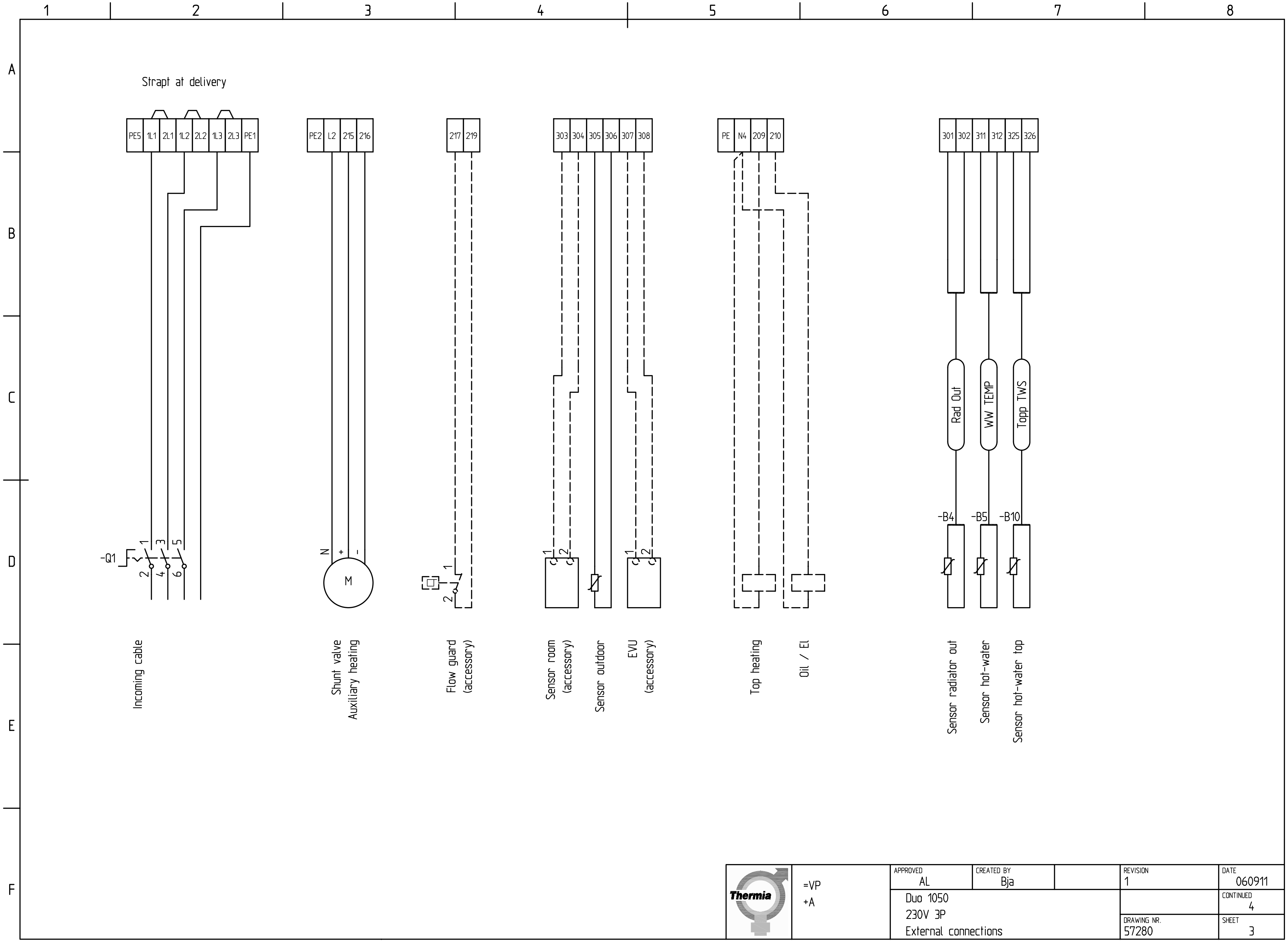
=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911

Duo 1050
230V 3P
Circuit diagram

CONTINUED 2
DRAWING NR. 57280
SHEET 1


-E1
Control unit
Expansion unit
Ext. COM
-E3
Display
DISPLAY
L1 6KW 3KW L2 6KW 3KW
1 5 7 2 6 8
/1.3D /1.3D /1.4D /1.3D /1.3D /1.4D
211 212 213 214 215 216 217 218 219 220
gy-211 gy-212 gy-213 gy-214 gy-215 gy-216 gy-217,a gy-218 gy-219 bl
301 302 303 304 305 306 307 308 309 310
-301 -302 wh-303 rd-304 wh-305 rd-306 wh-307 rd-308 wh-309 rd-310
311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326
wh-311 rd-312 wh-325 rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D gy -217,b
/1.8D E gy-104
/1.8D F bl
217
219
Rad In
Brine Ut
Brine In
HP
LP
TRYCKROR
-B6 -B7 -B8 -B2 -B3 -B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4 209 210 L2 215 216 301 302 303 304 305 306 307 308 311 312 325 326
Auxiliary heating
Top heating
Oil / El
1 2 /1.5C
3 4 /1.5C
5 6 /1.5C
13 14 -
21 22 -
-K1
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor reurn line radiator
Sensor return line Brine
Sensor feed line
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
1 2 3 4 5 6 7 8
A B C D E F
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
Duo 1050
230V 3P
Circuit diagram
CONTINUED 3
DRAWING NR. 57280
SHEET 2


1
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7
8
A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
PE2
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia

=VP
+A

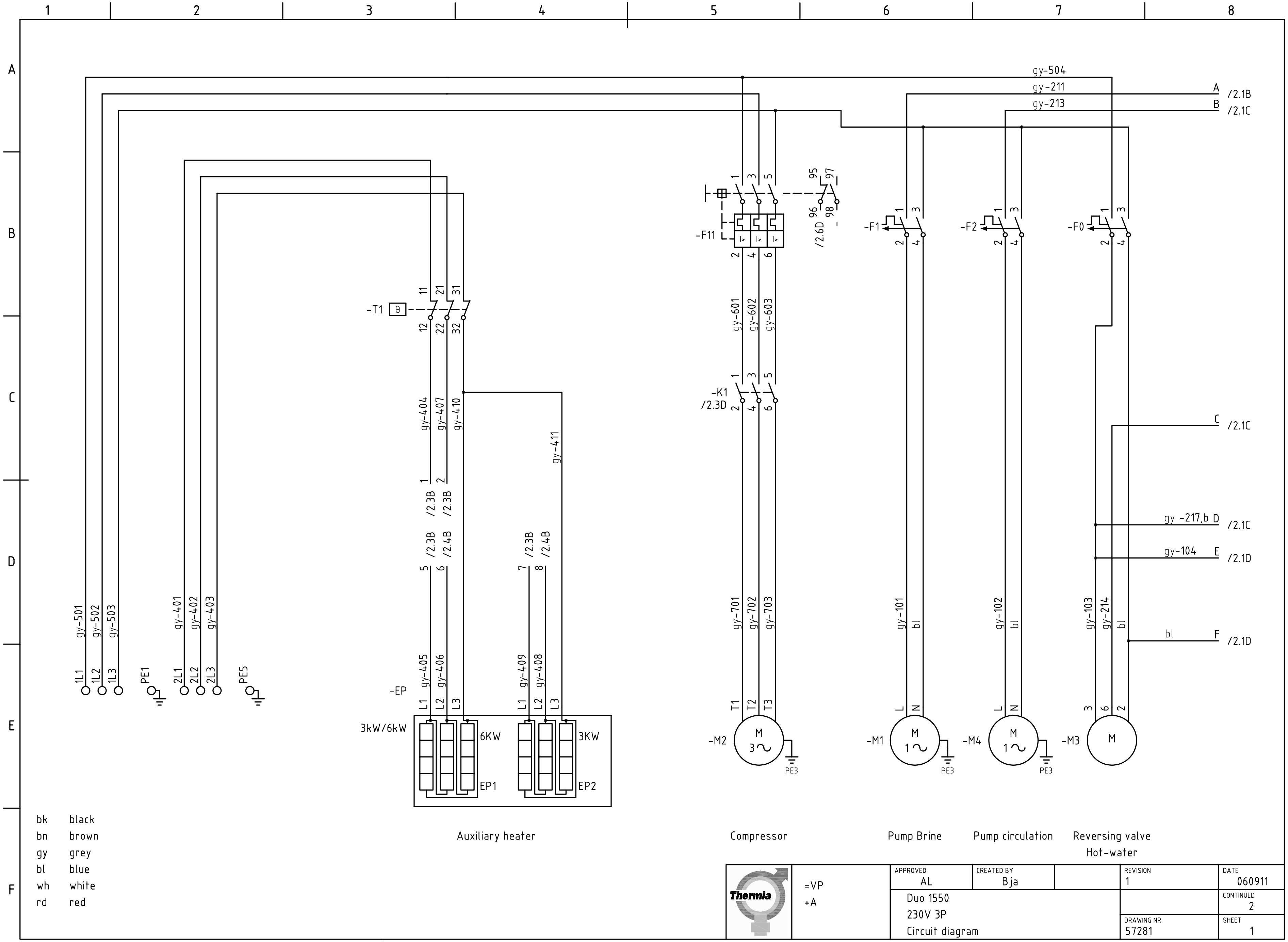
APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
Duo 1050
230V 3P
External connections
CONTINUED 4
DRAWING NR. 57280
SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	1,5 bar	9578-37142001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	13-18A	9588-54715001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-53865003	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51335002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 1050 230V 3P Part list			DRAWING NR. 57280	CONTINUED - SHEET 4


1
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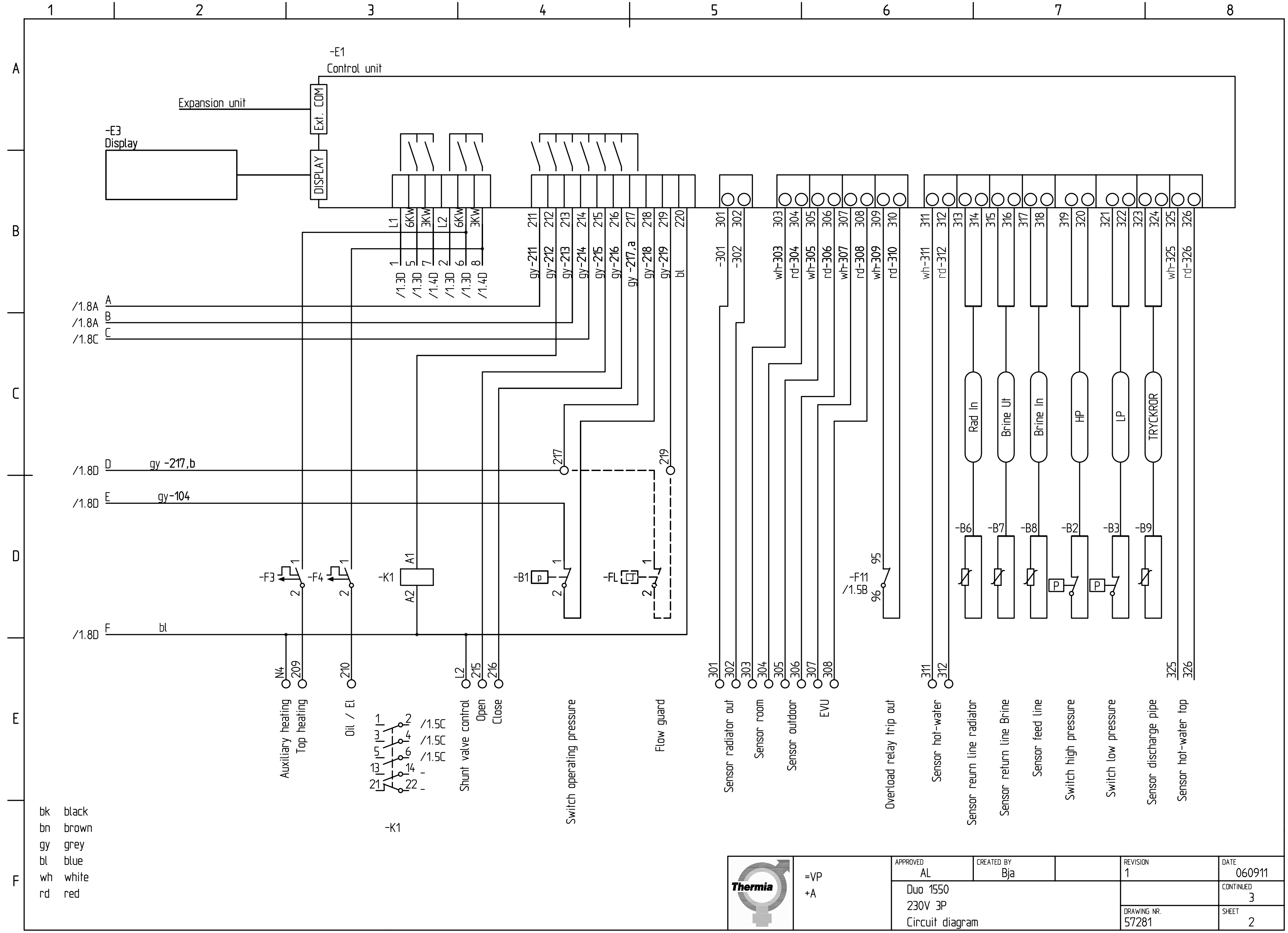
3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-T1 θ
11 21 31
12 22 32
gy-404 gy-407 gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-501 gy-502 gy-503
1L1 1L2 1L3
PE1
gy-401 gy-402 gy-403
2L1 2L2 2L3
PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
L1 L2 L3
3kW/6kW
6KW
EP1
3KW
EP2
-F11
95 97
96 98
/2.6D
gy-601 gy-602 gy-603
-K1
/2.3D
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M
3~
PE3
-F1
-F2
-F0
gy-101 bl
L N
-M1
M
1~
PE3
gy-102 bl
L N
-M4
M
1~
PE3
gy-103 gy-214 bl
3 6 2
-M3
M
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 1550				CONTINUED 2
230V 3P			DRAWING NR.	SHEET
Circuit diagram			57281	1

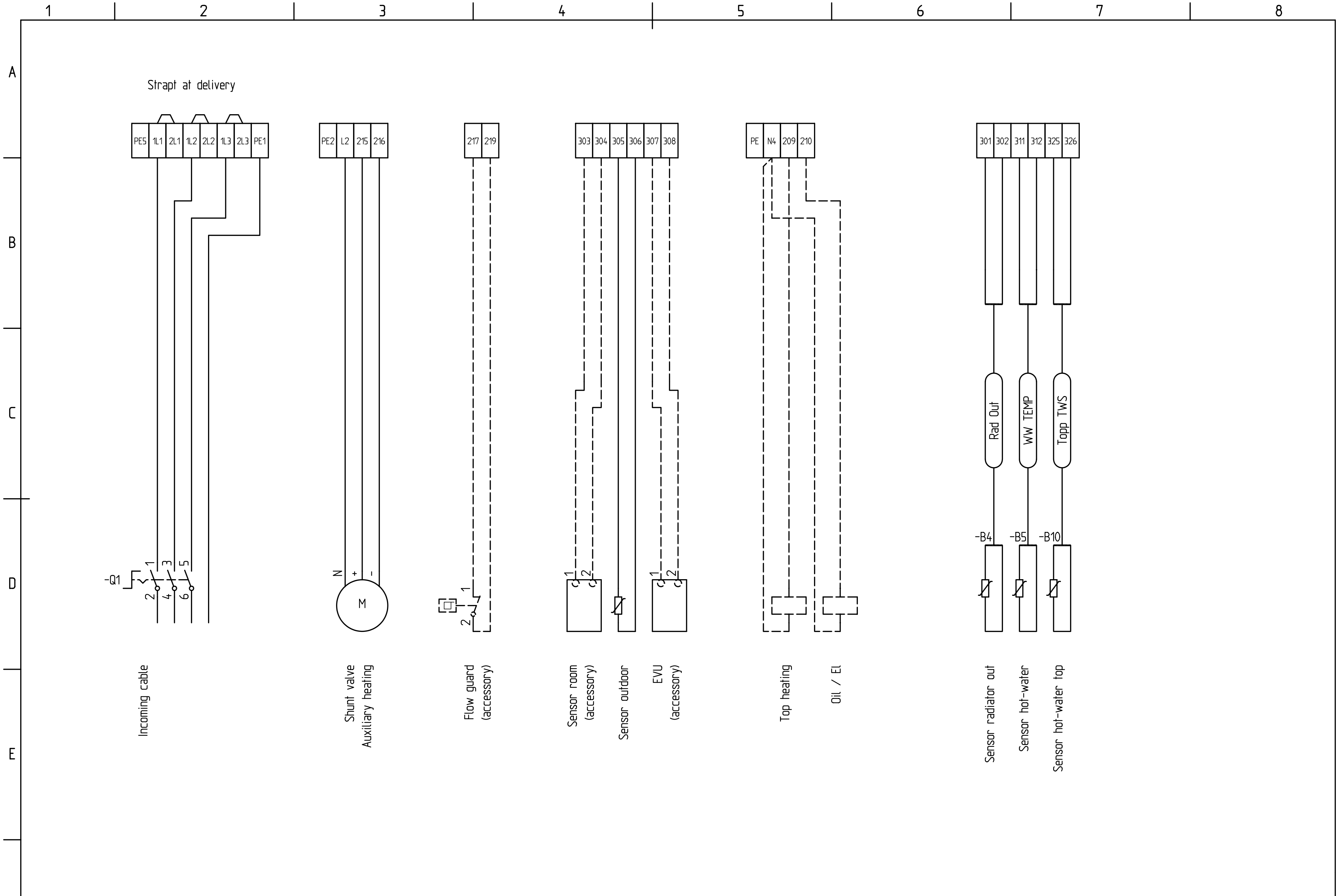

1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Expansion unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
gy-211
gy-212
gy-213
gy-214
gy-215
gy-216
gy-217,a
gy-218
gy-219
bl
301
302
-301
-302
303
304
305
306
307
308
309
310
wh-303
rd-304
wh-305
rd-306
wh-307
rd-308
wh-309
rd-310
311
312
wh-311
rd-312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
gy -217,b
/1.8D E
gy-104
/1.8D F
bl
217
219
Rad In
Brine Ut
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4
209
210
L2
215
216
Auxiliary heating
Top heating
Oil / El
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor reurn line radiator
Sensor return line Brine
Sensor feed line
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
1 2 /1.5C
3 4 /1.5C
5 6 /1.5C
13 14 -
21 22 -
-K1

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
CONTINUED 3
Duo 1550
230V 3P
Circuit diagram
DRAWING NR. 57281
SHEET 2


1
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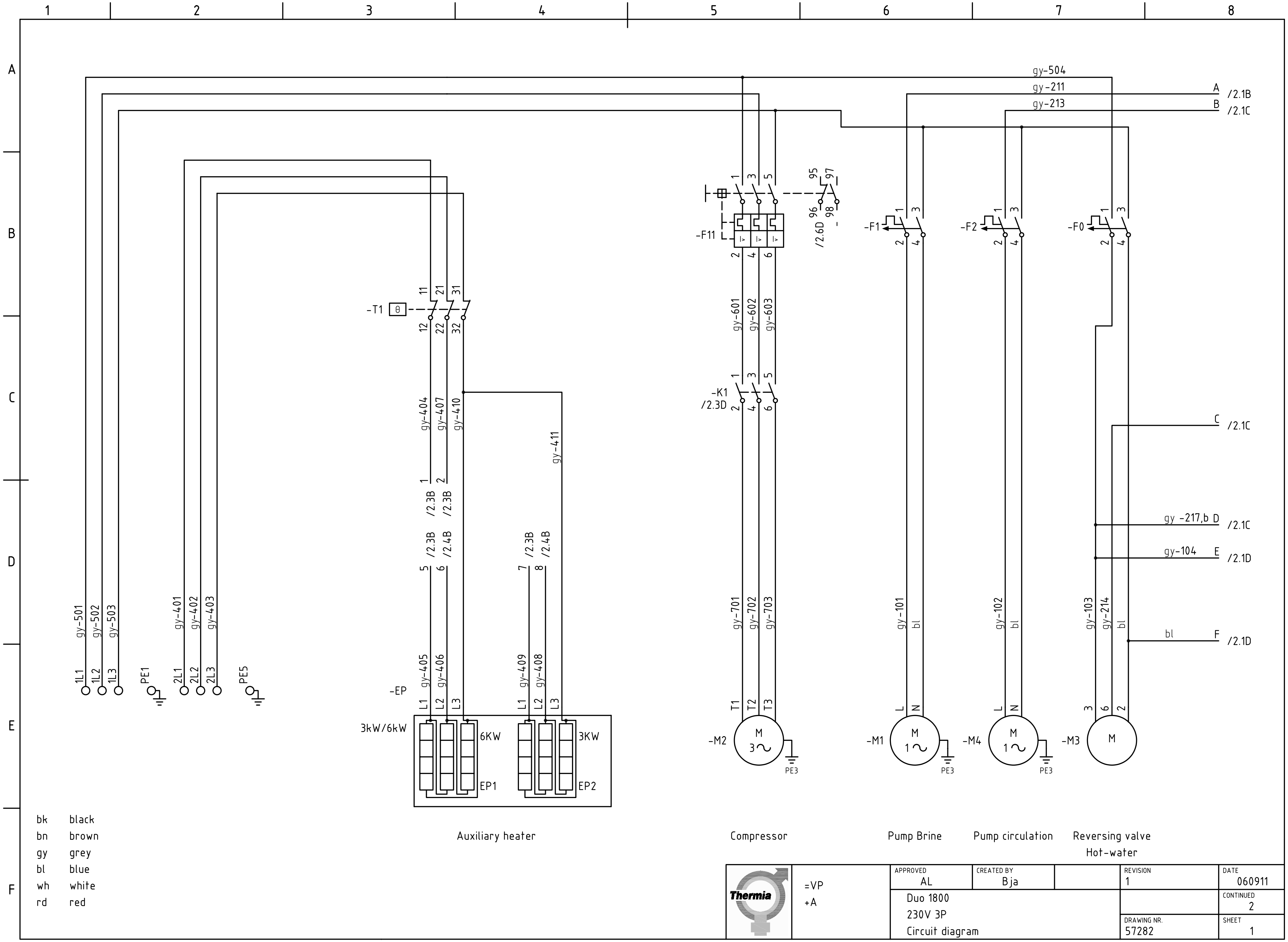
7
8
A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
PE2
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
		Duo 1550 230V 3P External connections		DRAWING NR. 57281	CONTINUED 4 SHEET 3

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	1,5 bar	9578-37142001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	24-32A	9588-52640004	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	32A 15KW 230VA	9588-38378005	/2.3D
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51337002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865003	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
	Duo 1550 230V 3P Part list			CONTINUED -
			DRAWING NR. 57281	SHEET 4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7
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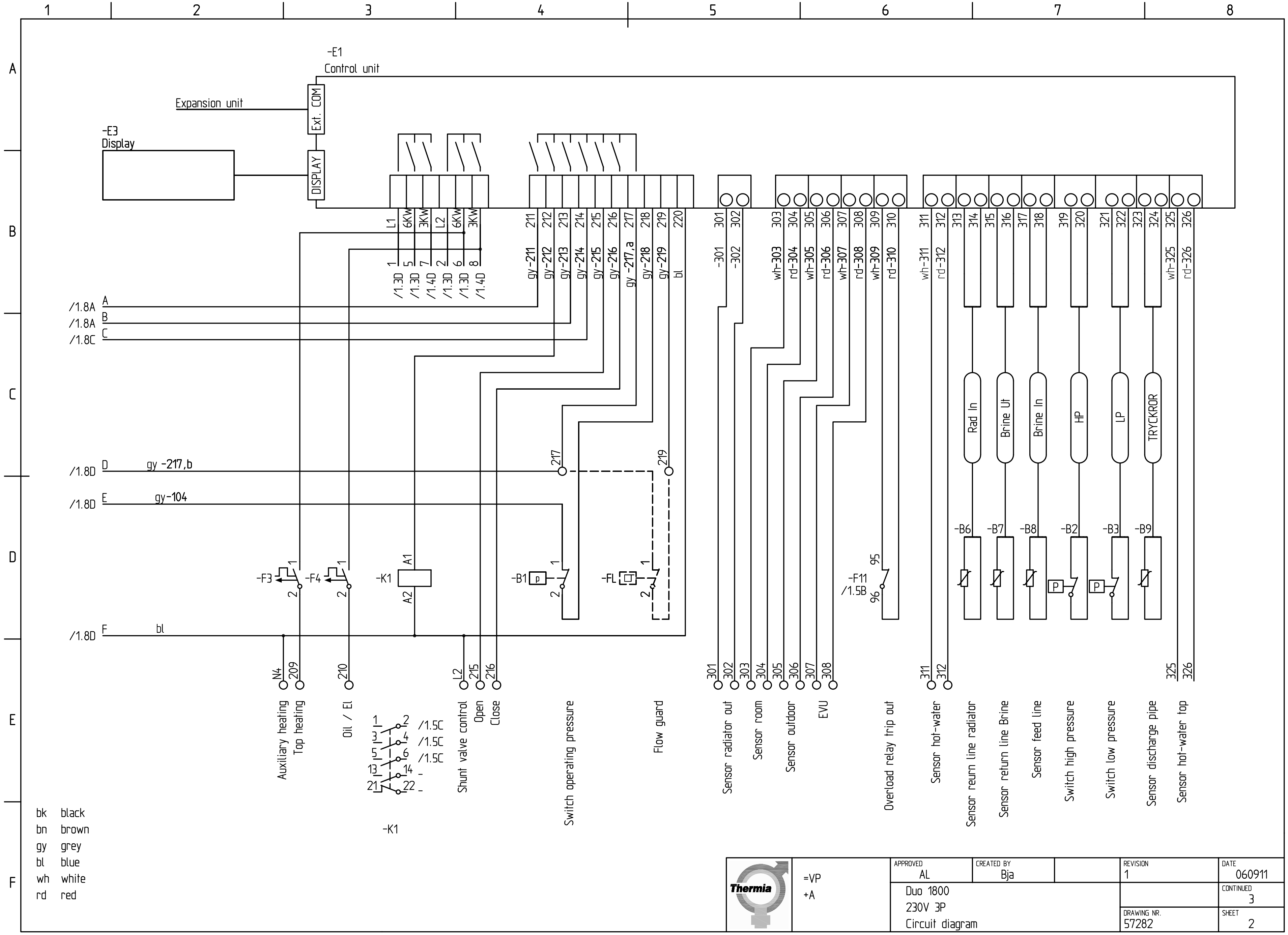
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-T1 θ
11 21 31
12 22 32
gy-404 gy-407 gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-405 gy-406
gy-409 gy-408
-EP
L1 L2 L3
L1 L2 L3
3kW/6kW
6KW
EP1
3KW
EP2
gy-501 gy-502 gy-503
1L1 1L2 1L3
PE1
gy-401 gy-402 gy-403
2L1 2L2 2L3
PE5
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
gy-601 gy-602 gy-603
-K1
/2.3D
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M
3~
PE3
-F1
-F2
-F0
1 3
2 4
gy-101 bl
L N
-M1
M
1~
PE3
gy-102 bl
-M4
M
1~
PE3
gy-103 gy-214 bl
3 6 2
-M3
M
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
Duo 1800
230V 3P
Circuit diagram
CONTINUED 2
DRAWING NR. 57282
SHEET 1

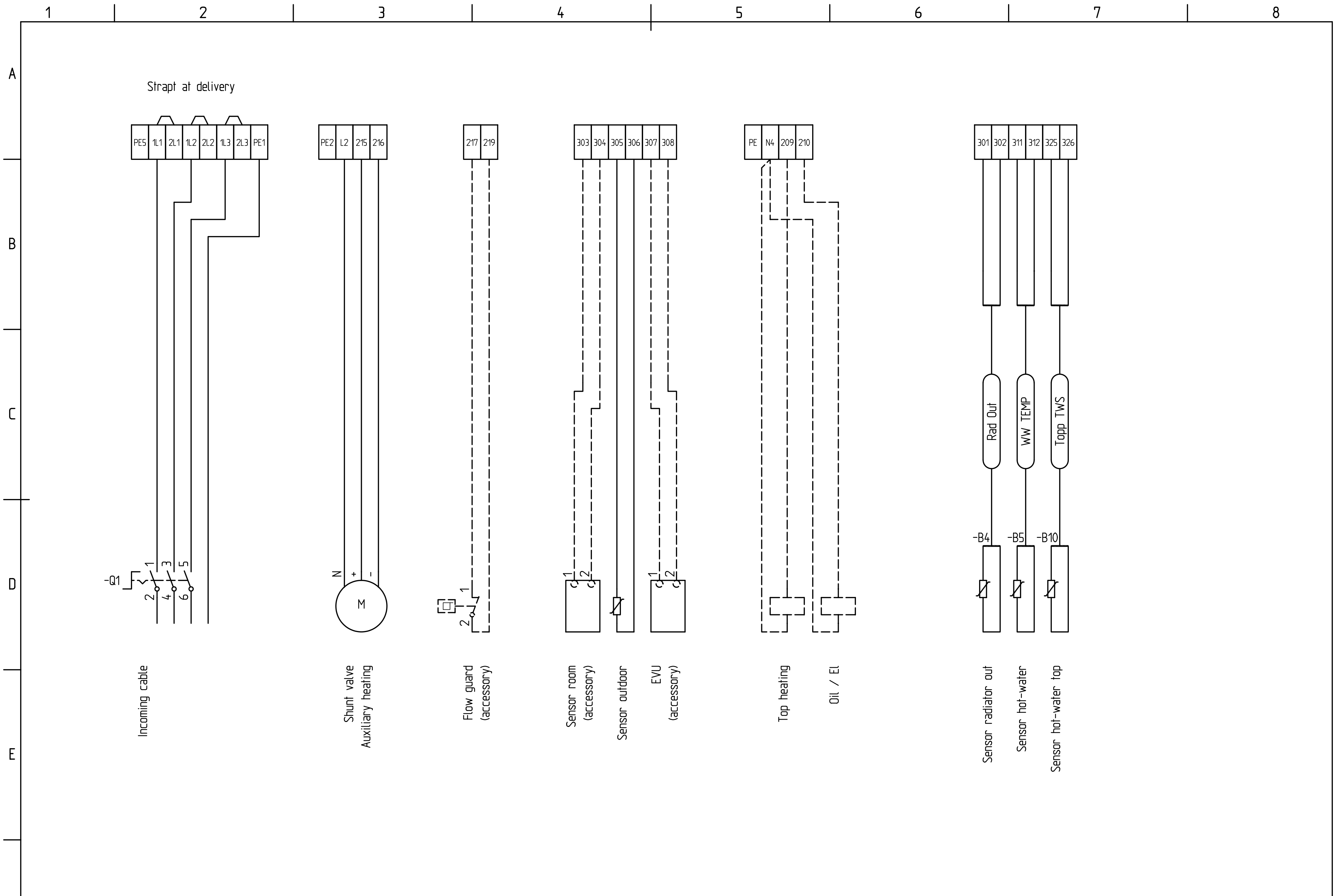

1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Expansion unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
gy-211
gy-212
gy-213
gy-214
gy-215
gy-216
gy -217,a
gy-218
gy-219
bl
301
302
-301
-302
303
304
305
306
307
308
309
310
wh-303
rd-304
wh-305
rd-306
wh-307
rd-308
wh-309
rd-310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
gy -217,b
/1.8D E
gy-104
/1.8D F
bl
217
219
Rad In
Brine Ut
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
1
2
-F11
/1.5B
95
96
N4
209
210
L2
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
1
2
/1.5C
3
4
/1.5C
5
6
/1.5C
13
14
-
21
22
-
-K1
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor reurn line radiator
Sensor return line Brine
Sensor feed line
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	060911
Duo 1800				CONTINUED 3
230V 3P			DRAWING NR. 57282	SHEET 2
Circuit diagram				


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
PE2
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
-Q1
1
3
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2
4
6
N
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-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 060911
		Duo 1800 230V 3P External connections			CONTINUED 4
				DRAWING NR. 57282	SHEET 3

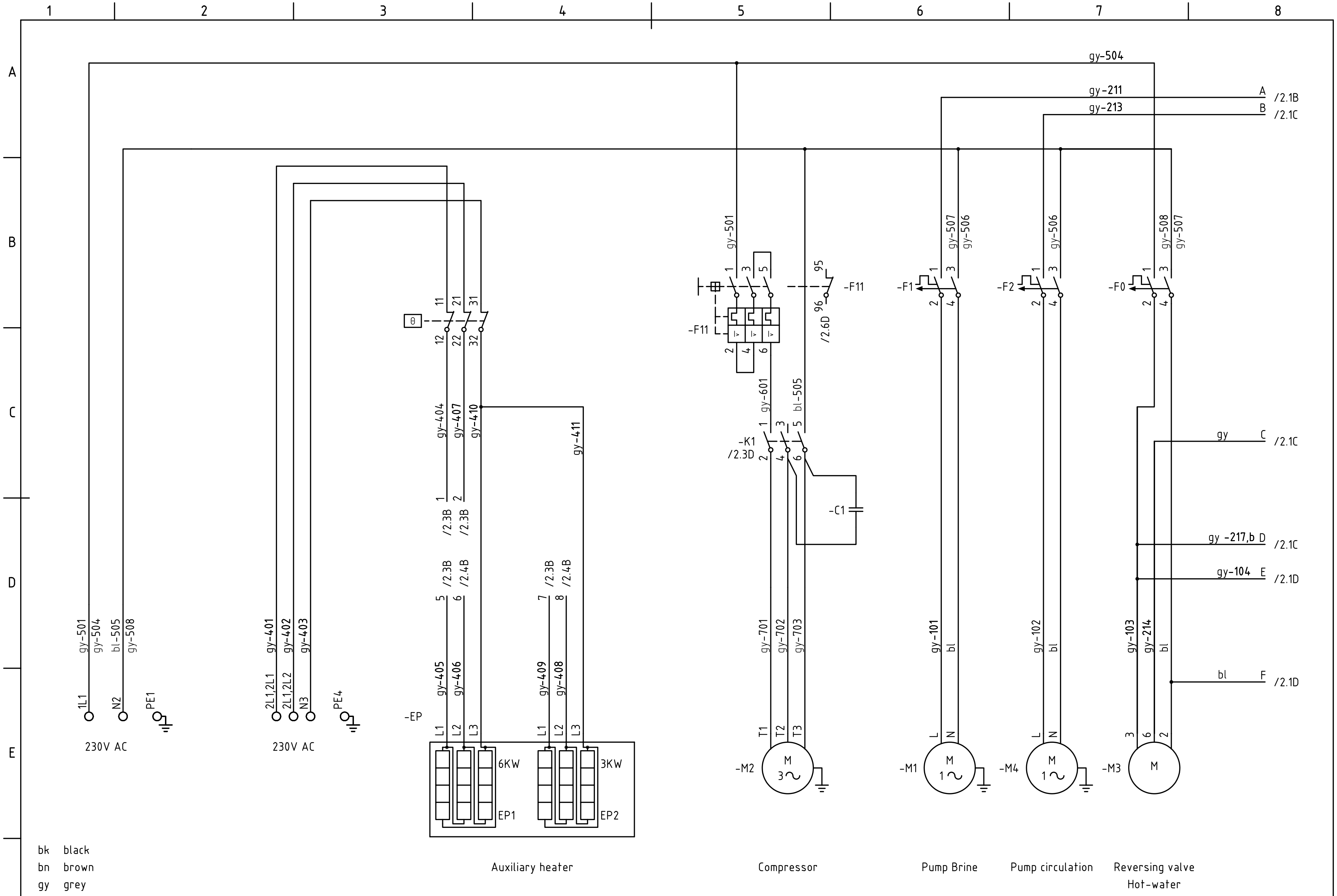
Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B2	1	Switch high pressure	Saginomiya	29,5 bar	9578-55891001	/2.7B
-B3	1	Switch low pressure	Saginomiya	1,5 bar	9578-37142001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/3.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-55701001	/2.3B
-E3	1	Display	Kimsafe AB		9588-55702001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	24-32A	9588-52640004	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	32A 15KW 230VA	9588-38378005	/2.3D
-M1	1	Pump circulation	Wilo		9675-54178001	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51338002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-53865003	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

Thermia

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 060911
CONTINUED -
DRAWING NR. 57282
SHEET 4

Duo 1800
230V 3P
Part list


1
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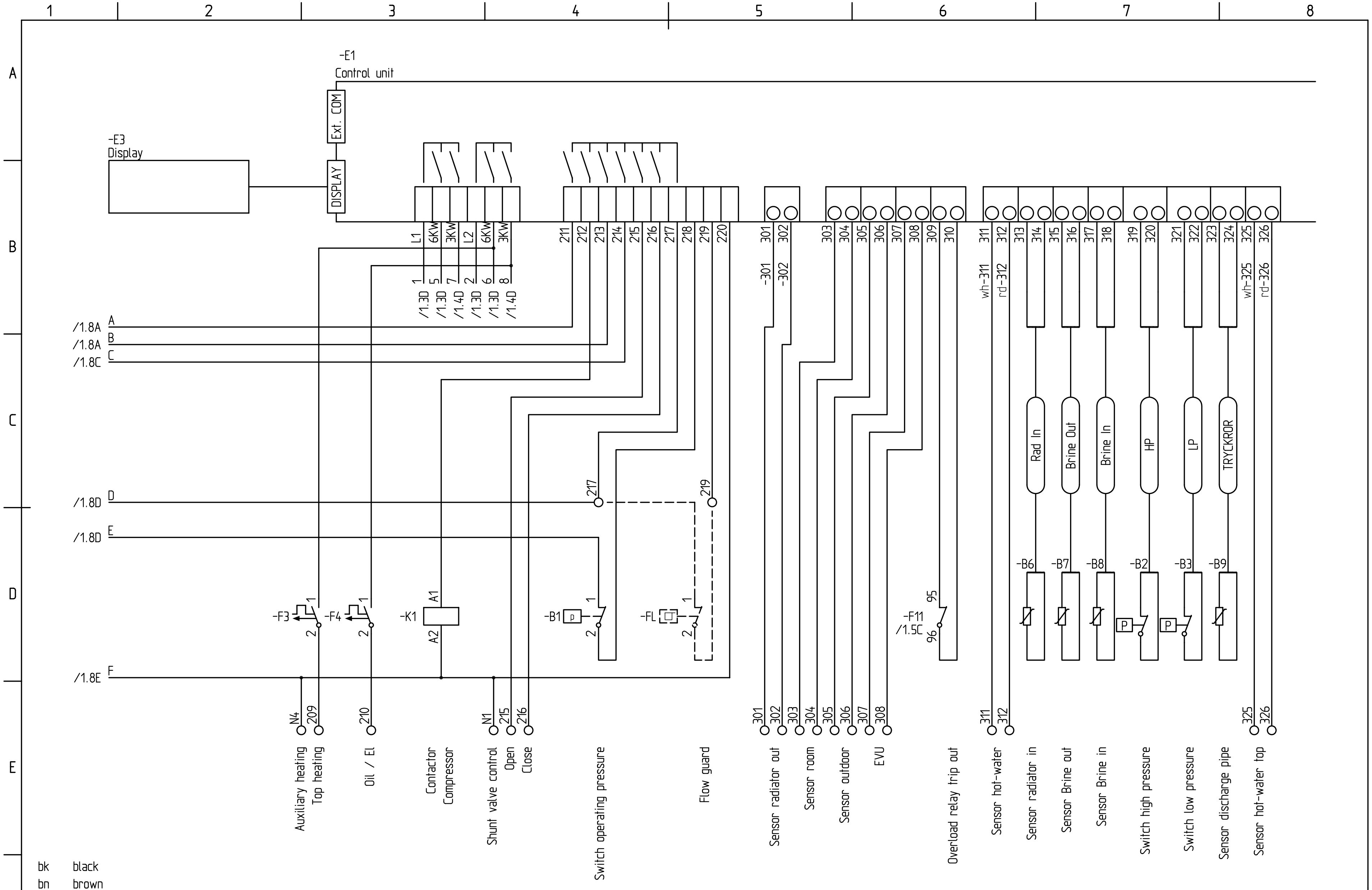
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
-F11
95
96
/2.6D
-F1
-F2
-F0
gy-507
gy-506
gy-508
θ
11 21 31
12 22 32
gy-404
gy-407
gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-504
bl-505
gy-508
1L1
N2
PE1
230V AC
gy-401
gy-402
gy-403
2L1,2L1
2L1,2L2
N3
PE4
230V AC
-EP
gy-405
gy-406
gy-409
gy-408
L1 L2 L3
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M 3~
gy-101
bl
L N
-M1
M 1~
gy-102
bl
L N
-M4
M 1~
gy-103
gy-214
bl
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003
Duo Optimum 6ES				CONTINUED 2
230V 1N			DRAWING NR. 57245	SHEET 1
Circuit diagram				


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1 /1.3D
5 /1.3D
7 /1.4D
2 /1.3D
6 /1.3D
8 /1.4D
211
212
213
214
215
216
217
218
219
220
301
302
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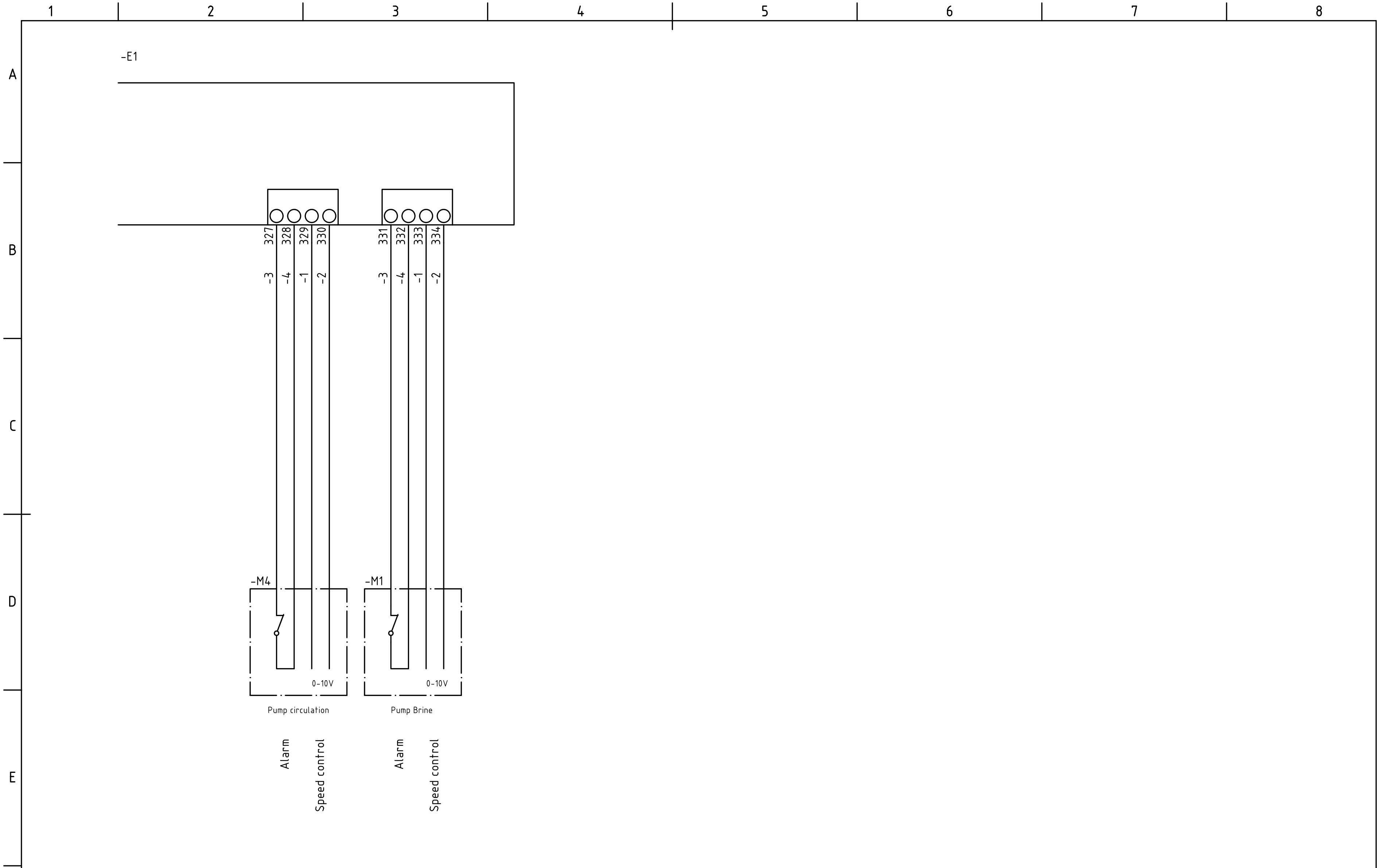
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
217
219
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 061003
Duo Optimum 6ES
230V 1N
Circuit diagram
CONTINUED 3
DRAWING NR. 57245
SHEET 2


1
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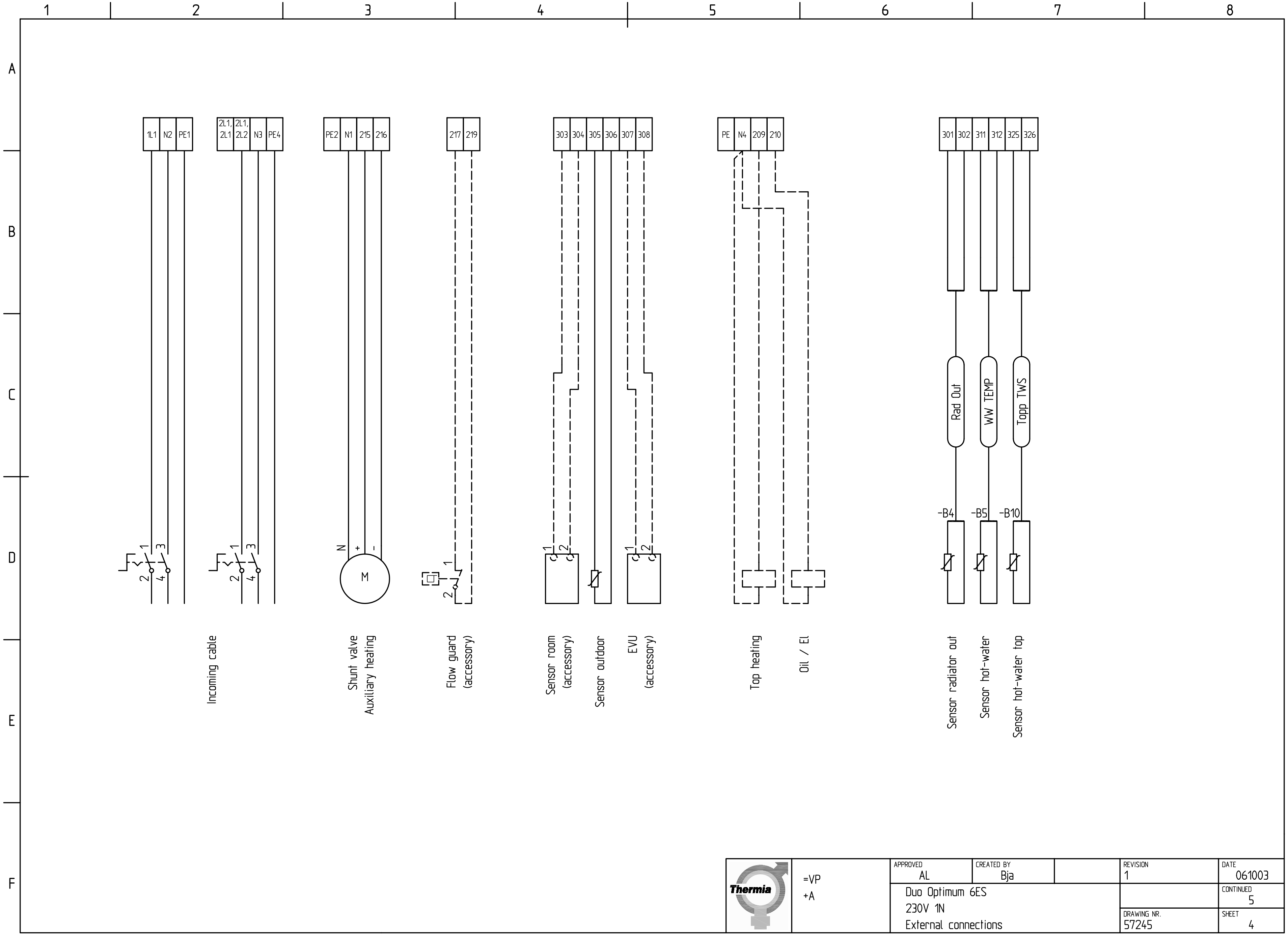
7
8
A
B
C
D
E
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

F

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003
Duo Optimum 6ES				CONTINUED 4
230V 1N			DRAWING NR. 57245	SHEET 3
Circuit diagram				


1L1 N2 PE1
2L1, 2L1 2L1, 2L2 N3 PE4
PE2 N1 215 216
217 219
303 304 305 306 307 308
PE N4 209 210
301 302 311 312 325 326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
M
N + -
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia
=VP
+A
APPROVED
AL
CREATED BY
Bja
REVISION
1
DATE
061003
Duo Optimum 6ES
230V 1N
External connections
CONTINUED
5
DRAWING NR.
57245
SHEET
4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	40 micF	9588-55265002	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	13-18A	9588-54715001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53310002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

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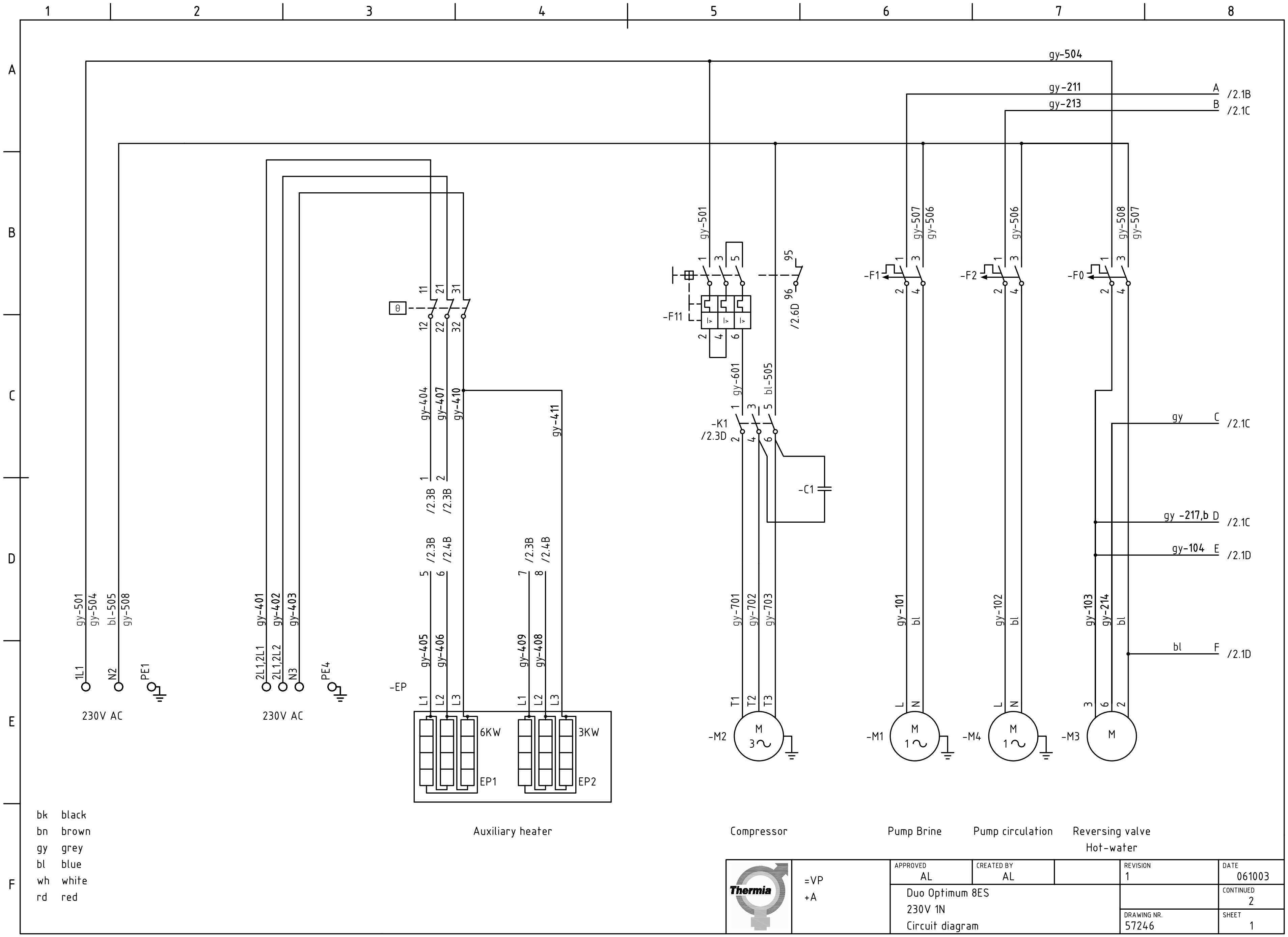
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	061003
Duo Optimum 6ES 230V 1N Part list			DRAWING NR. 57245	CONTINUED - SHEET 5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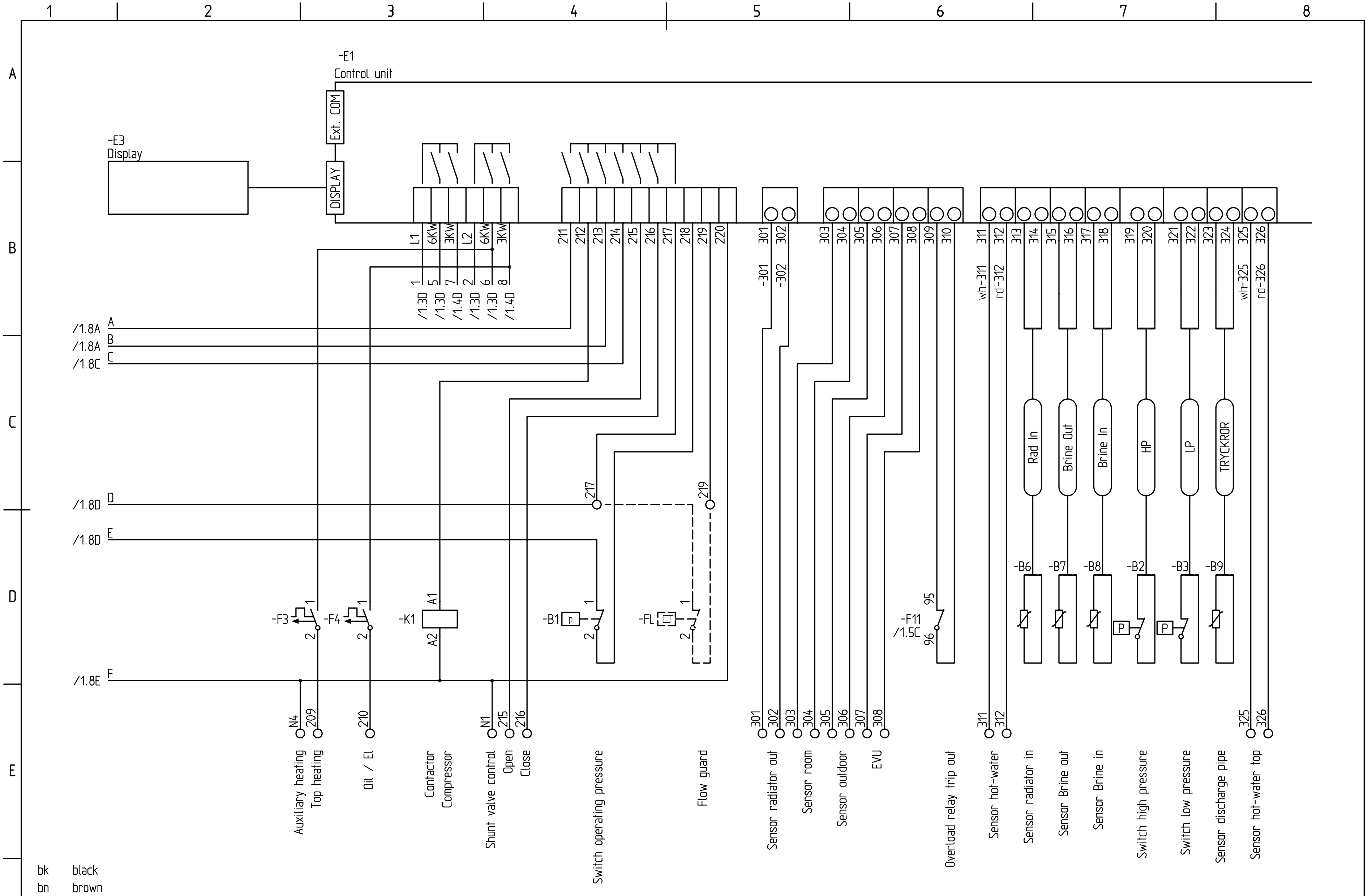
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
gy-504
bl-505
gy-508
1L1
N2
PE1
230V AC
gy-401
gy-402
gy-403
2L1,2L1
2L1,2L2
N3
PE4
230V AC
θ
11 21 31
12 22 32
gy-404
gy-407
gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-405
gy-406
gy-409
gy-408
-EP
L1 L2 L3
6KW
EP1
3KW
EP2
Auxiliary heater
-F11
95
96
/2.6D
gy-601
bl-505
-K1
/2.3D
-C1
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M
3~
Compressor
-F1
gy-507
gy-506
gy-101
bl
L N
-M1
M
1~
Pump Brine
-F2
gy-506
gy-102
bl
-M4
M
1~
Pump circulation
-F0
gy-508
gy-507
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-103
gy-214
bl
3 6 2
-M3
M
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL	CREATED BY AL		REVISION 1	DATE 061003
Duo Optimum 8ES 230V 1N Circuit diagram				CONTINUED 2
			DRAWING NR. 57246	SHEET 1


1
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3
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5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
303
304
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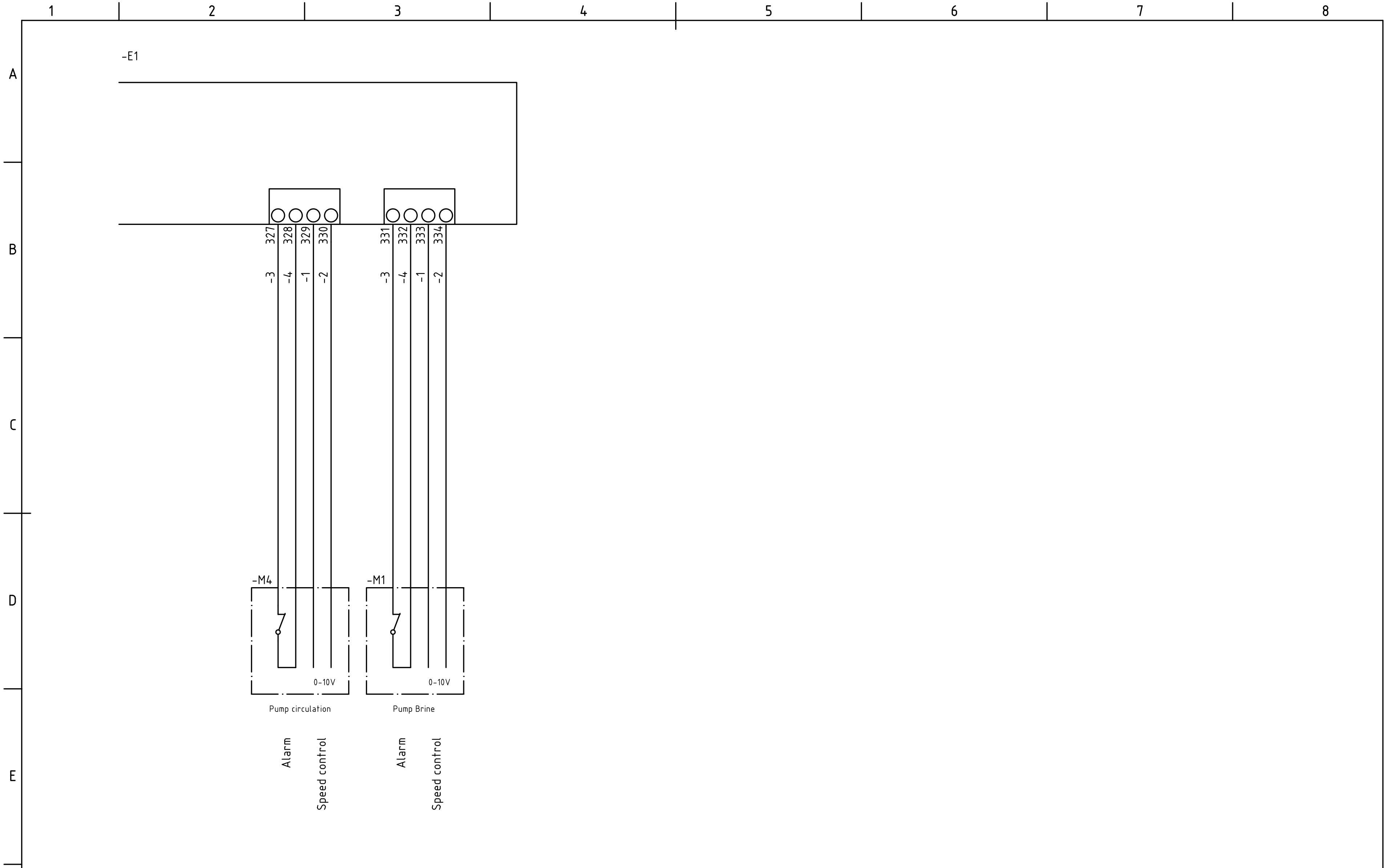
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
217
219
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
-F3
-F4
-K1
A1
A2
-B1
-FL
-F11
/1.5C
95
96
N4
209
210
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	061003
Duo Optimum 8ES				CONTINUED 3
230V 1N			DRAWING NR. 57246	SHEET 2
Circuit diagram				


1
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8
A
B
C
D
E
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

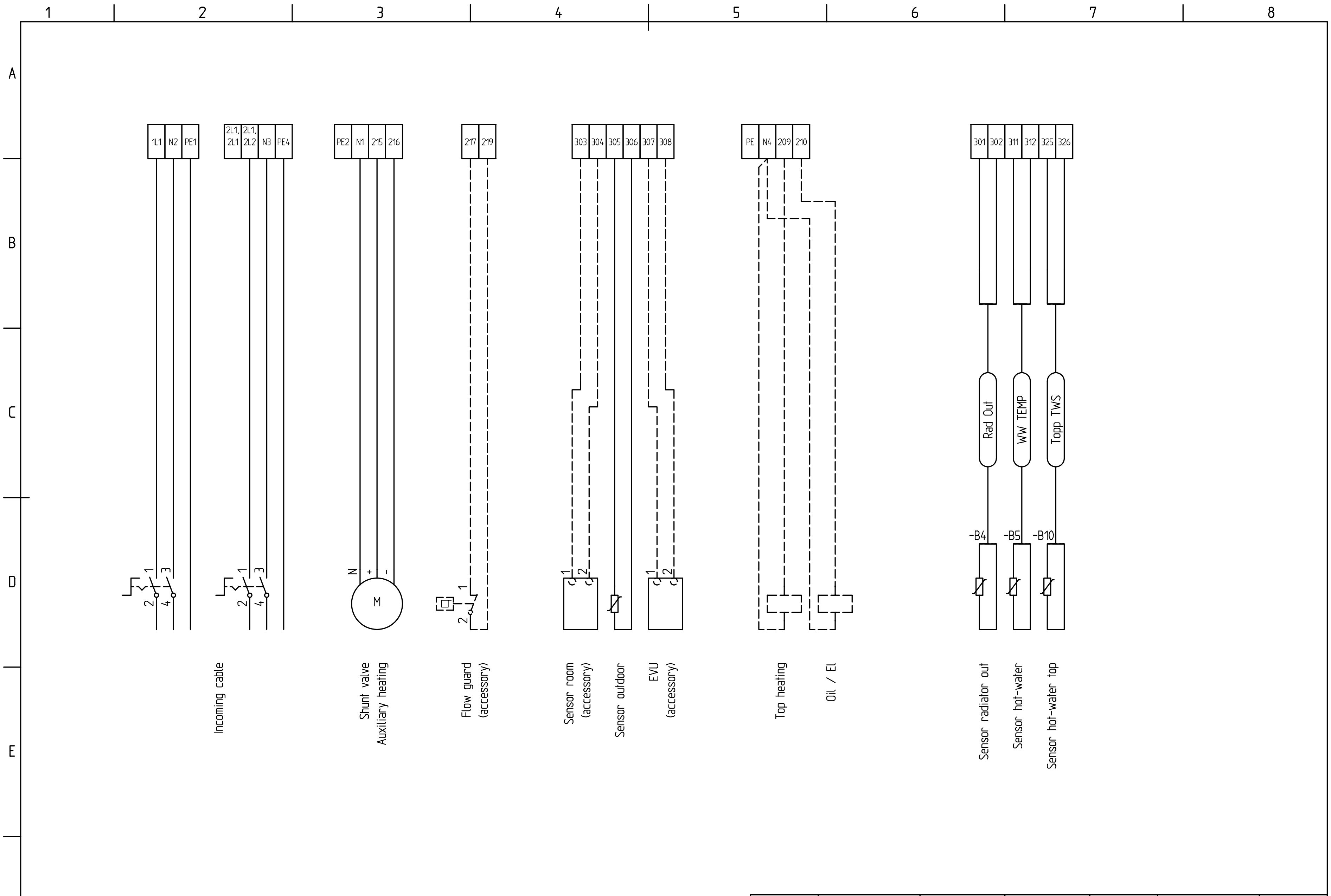
F

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003
Duo Optimum 8ES 230V 1N Circuit diagram			DRAWING NR. 57246	CONTINUED 4 SHEET 3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5
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7
8
A
B
C
D
E
F
1L1
N2
PE1
2L1, 2L1
2L1, 2L2
N3
PE4
PE2
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
M
N
+
-
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

=VP
+A

APPROVED AL
CREATED BY Bja
REVISION 1
DATE 061003
CONTINUED 5
DRAWING NR. 57246
SHEET 4

Duo Optimum 8ES
230V 1N
External connections

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	50 micF	9588-55265004	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	17-23A	9588-55363001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53312002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

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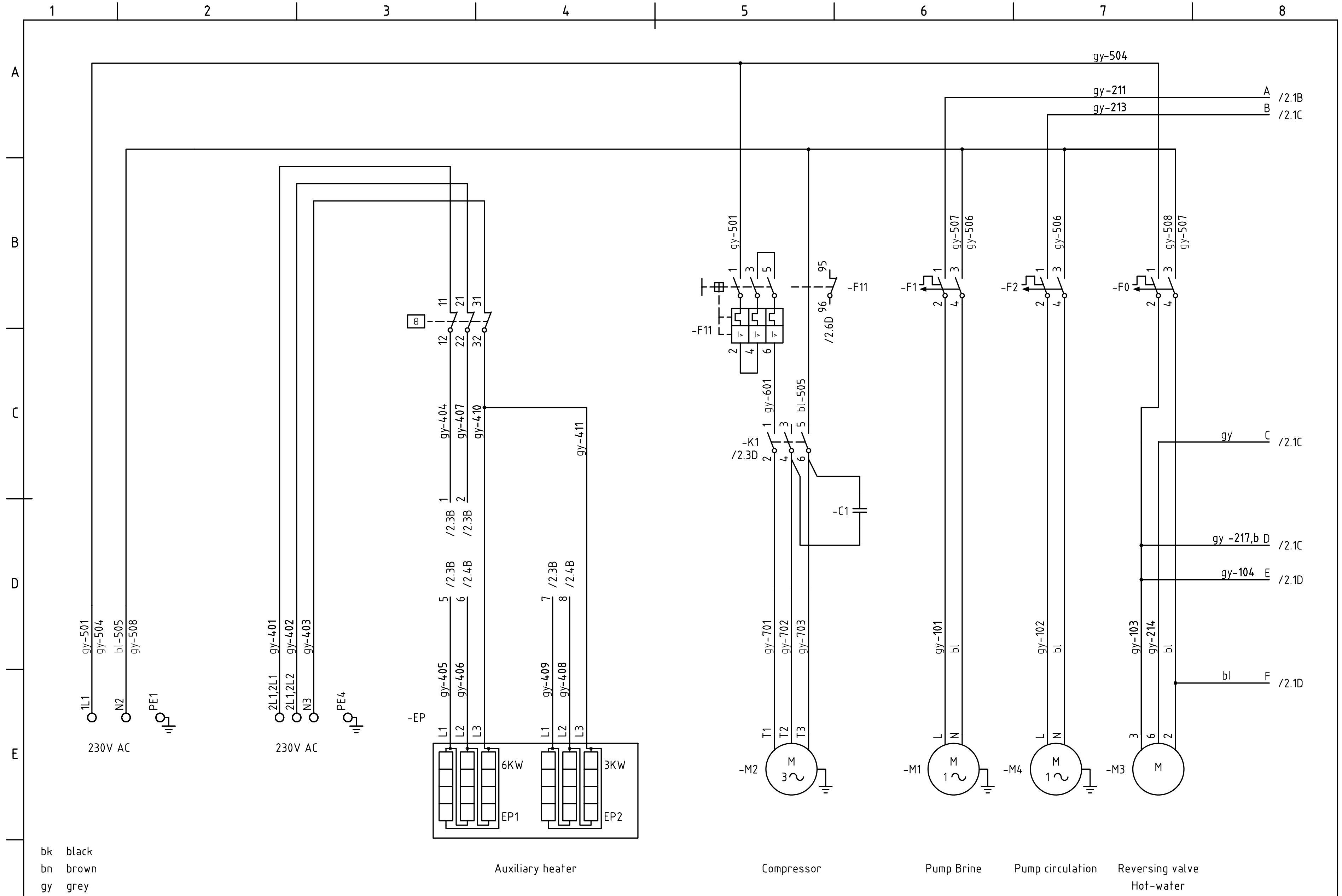
APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	061003
Duo Optimum 8ES 230V 1N Part list			DRAWING NR. 57246	CONTINUED - SHEET 5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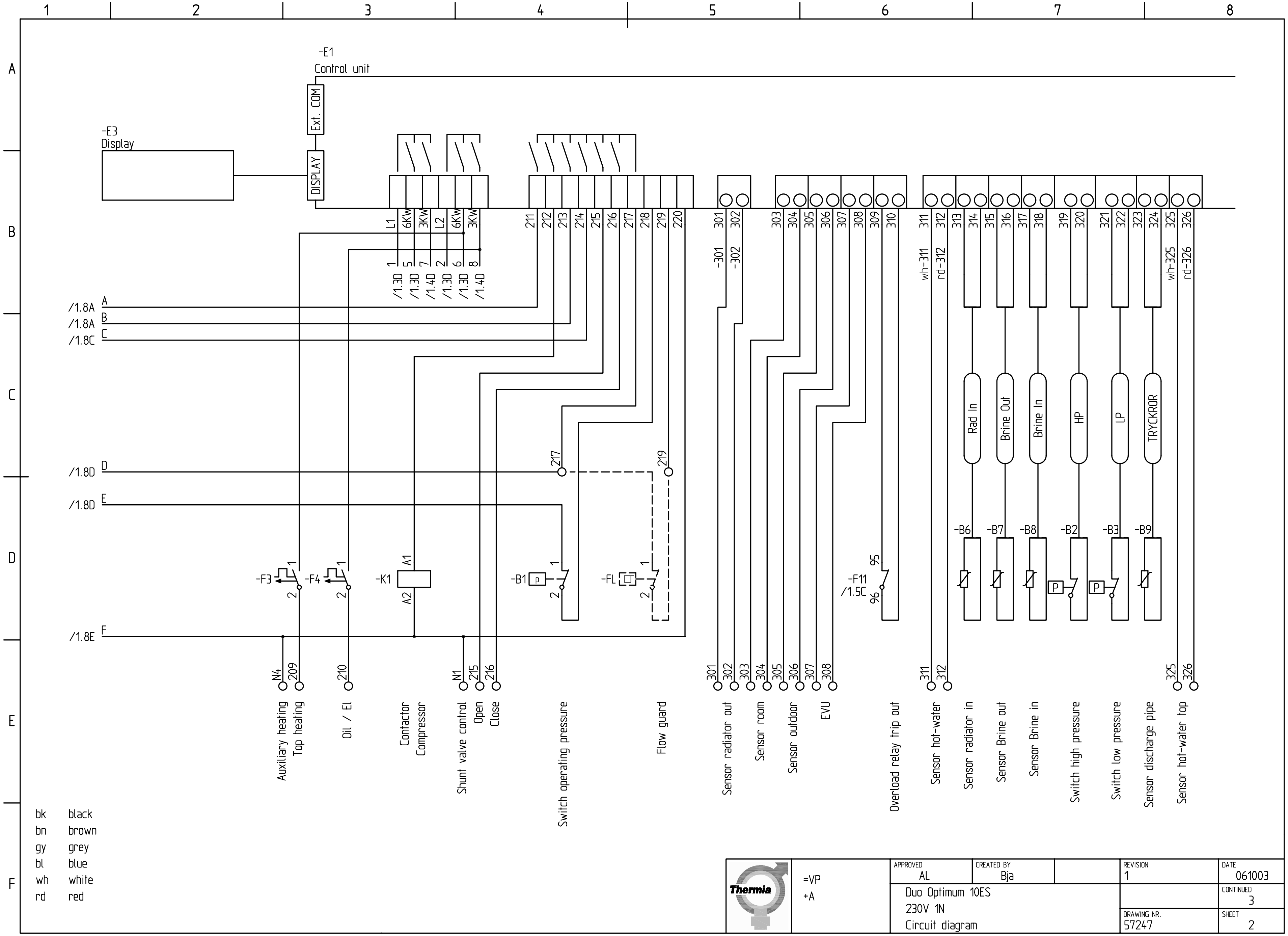
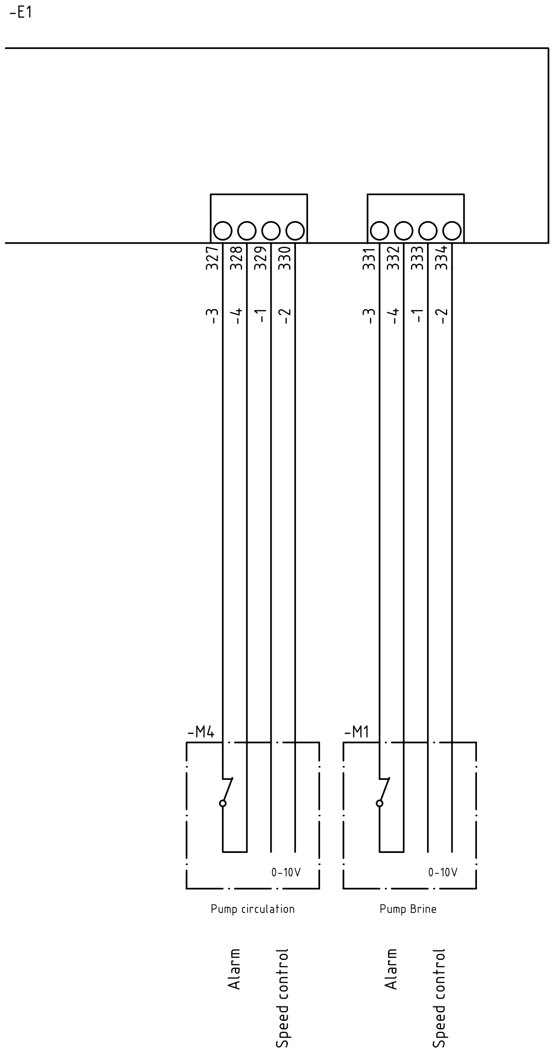
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
gy-507
gy-506
gy-508
-F11
95
96
/2.6D
-F1
-F2
-F0
θ
11
21
31
12
22
32
gy-404
gy-407
gy-410
gy-411
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
/2.3B
/2.4B
gy-501
gy-504
bl-505
gy-508
gy-401
gy-402
gy-403
gy-405
gy-406
gy-409
gy-408
gy-701
gy-702
gy-703
gy-101
gy-102
gy-103
gy-214
bl
1L1
N2
PE1
2L1,2L1
2L1,2L2
N3
PE4
-EP
L1
L2
L3
T1
T2
T3
L
N
230V AC
230V AC
6KW
EP1
3KW
EP2
-M2
M
3~
-M1
1~
-M4
-M3
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

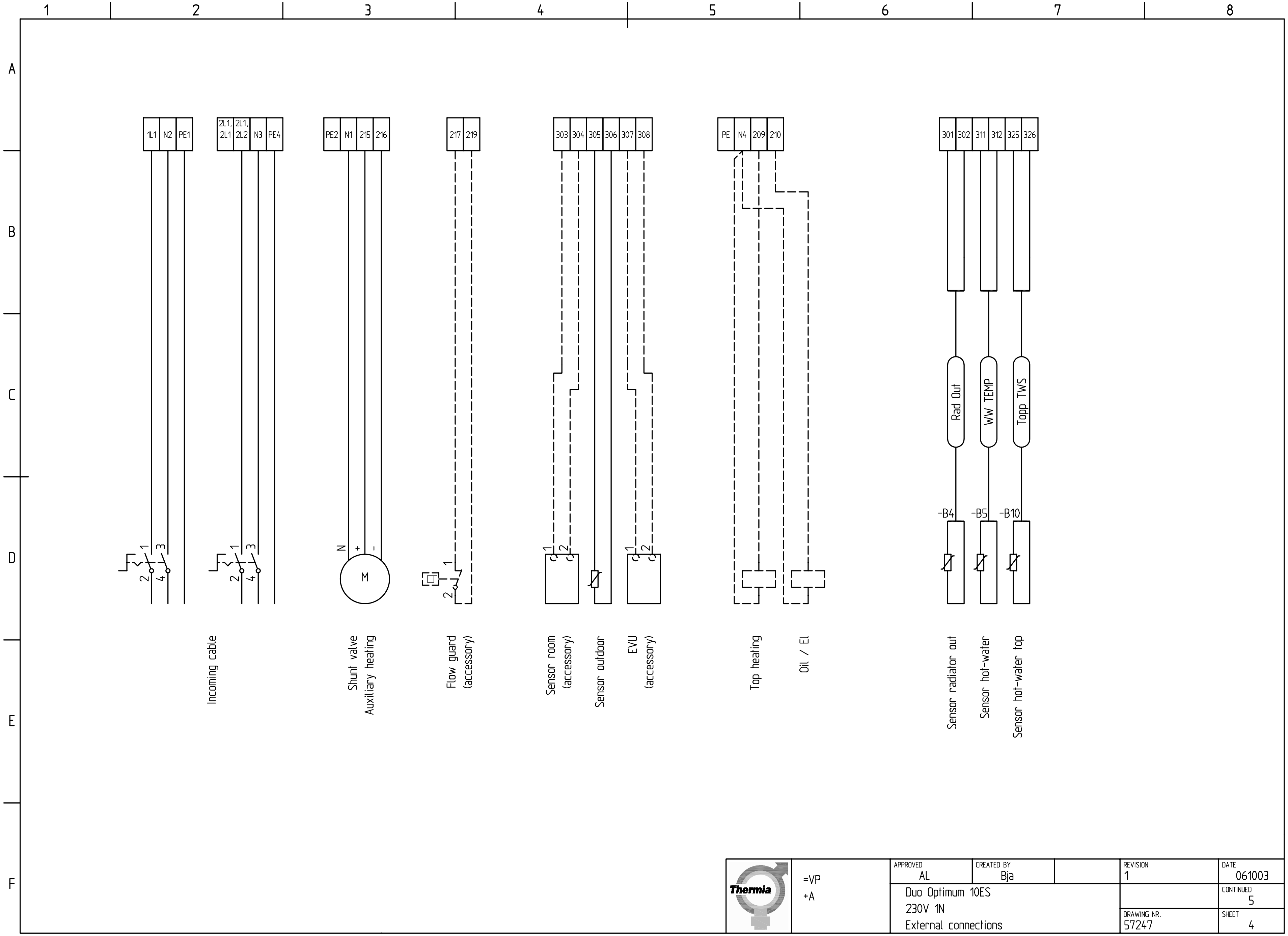
APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003
Duo Optimum 10ES 230V 1N Circuit diagram			DRAWING NR. 57247	CONTINUED 2 SHEET 1


-E1
Control unit
Ext. COM
DISPLAY
-E3
Display
L1 6KW 3KW L2 6KW 3KW
1 5 7 2 6 8
/1.3D /1.3D /1.4D /1.3D /1.3D /1.4D
211 212 213 214 215 216 217 218 219 220
301 302 303 304 305 306 307 308 309 310
311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326
-301 -302
wh-311 rd-312
wh-325 rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6 -B7 -B8 -B2 -B3 -B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
217
219
N4 209 210 N1 215 216
301 302 303 304 305 306 307 308
311 312 325 326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
1 2 3 4 5 6 7 8
A B C D E F
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED AL
CREATED BY Bja
REVISION 1
DATE 061003
Duo Optimum 10ES
230V 1N
Circuit diagram
CONTINUED 3
DRAWING NR. 57247
SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

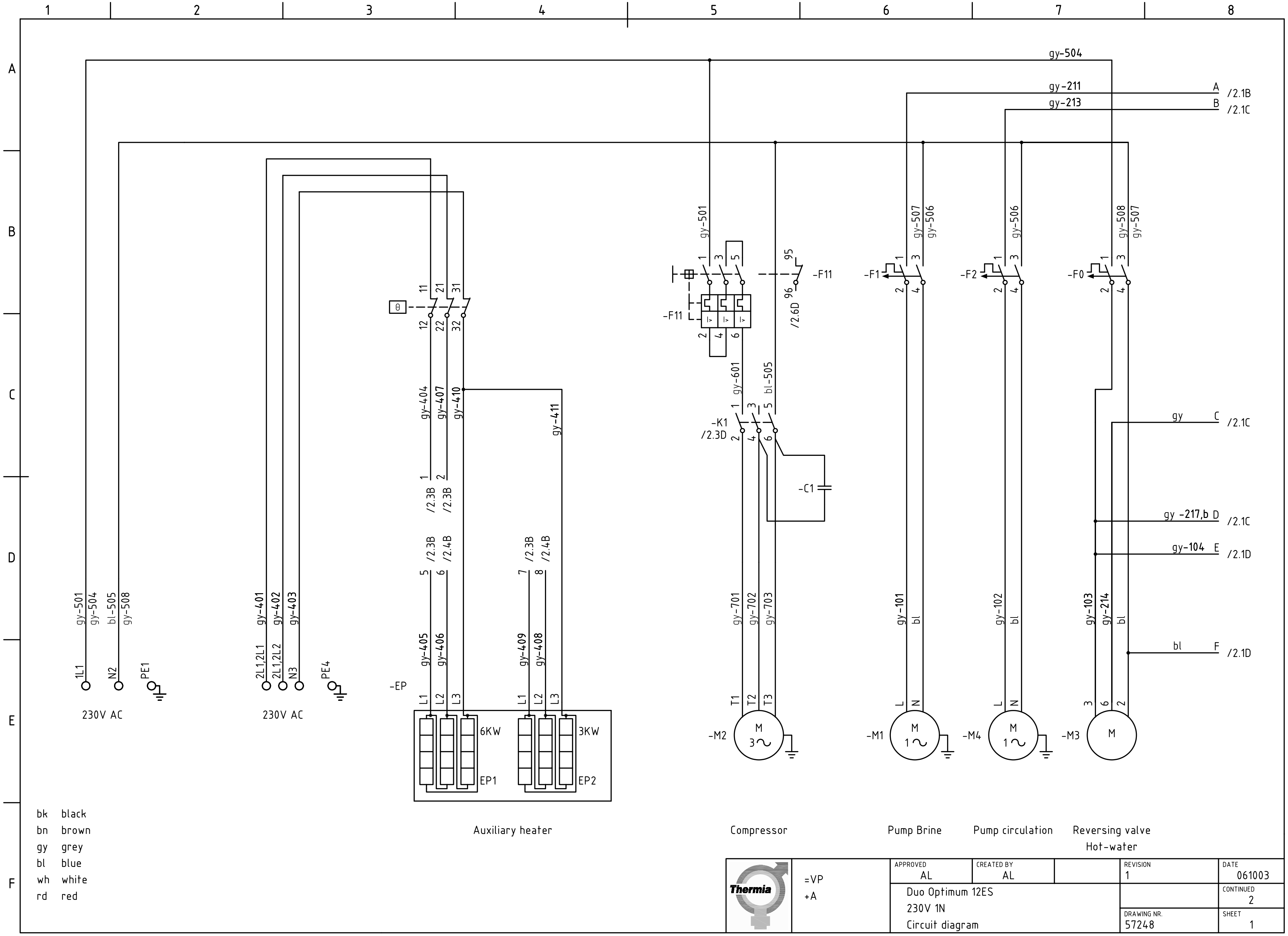
=VP +A	APPROVED AL	CREATED BY AL		REVISION 1	DATE 061003
	Duo Optimum 10ES 230V 1N Circuit diagram			DRAWING NR. 57247	CONTINUED 4 SHEET 3


1
2
3
4
5
6
7
8
A
B
C
D
E
F
1L1
N2
PE1
2L1, 2L1
2L1, 2L2
N3
PE4
PE2
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
M
N
+
-
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia

=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 061003
	Duo Optimum 10ES 230V 1N External connections		DRAWING NR. 57247	CONTINUED 5 SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	60 micF	9588-55265005	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	20-25A	9588-55546001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53313002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

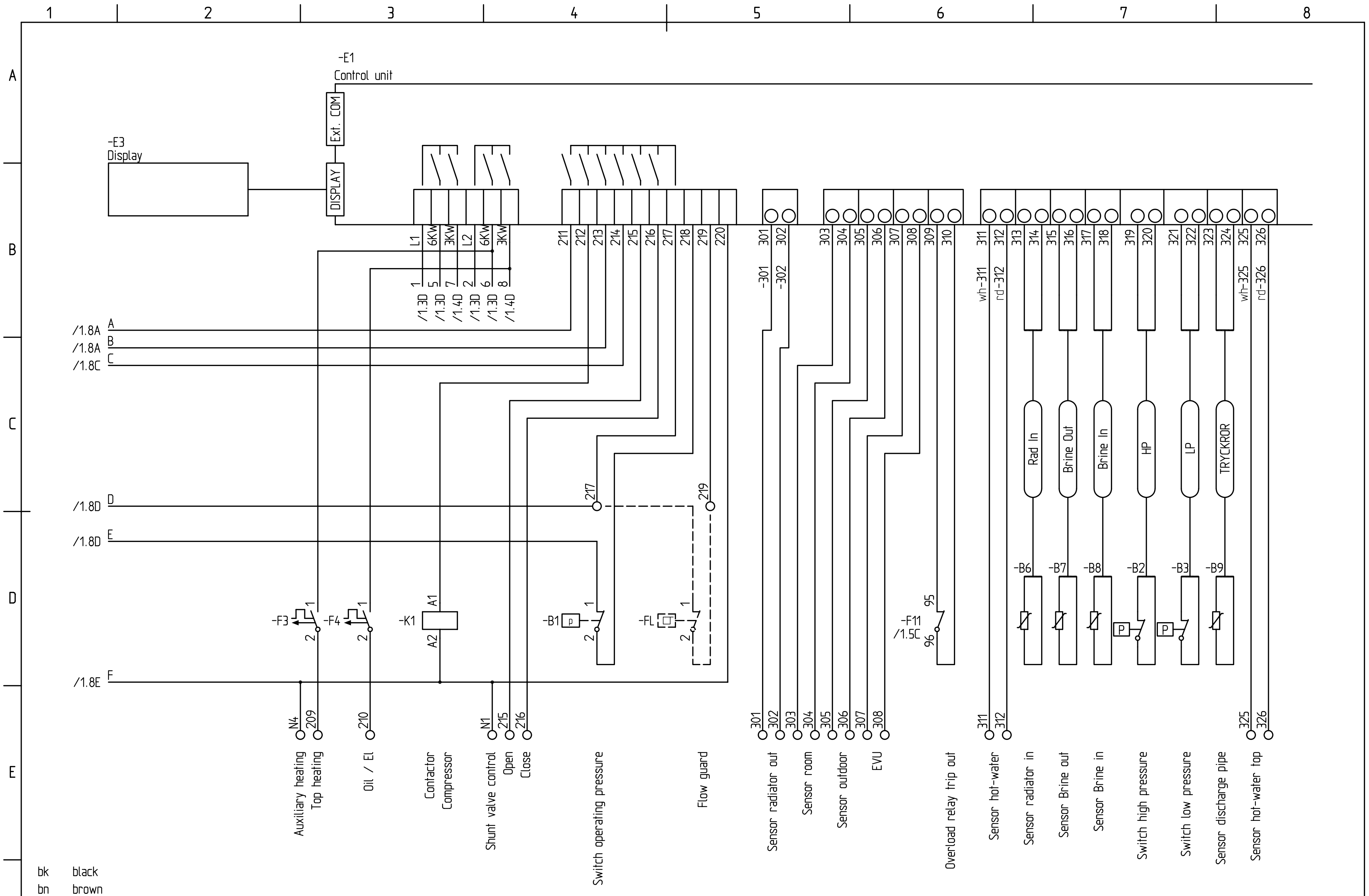
=VP +A	APPROVED AL	CREATED BY Bja	REVISION 1	DATE 061003
	Duo Optimum 10ES 230V 1N Part list			CONTINUED -
			DRAWING NR. 57247	SHEET 5


1
2
3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
gy-501
gy-507
gy-506
gy-506
gy-508
gy-507
-F11
95
96
/2.6D
-F1
-F2
-F0
θ
11
21
31
12
22
32
gy-404
gy-407
gy-410
gy-411
gy-601
bl-505
-K1
/2.3D
-C1
gy C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-501
gy-504
bl-505
gy-508
gy-401
gy-402
gy-403
gy-405
gy-406
gy-409
gy-408
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
gy-103
gy-214
bl
1L1
N2
PE1
2L1,2L1
2L1,2L2
N3
PE4
230V AC
230V AC
-EP
L1
L2
L3
6KW
3KW
EP1
EP2
T1
T2
T3
-M2
M
3~
-M1
M
1~
-M4
M
1~
-M3
M
L
N
3
6
2
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003
Duo Optimum 12ES 230V 1N Circuit diagram			DRAWING NR. 57248	CONTINUED 2 SHEET 1

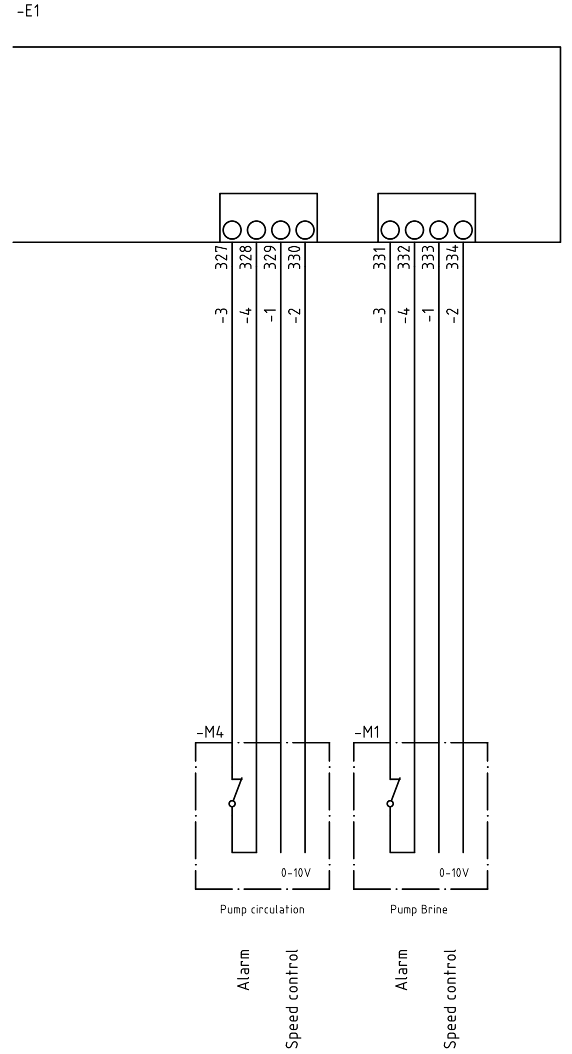

1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6Kw
3Kw
L2
6Kw
3Kw
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8E F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
217
219
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
1
2
N4
209
210
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Contactor
Compressor
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061003
Duo Optimum 12ES 230V 1N Circuit diagram			DRAWING NR. 57248	CONTINUED 3 SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

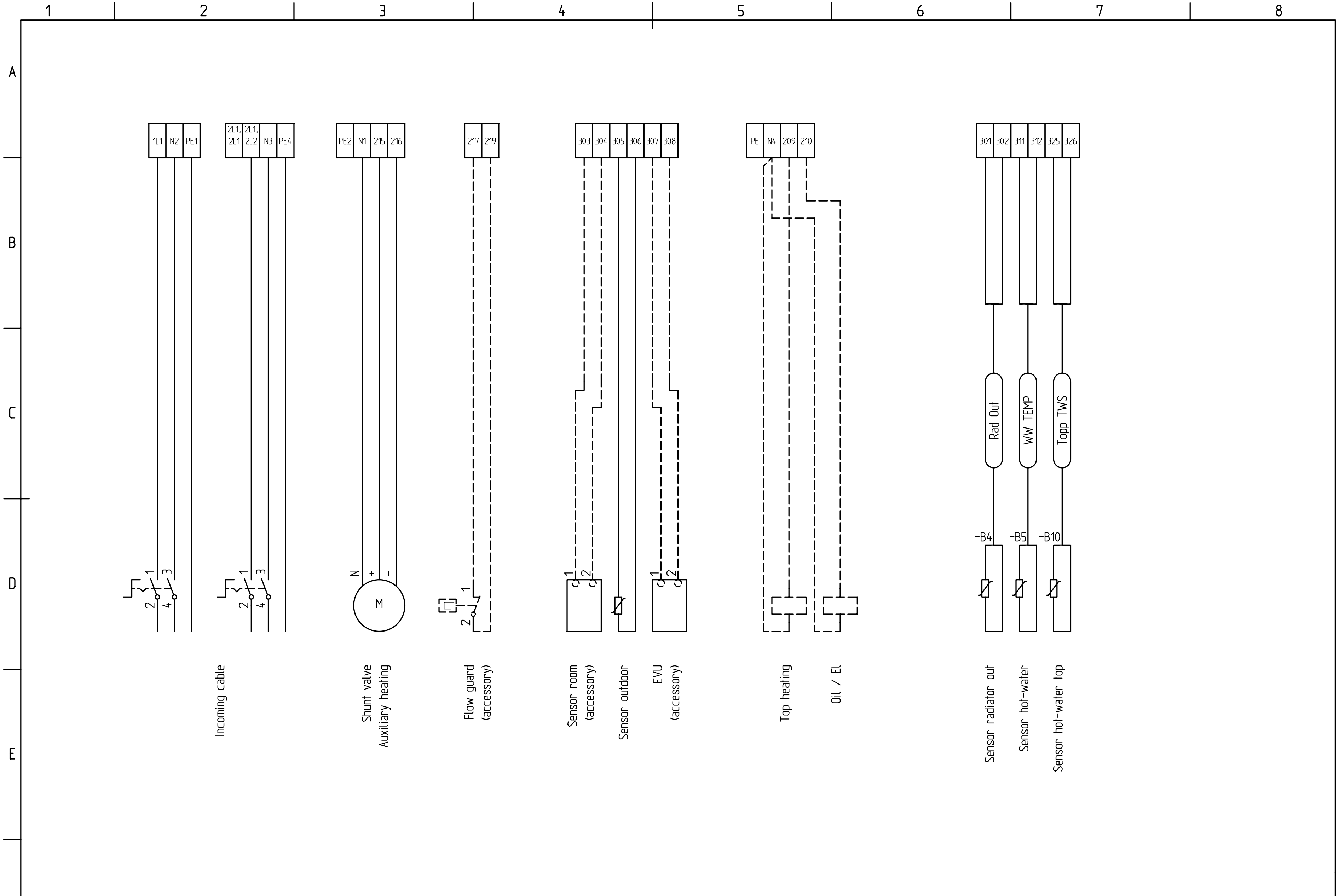
bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061003

Duo Optimum 12ES
230V 1N
Circuit diagram

CONTINUED 4
DRAWING NR. 57248
SHEET 3


1
2
3
4
5
6
7
8
A
B
C
D
E
F
1L1
N2
PE1
2L1, 2L1
2L1, 2L2
N3
PE4
PE2
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
-B5
-B10
N
+
-
M
Incoming cable
Shunt valve
Auxiliary heating
Flow guard (accessory)
Sensor room (accessory)
Sensor outdoor
EVU (accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia

=VP
+A

APPROVED
AL

CREATED BY
Bja

Duo Optimum 12ES
230V 1N
External connections

REVISION
1

DATE
061003

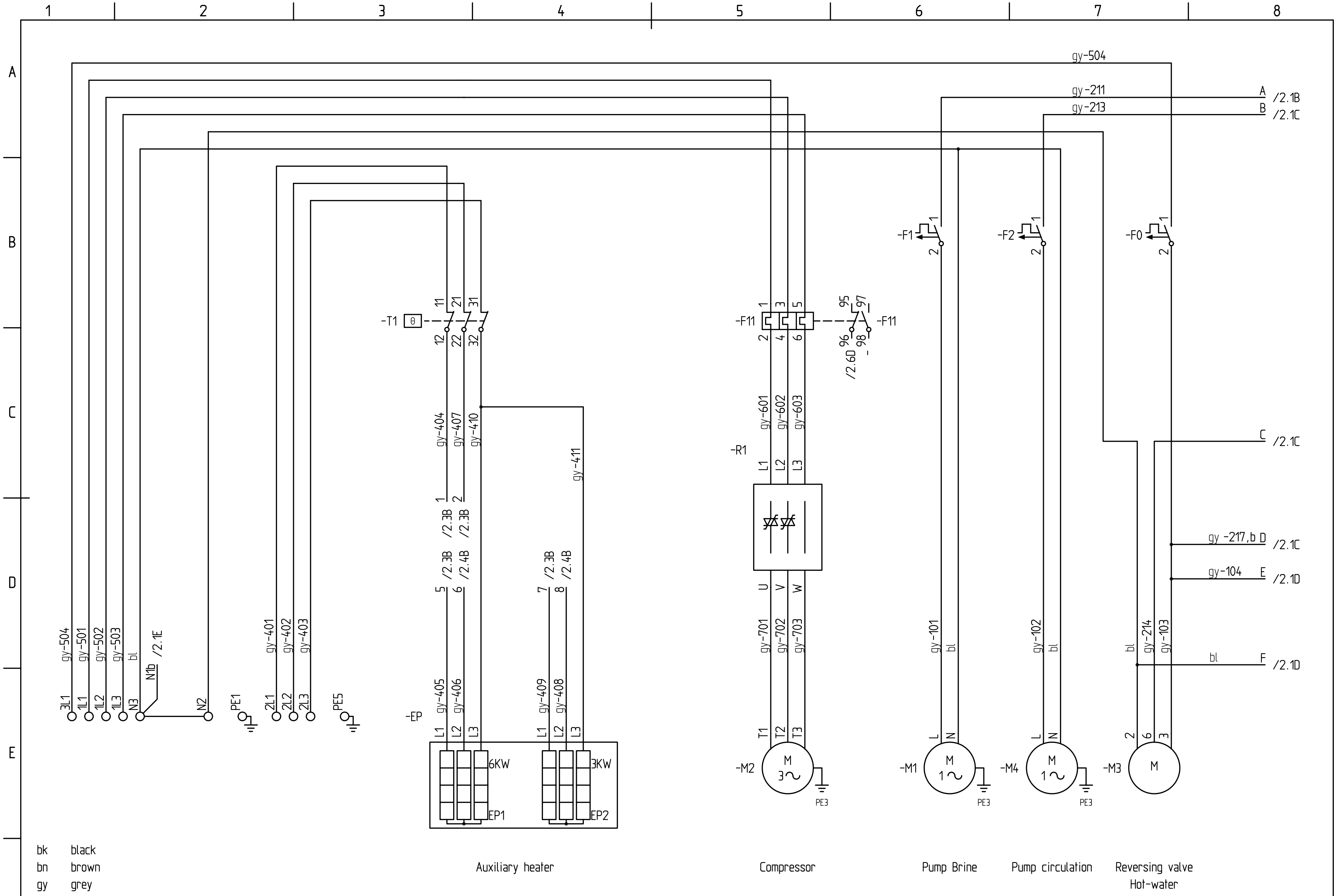
CONTINUED
5

DRAWING NR.
57248

SHEET
4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-C1	1	Capacitor	Ducati	45 micF	9588-55265003	/1.6D
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	20-25A	9588-55546001	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5C
-K1	1	Contactor	Telemecanique	25A 11KW 230VA	9588-38378004	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V	9686-53314002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	230V Es	9583-55529001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061003
	Duo Optimum 12ES 230V 1N Part list			DRAWING NR. 57248	CONTINUED - SHEET 5

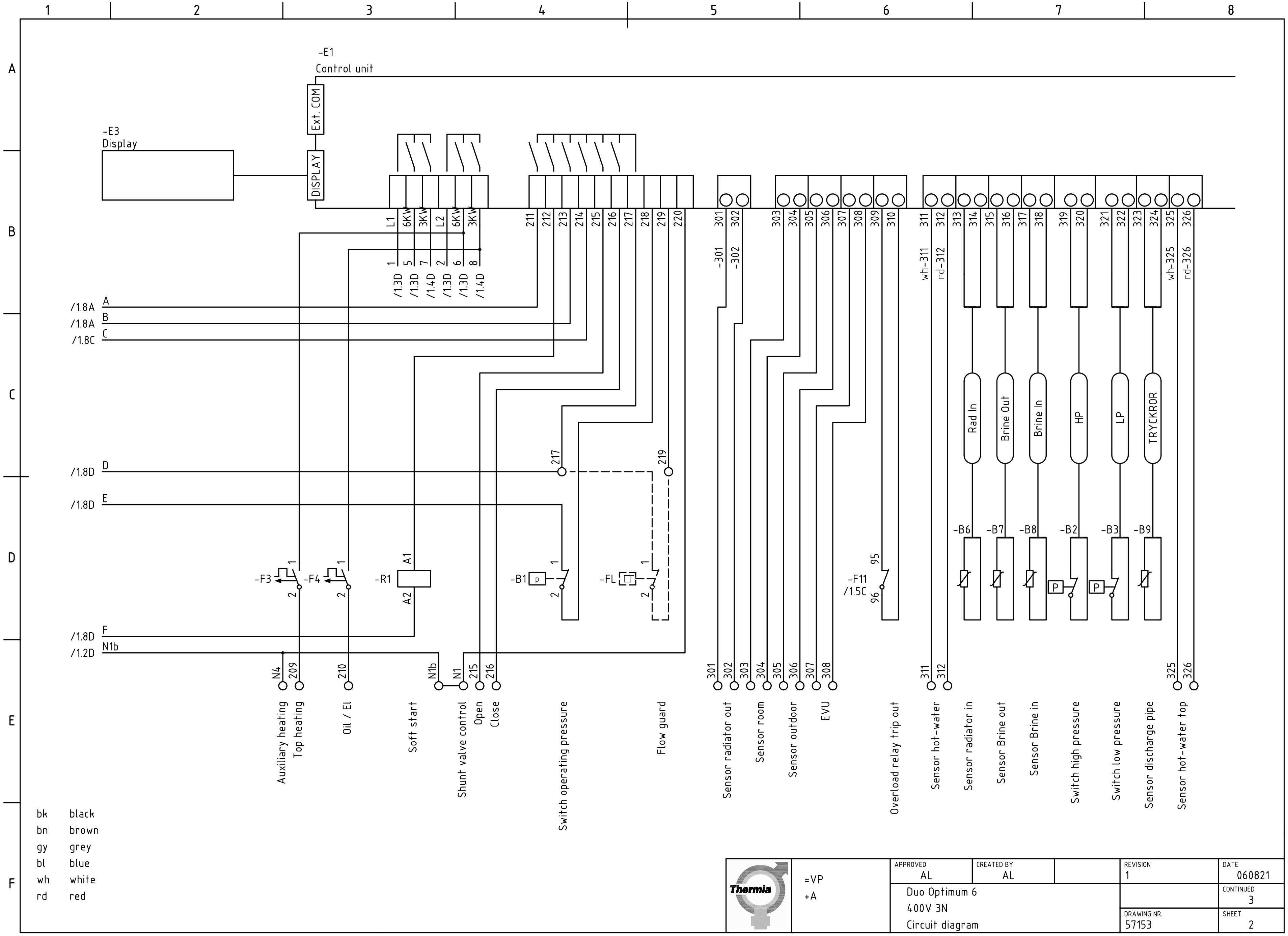

1
2
3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
-F11
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1 L2 L3
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
U V W
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1
1L1
1L2
1L3
N3
N2
PE1
gy-401
gy-402
gy-403
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M 3~
PE3
gy-101
bl
L N
-M1
M 1~
PE3
gy-102
bl
-M4
M 1~
PE3
bl
gy-214
gy-103
2 6 3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

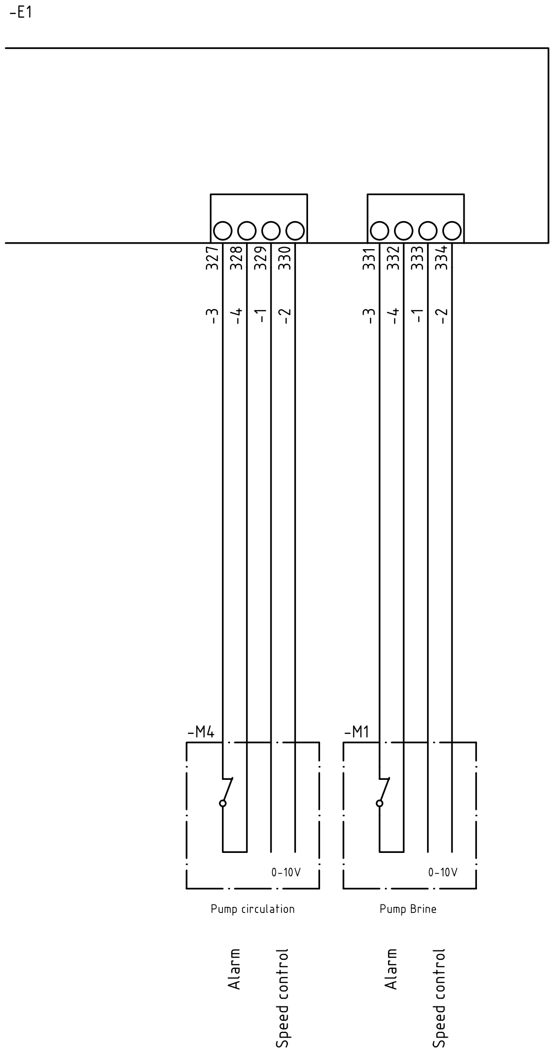
=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	060821
Duo Optimum 6				CONTINUED 2
400V 3N			DRAWING NR.	SHEET
Circuit diagram			57153	1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

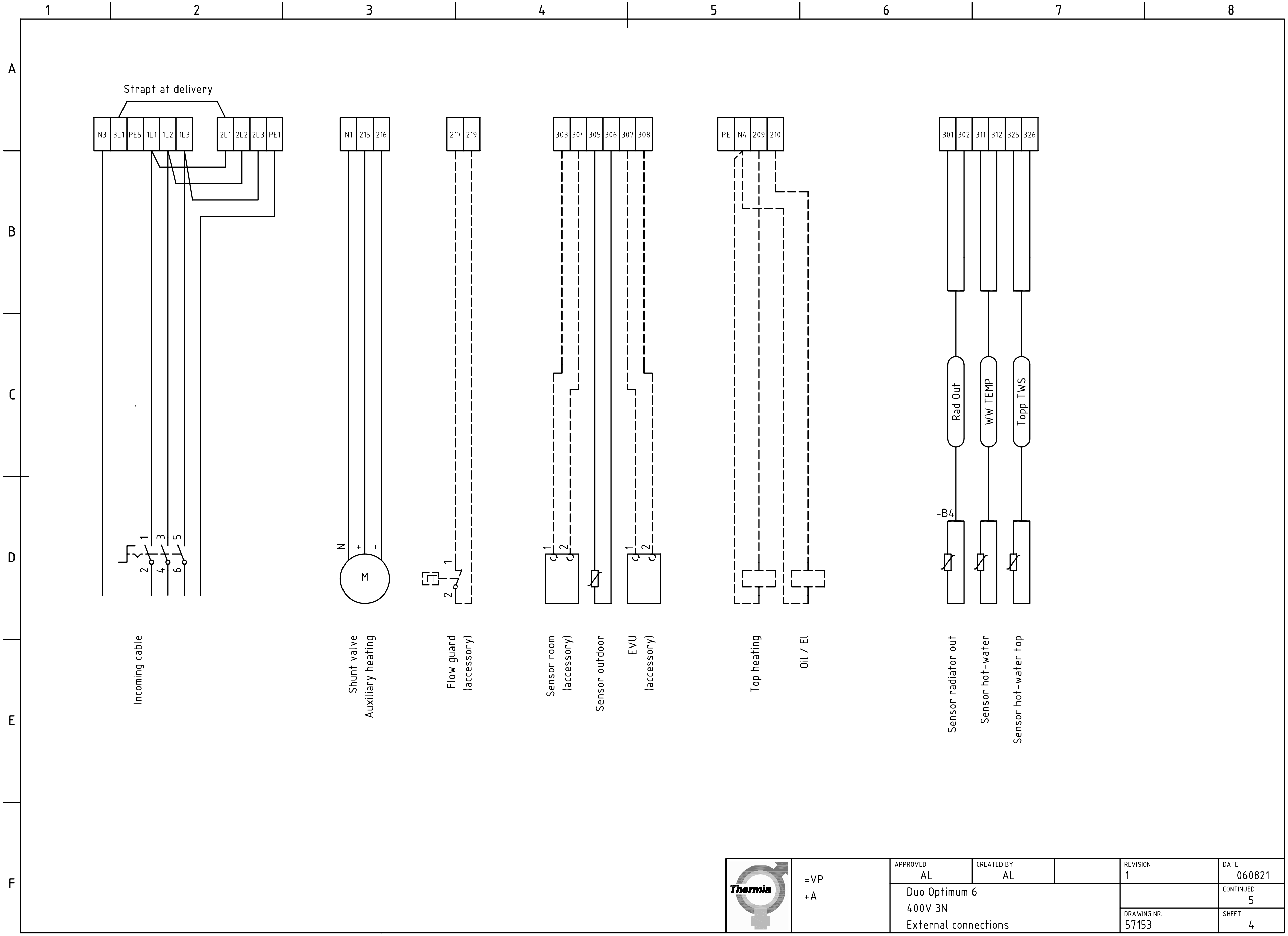
bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY AL	REVISION 1	DATE 060821
		Duo Optimum 6 400V 3N Circuit diagram		DRAWING NR. 57153	CONTINUED 3 SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY AL		REVISION 1	DATE 060821
	Duo Optimum 6 400V 3N Circuit diagram			DRAWING NR. 57153	CONTINUED 4 SHEET 3


1
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6
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	060821

Duo Optimum 6
400V 3N
External connections

CONTINUED 5
DRAWING NR. 57153
SHEET 4

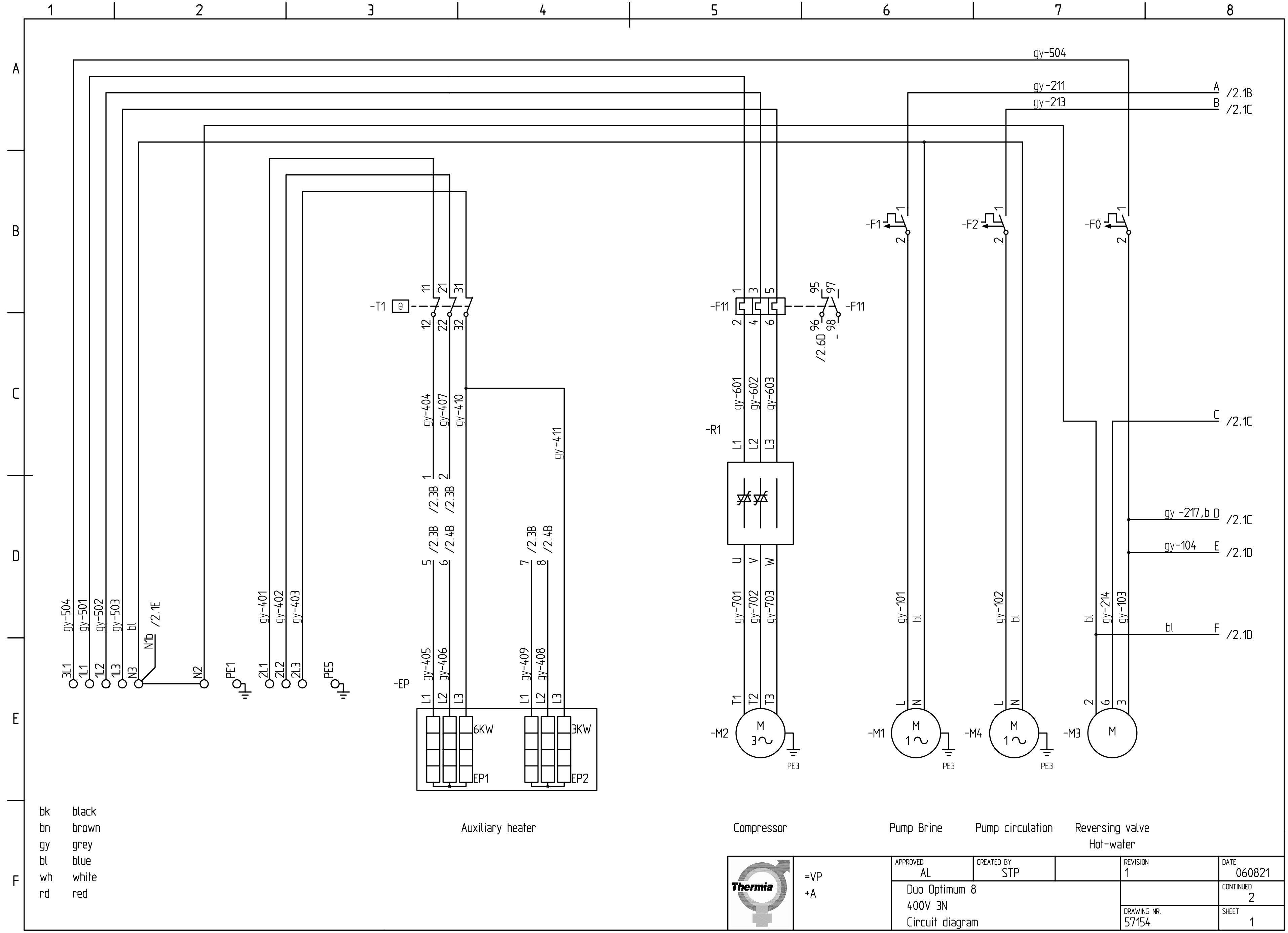
Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	4-6A	9588-38379009	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53310001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	060821
Duo Optimum 6 400V 3N Part list			DRAWING NR. 57153	CONTINUED - SHEET 5

THERMIA DUO OPTIMUM


1
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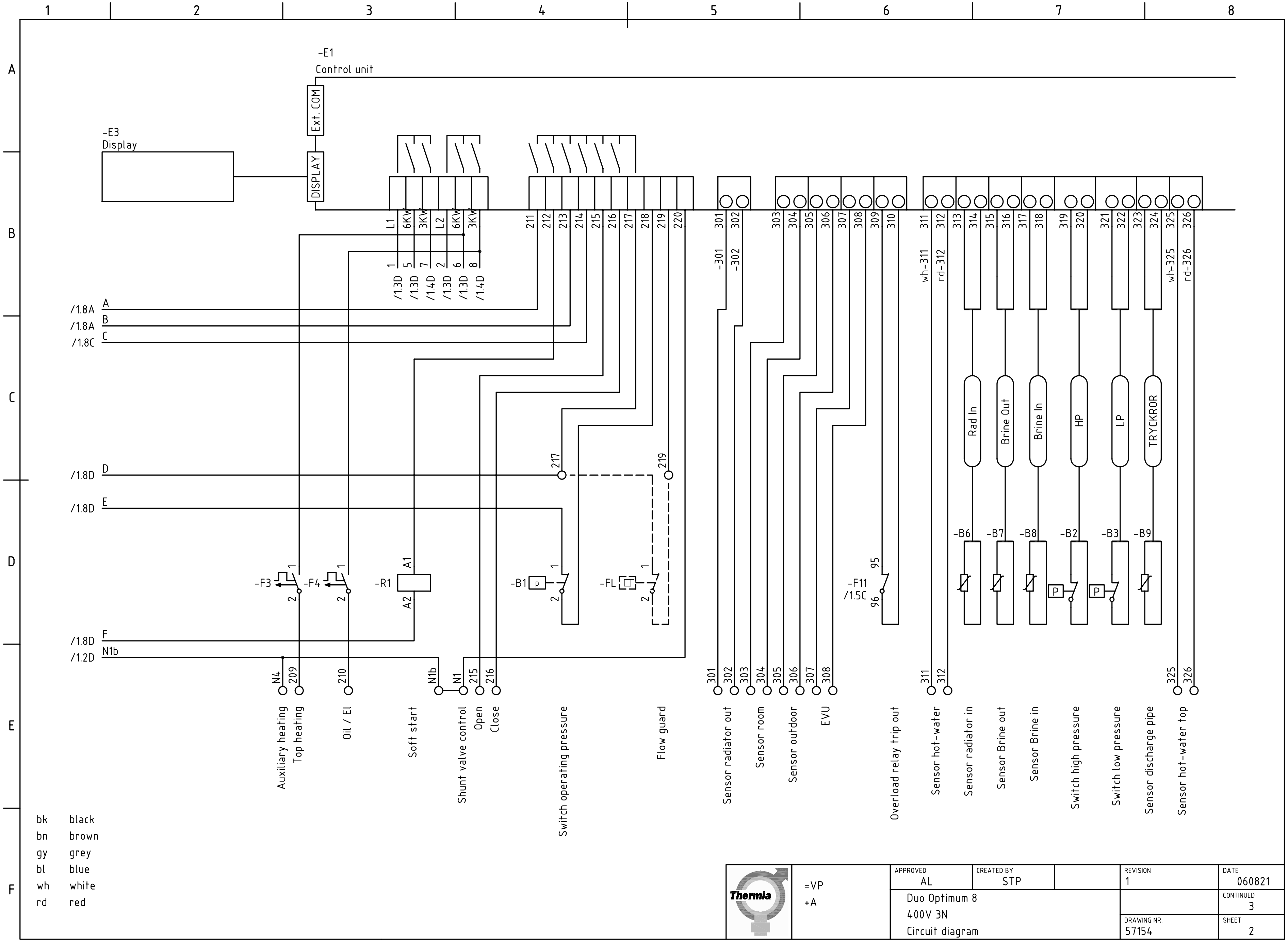
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
-F11
gy-601 gy-602 gy-603
-R1
L1 L2 L3
U V W
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M 3~
PE3
gy-404 gy-407 gy-410
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy-411
gy-405 gy-406
gy-409 gy-408
-EP
L1 L2 L3
6KW
EP1
3KW
EP2
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-101 bl
gy-102 bl
bl gy-214 gy-103
L N
-M1
M 1~
-M4
M 1~
-M3
M
2 6 3
gy-504 gy-501 gy-502 gy-503 bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3 N2
PE1
gy-401 gy-402 gy-403
2L1 2L2 2L3
PE5
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL
CREATED BY STP
REVISION 1
DATE 060821
CONTINUED 2
Duo Optimum 8
400V 3N
Circuit diagram
DRAWING NR. 57154
SHEET 1

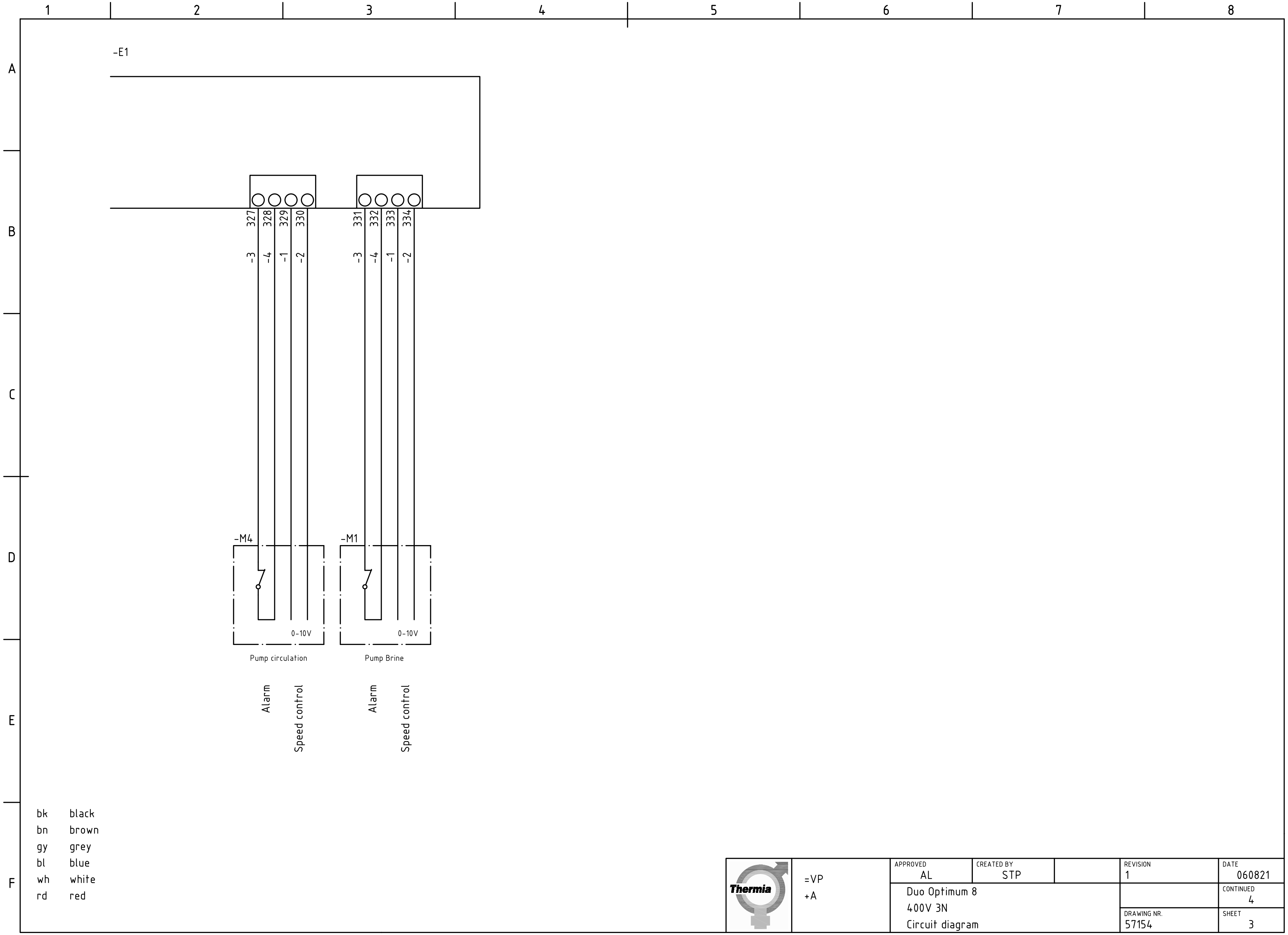

1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
217
219
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
1
2
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	STP		1	060821
Duo Optimum 8				CONTINUED 3
400V 3N			DRAWING NR. 57154	SHEET 2
Circuit diagram				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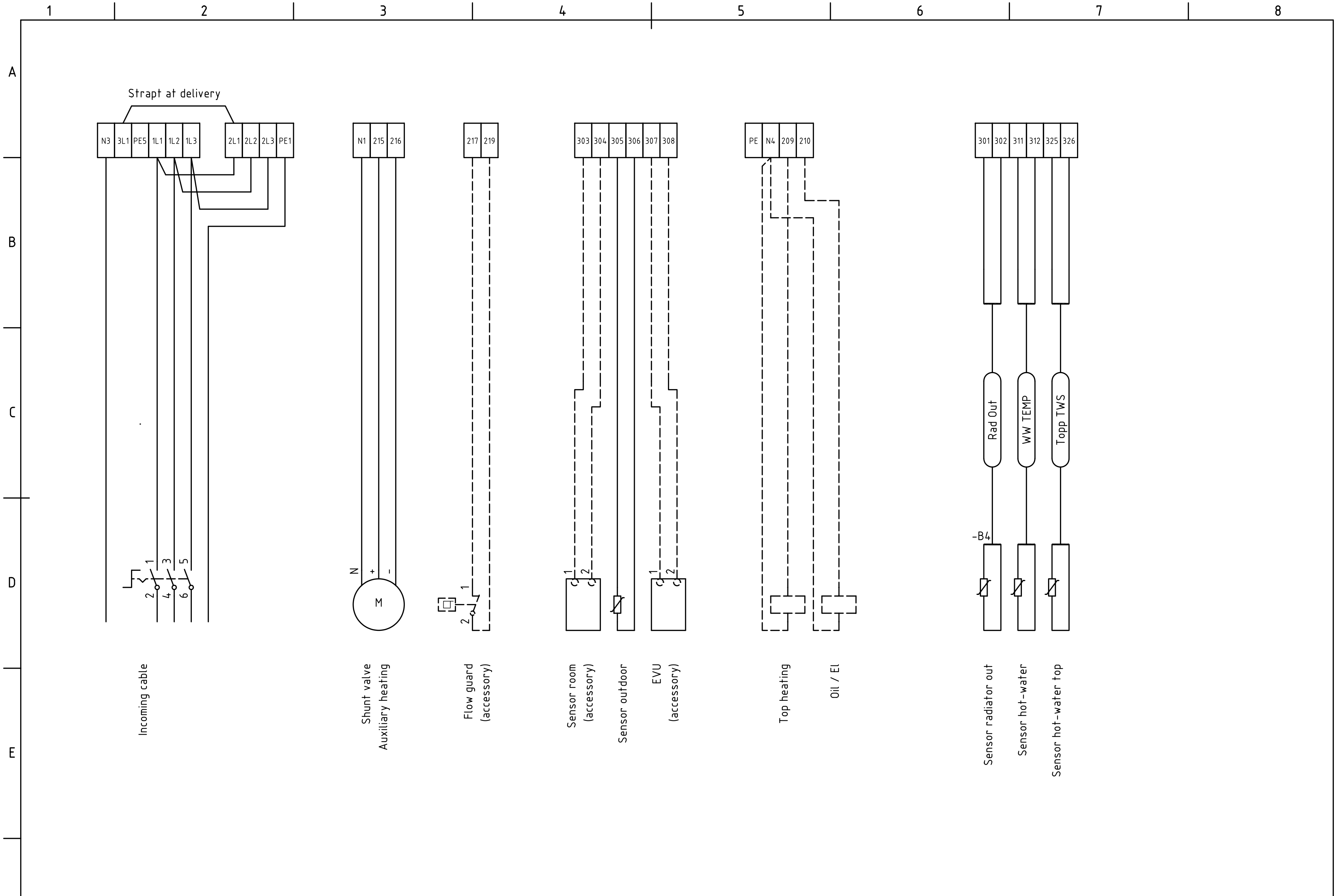
A
B
C
D
E
F
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	STP		1	060821
Duo Optimum 8 400V 3N Circuit diagram				CONTINUED 4
			DRAWING NR. 57154	SHEET 3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8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
1
3
5
2
4
6
N
+
-
M
1
2
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

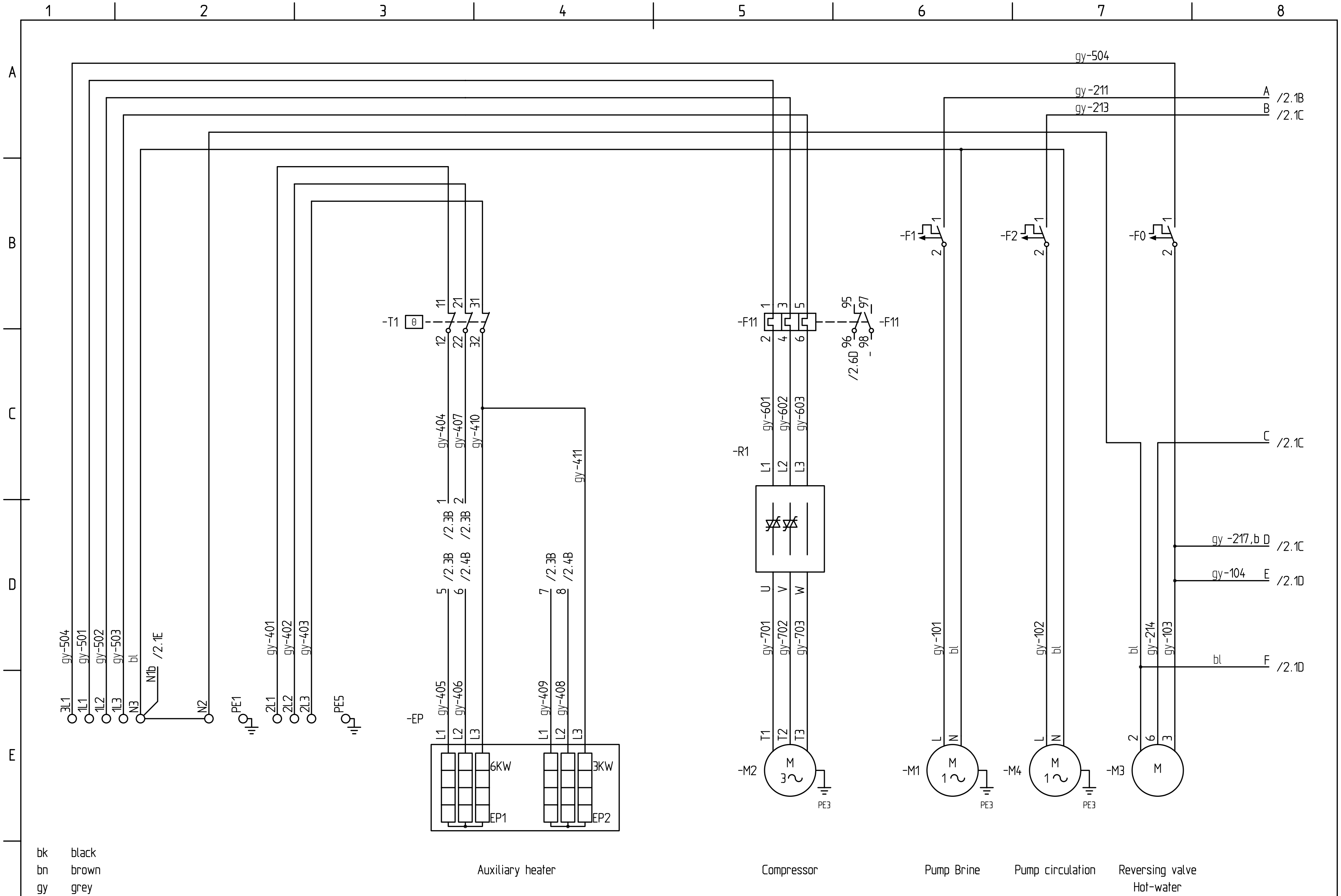
Thermia	=VP +A	APPROVED AL	CREATED BY STP	REVISION 1	DATE 060821
		Duo Optimum 8 400V 3N External connections		DRAWING NR. 57154	CONTINUED 5 SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	5,5-8A	9588-38379010	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53312001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	STP		1	060821
Duo Optimum 8 400V 3N Part list			DRAWING NR. 57154	CONTINUED - SHEET 5


1
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3
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5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
1
2
-T1
θ
11
21
31
12
22
32
-F11
1
3
5
2
4
6
95
97
96
98
/2.6D
gy-404
gy-407
gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
-R1
gy-601
gy-602
gy-603
L1
L2
L3
U
V
W
gy-701
gy-702
gy-703
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-101
gy-102
bl
gy-214
gy-103
gy-504
gy-501
gy-502
gy-503
N1b /2.1E
3L1
1L1
1L2
1L3
N3
N2
PE1
gy-401
gy-402
gy-403
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
6KW
3KW
EP1
EP2
T1
T2
T3
-M2
M
3~
PE3
L
N
-M1
1~
-M4
-M3
2
6
3
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED
AL

CREATED BY
STP

REVISION
1

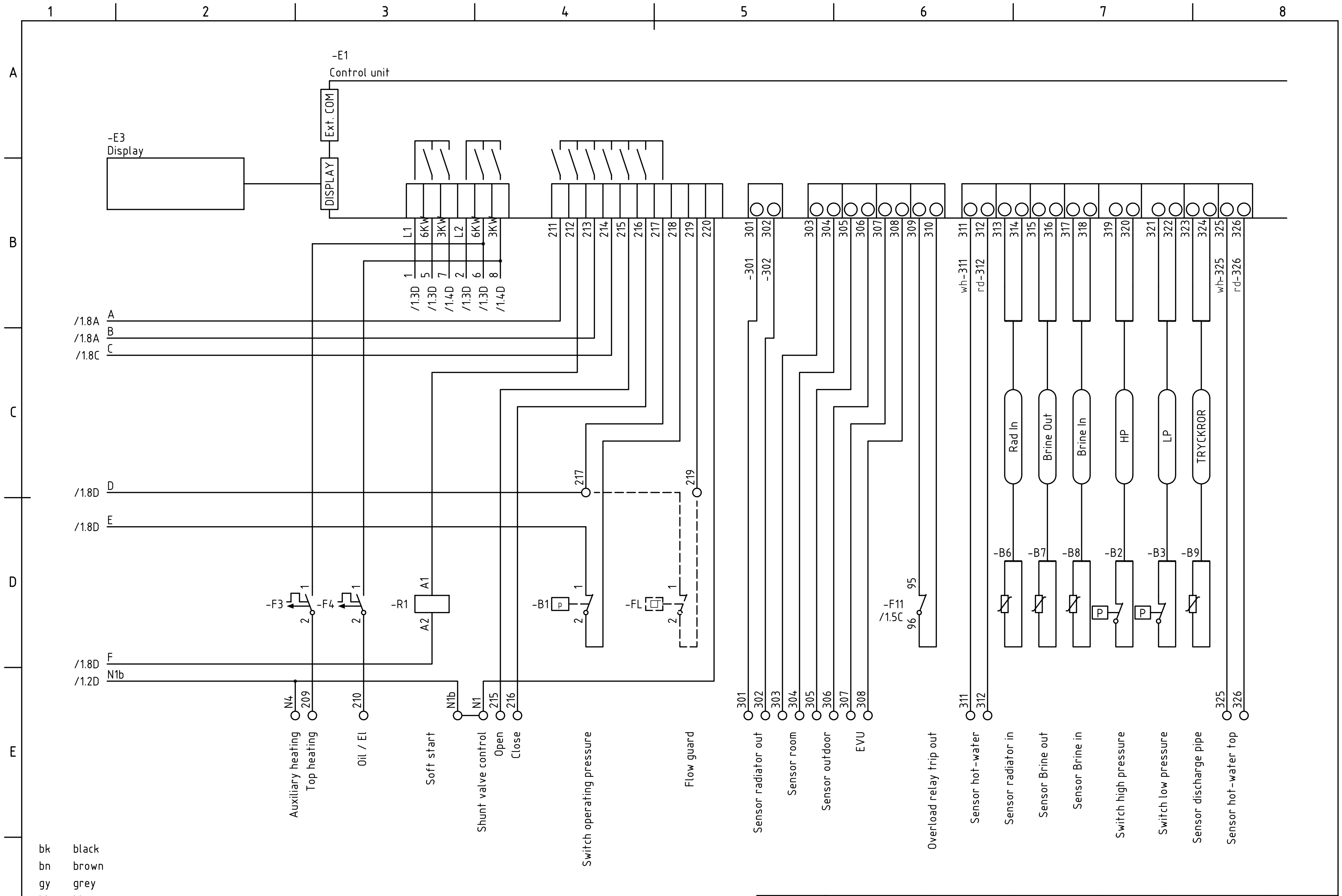
DATE
060821

Duo Optimum 10
400V 3N
Circuit diagram

CONTINUED
2

DRAWING NR.
57155

SHEET
1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1 /1.3D
5 /1.3D
7 /1.4D
2 /1.3D
6 /1.3D
8 /1.4D
211
212
213
214
215
216
217
218
219
220
301
302
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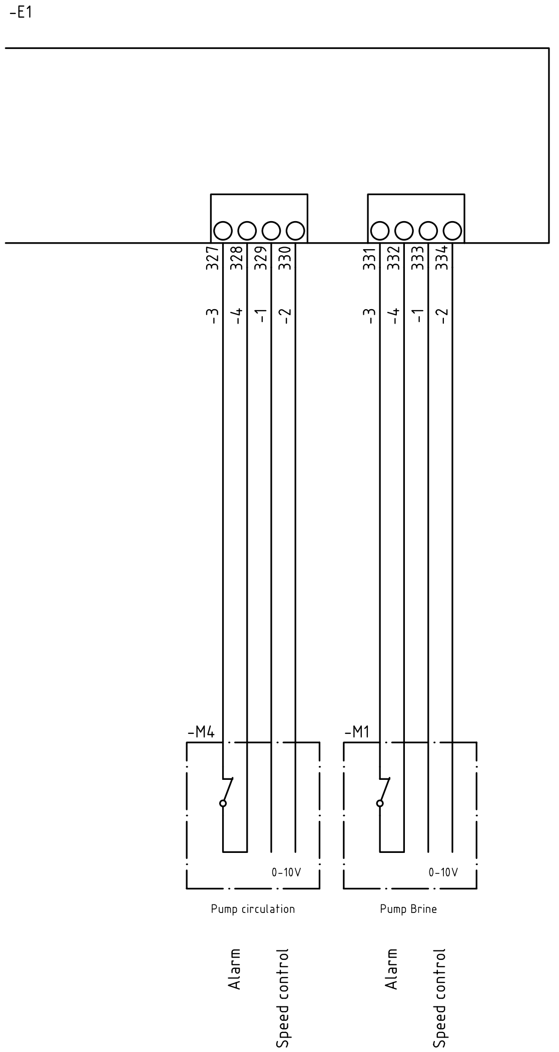
318
319
320
321
322
323
324
325
326
-301
-302
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1b
N1
215
216
217
219
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

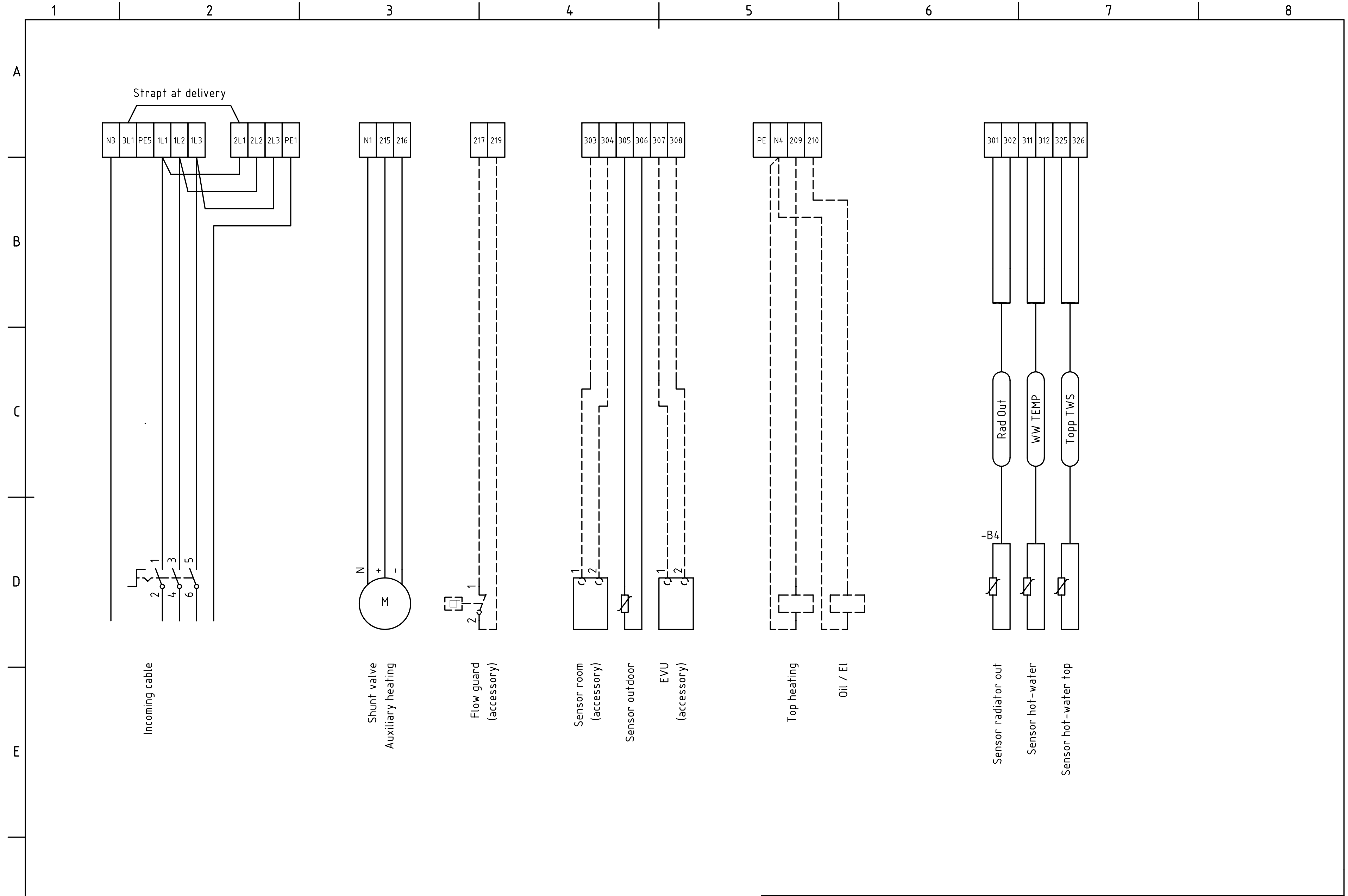
Thermia	=VP +A	APPROVED AL	CREATED BY STP	REVISION 1	DATE 060821
		Duo Optimum 10 400V 3N Circuit diagram		DRAWING NR. 57155	CONTINUED 3 SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL	CREATED BY STP		REVISION 1	DATE 060821
Duo Optimum 10 400V 3N Circuit diagram			DRAWING NR. 57155	CONTINUED 4 SHEET 3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A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

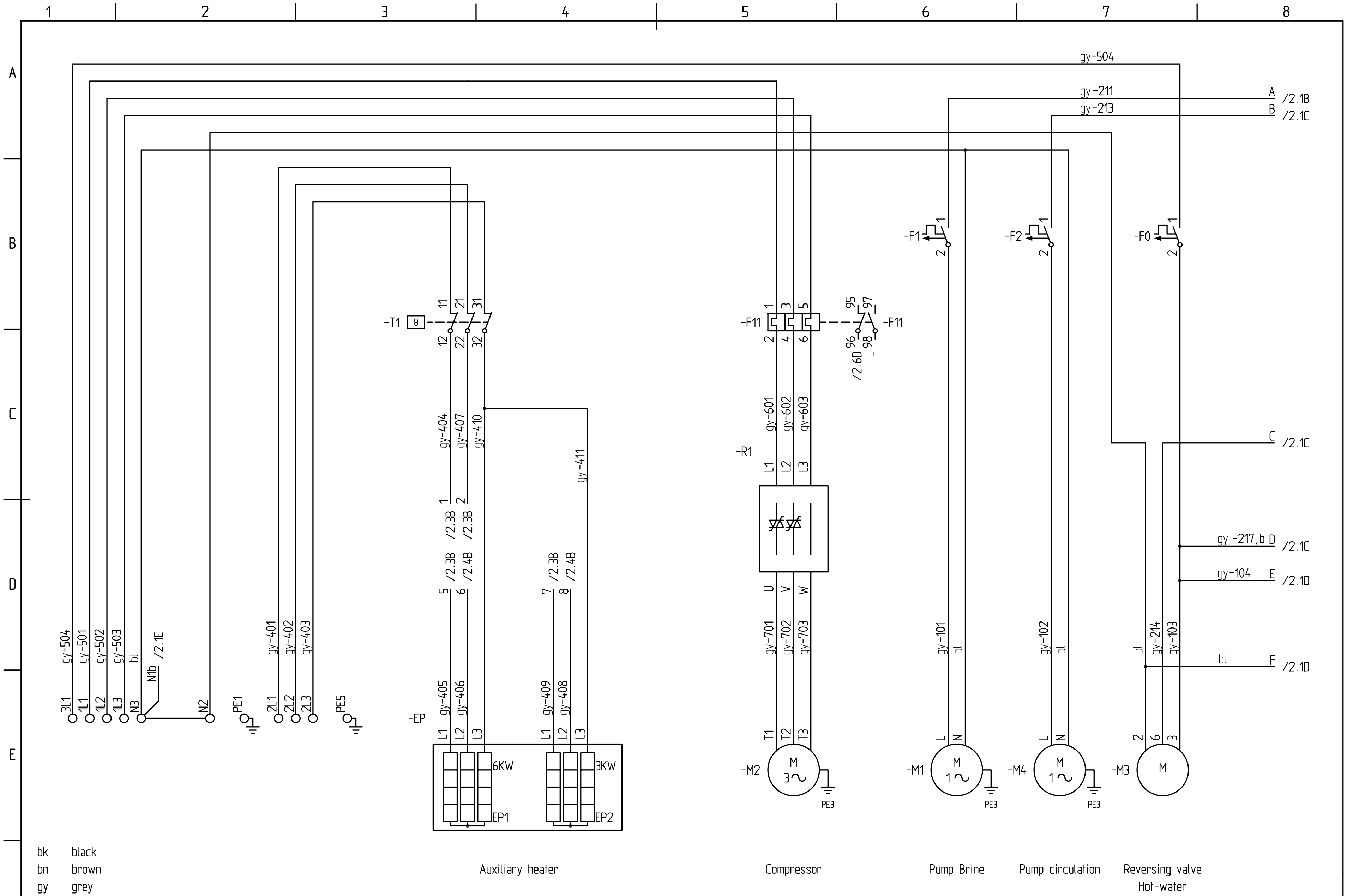
Thermia	=VP +A	APPROVED AL	CREATED BY STP		REVISION 1	DATE 060821
		Duo Optimum 10				CONTINUED 5
		400V 3N			DRAWING NR. 57155	SHEET 4
		External connections				

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	9-13A	9588-38379012	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53313001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED AL
CREATED BY STP
REVISION 1
DATE 060
Duo Optimum 10
400V 3N
Part list
CONTINUED -
DRAWING NR. 57155
SHEET 5


1
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7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
-F11
gy-601
gy-602
gy-603
-R1
L1 L2 L3
U V W
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M
3~
PE3
gy-404
gy-407
gy-410
gy-411
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3
N2
PE1
gy-401
gy-402
gy-403
2L1 2L2 2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-101
bl
L N
-M1
M
1~
PE3
gy-102
bl
L N
-M4
M
1~
PE3
bl
gy-214
gy-103
2 6 3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED
AL

CREATED BY
STP

REVISION
1

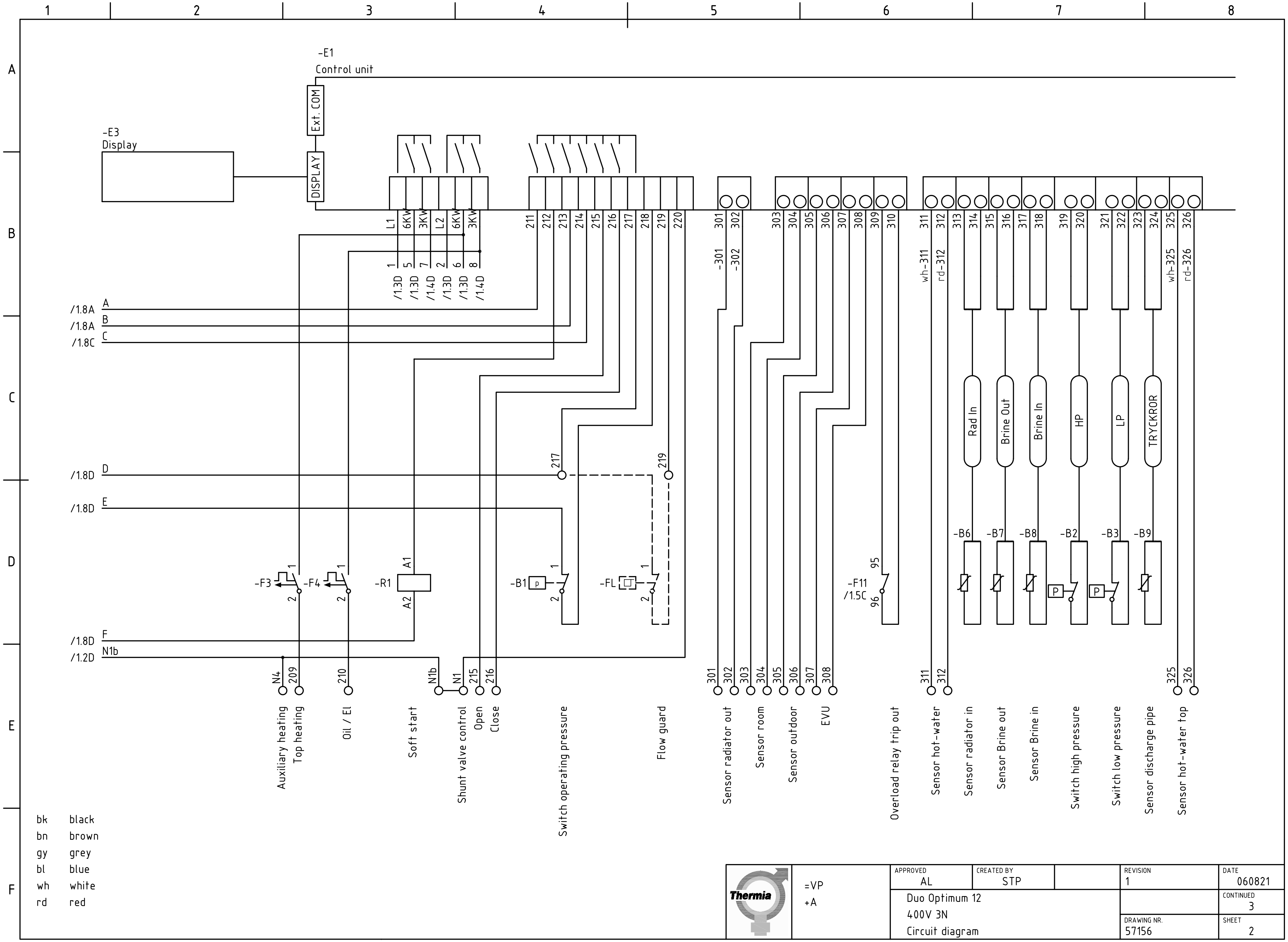
DATE
060821

Duo Optimum 12
400V 3N
Circuit diagram

CONTINUED
2

DRAWING NR.
57156

SHEET
1


1
2
3
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6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
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308
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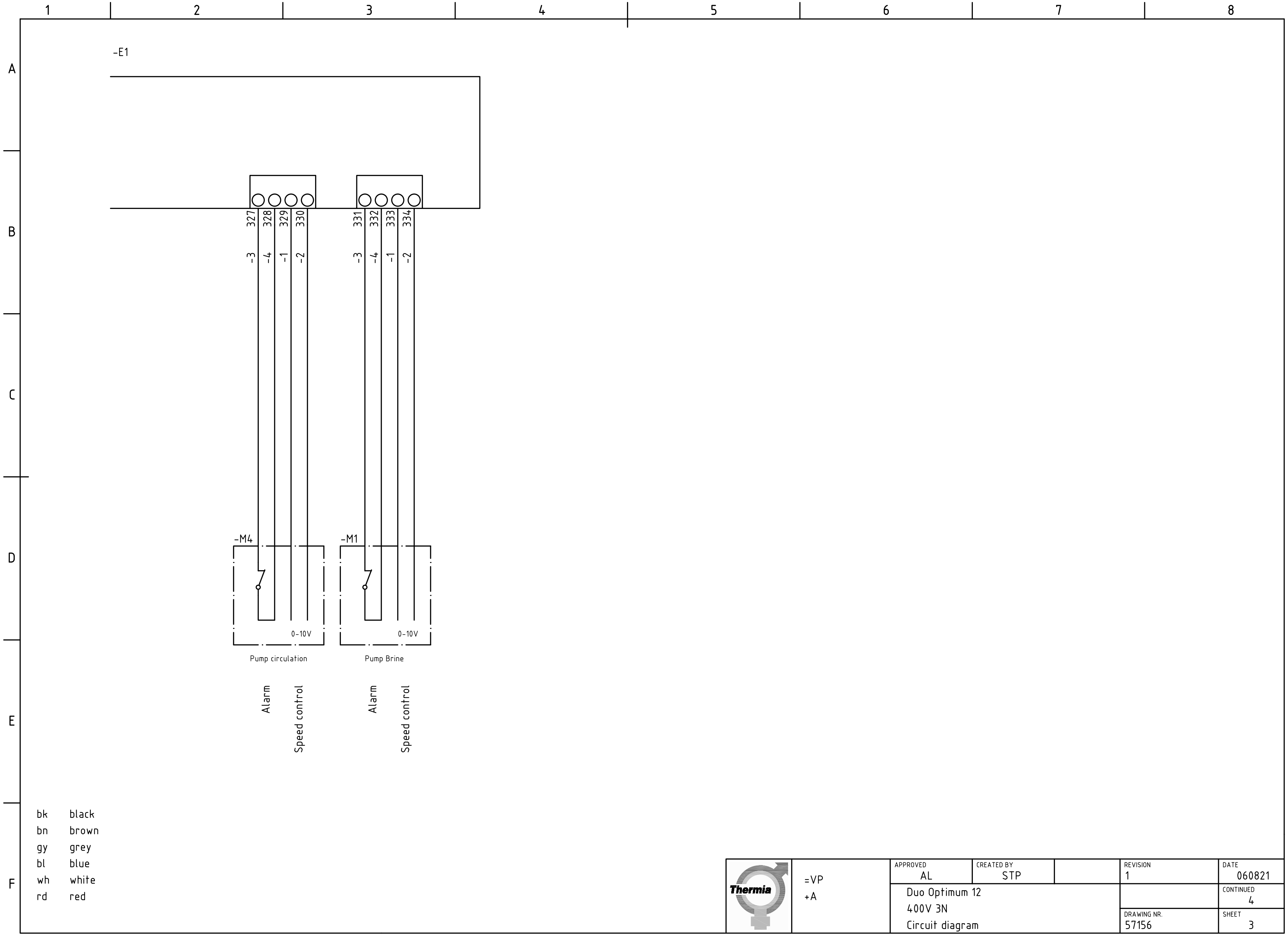
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
217
219
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
1
2
N4
209
210
N1b
N1
215
216
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	STP		1	060821
Duo Optimum 12				CONTINUED 3
400V 3N			DRAWING NR. 57156	SHEET 2
Circuit diagram				


1
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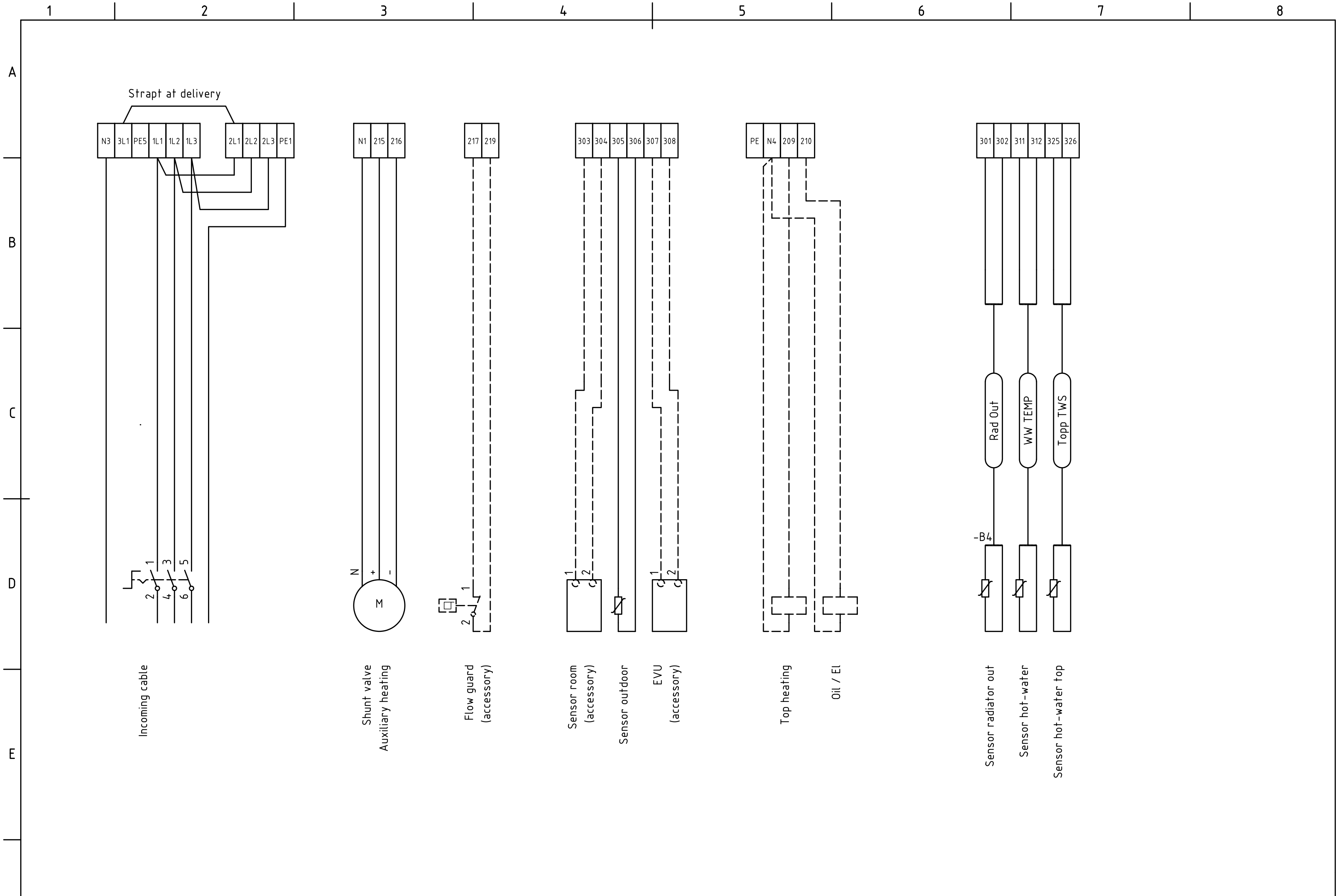
7
8
A
B
C
D
E
F
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia	=VP +A	APPROVED AL	CREATED BY STP		REVISION 1	DATE 060821
		Duo Optimum 12 400V 3N Circuit diagram			DRAWING NR. 57156	CONTINUED 4 SHEET 3


1
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4
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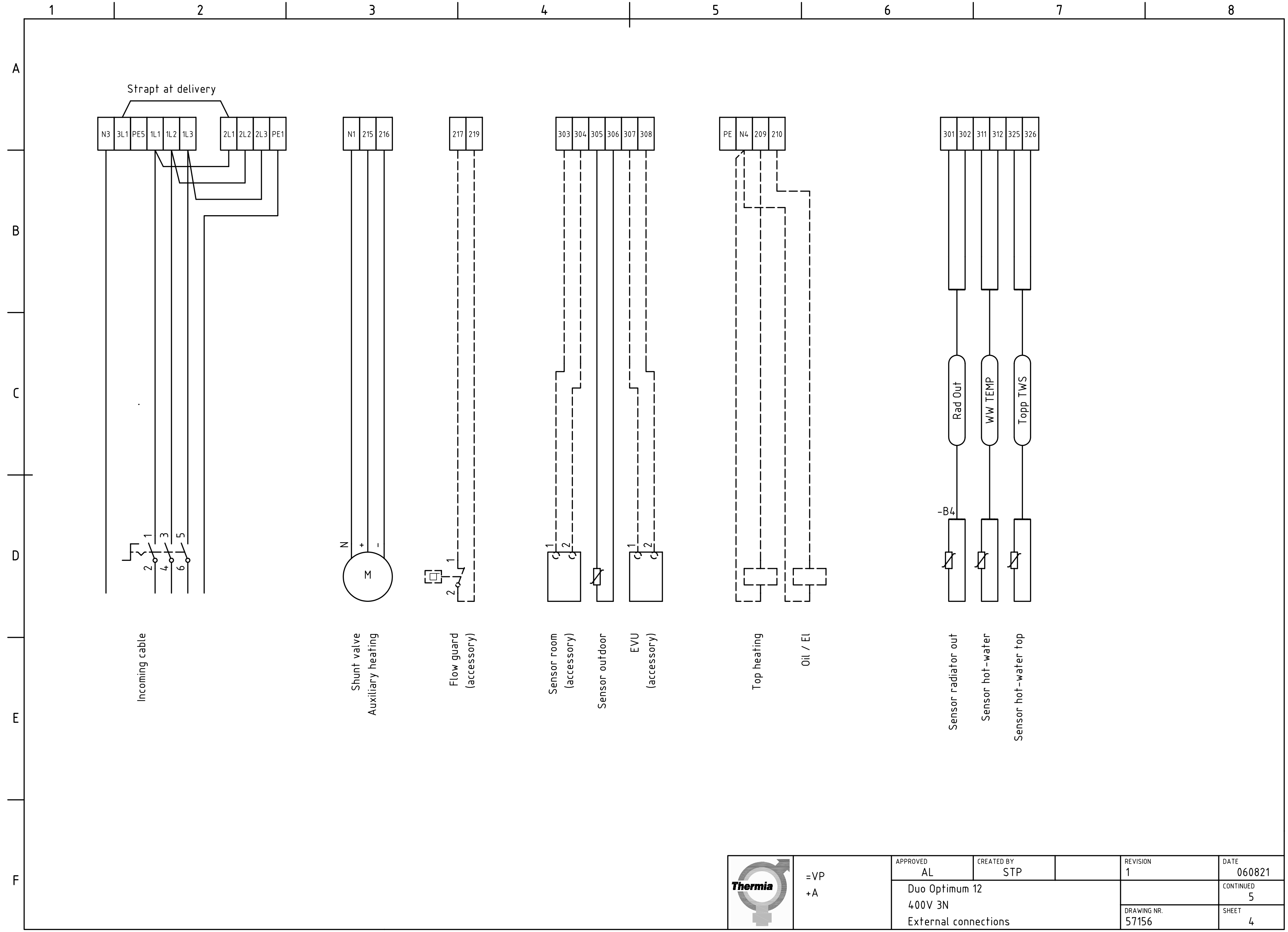
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

=VP +A	APPROVED AL	CREATED BY STP	REVISION 1	DATE 060821
	Duo Optimum 12 400V 3N External connections		DRAWING NR. 57156	CONTINUED 5 SHEET 4


1
2
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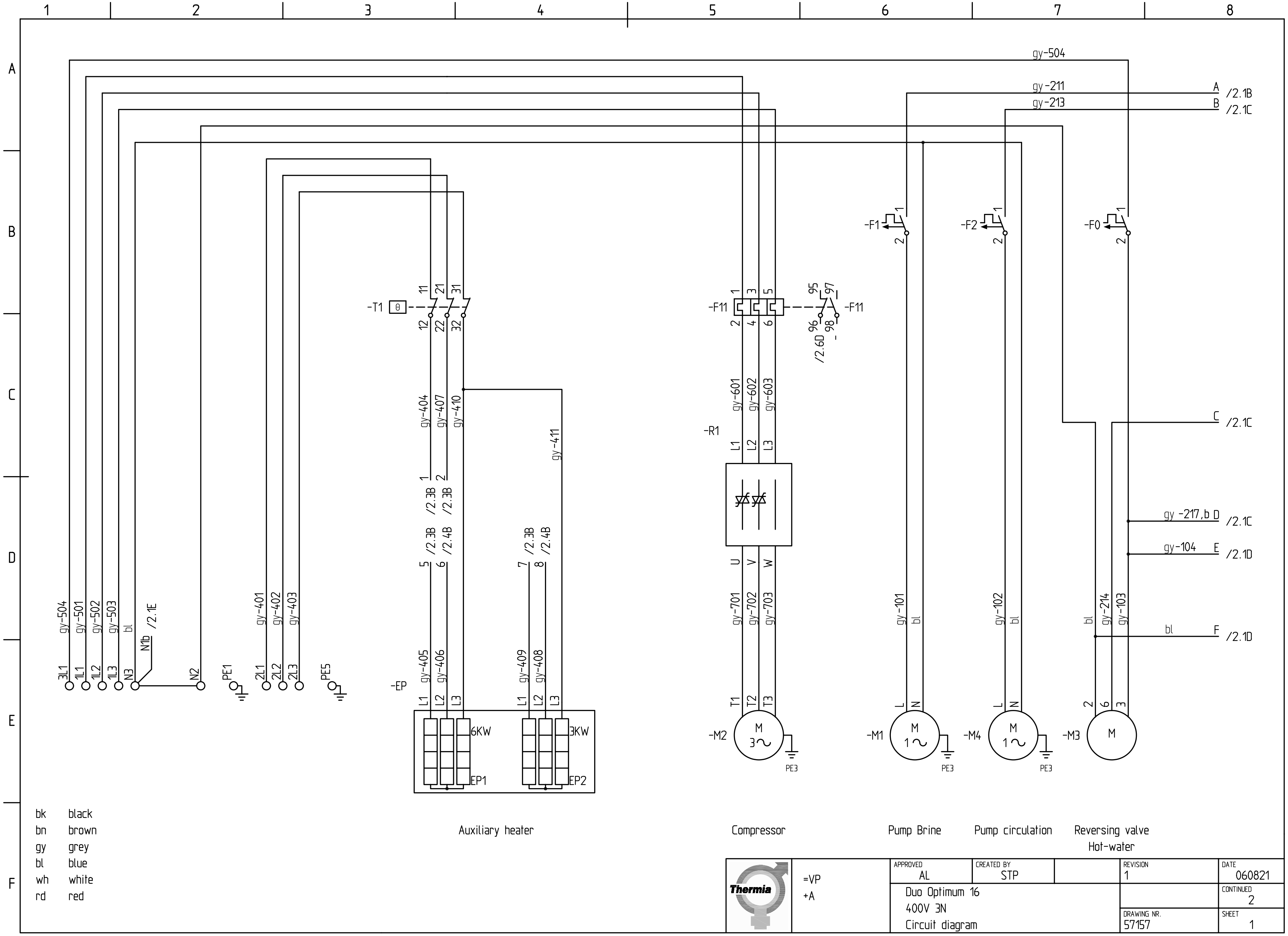
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
N
+
-
M
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia
=VP
+A
APPROVED
AL
CREATED BY
STP
REVISION
1
DATE
060821
Duo Optimum 12
400V 3N
External connections
CONTINUED
5
DRAWING NR.
57156
SHEET
4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	9-13A	9588-38379012	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53314001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	STP		1	060821
Duo Optimum 12 400V 3N Part list				CONTINUED -
			DRAWING NR. 57156	SHEET 5


1
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3
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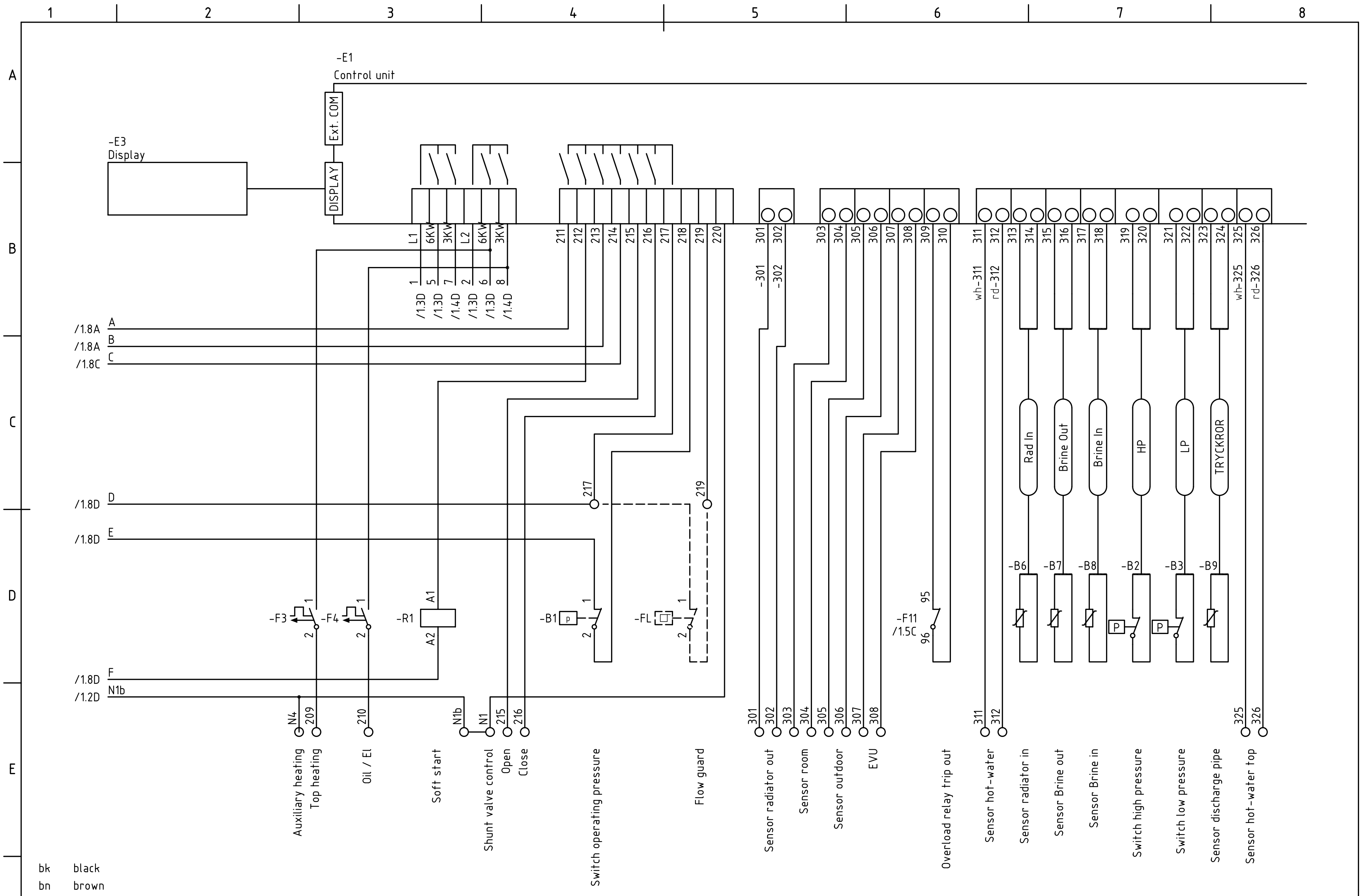
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F1
-F2
-F0
-T1
-F11
95
97
96
98
/2.6D
11 21 31
12 22 32
1 3 5
2 4 6
gy-404
gy-407
gy-410
gy-411
gy-601
gy-602
gy-603
-R1
L1 L2 L3
U V W
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-504
gy-501
gy-502
gy-503
bl
N1b /2.1E
3L1 1L1 1L2 1L3 N3
N2
PE1
gy-401
gy-402
gy-403
2L1 2L2 2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
6KW
3KW
EP1
EP2
gy-701
gy-702
gy-703
T1 T2 T3
-M2
M 3~
PE3
gy-101
bl
L N
-M1
M 1~
gy-102
-M4
bl
gy-214
gy-103
2 6 3
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve Hot-water
bk black
bn brown
gy grey
bl blue
wh white
rd red
Thermia
=VP
+A
APPROVED AL
CREATED BY STP
REVISION 1
DATE 060821
CONTINUED 2
Duo Optimum 16
400V 3N
Circuit diagram
DRAWING NR. 57157
SHEET 1


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
/1.2D N1b
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-R1
A1
A2
-B1
p
-FL
-F11
/1.5C
95
96
N4
209
210
N1b
N1
215
216
217
219
301
302
303
304
305
306
307
308
311
312
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

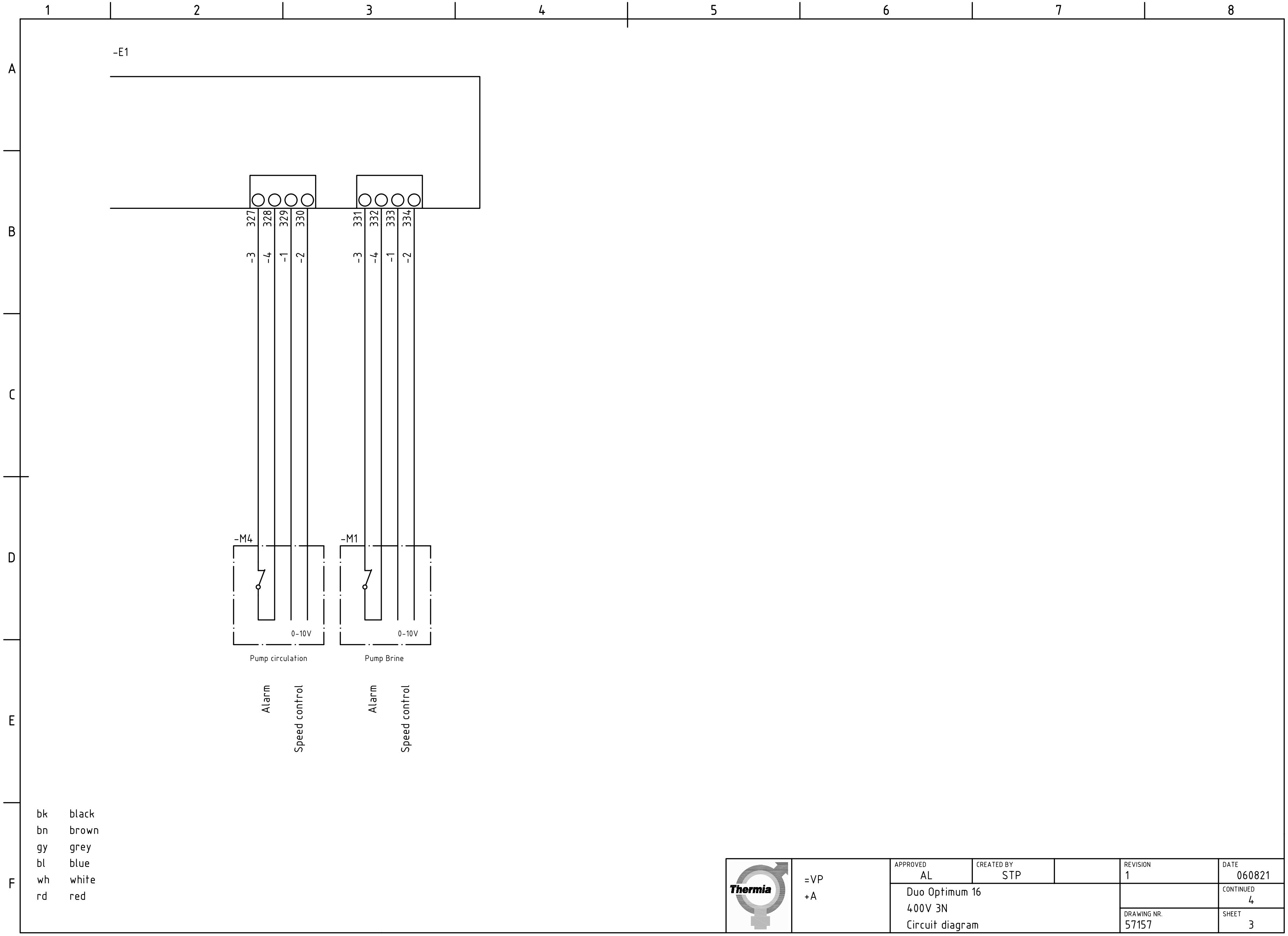
Thermia

=VP
+A

APPROVED AL
CREATED BY STP
REVISION 1
DATE 060821

Duo Optimum 16
400V 3N
Circuit diagram

CONTINUED 3
DRAWING NR. 57157
SHEET 2


1
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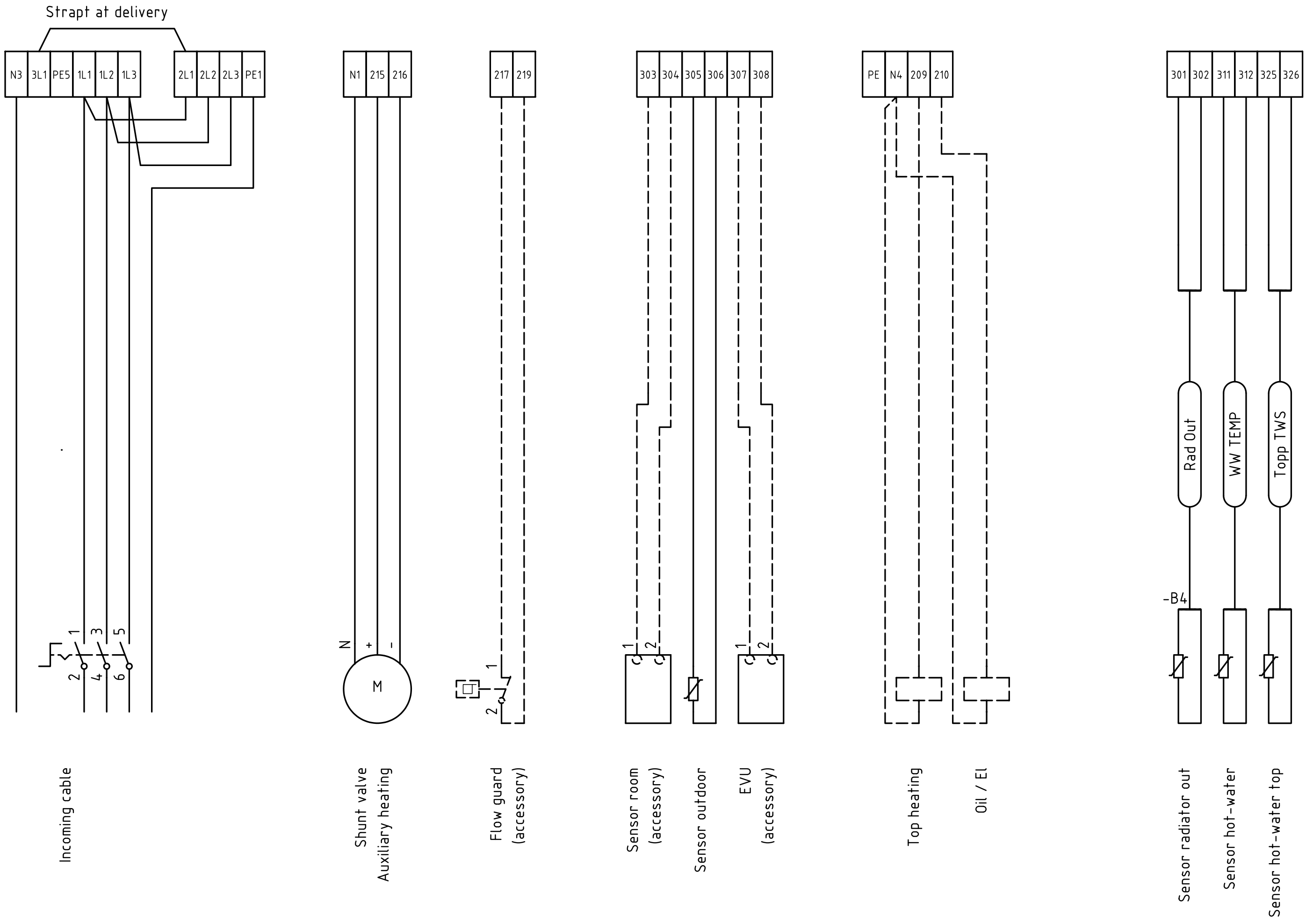
A
B
C
D
E
F
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL
CREATED BY STP
REVISION 1
DATE 060821
Duo Optimum 16
400V 3N
Circuit diagram
CONTINUED 4
DRAWING NR. 57157
SHEET 3


Strapt at delivery
N3
3L1
PE5
1L1
1L2
1L3
2L1
2L2
2L3
PE1
N1
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

=VP
+A

APPROVED AL
CREATED BY STP
REVISION 1
DATE 060821
Duo Optimum 16
400V 3N
External connections
CONTINUED 5
DRAWING NR. 57157
SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B4	1	Sensor radiator out	Betatherm	NTC	9588-54680001	/4.6B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-53938A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	1P C10A	9588-51463001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	1P C3A	9585-53791001	/2.3D
-F11	1	Thermal overload relay	Telemecanique	12-18A	9588-38379013	/1.5C
-F11	1	Auxiliary switch block	Telemecanique		9575-54492001	/1.5C
-M1	1	Pump circulation	Wilo		9675-56385A00	/1.6E
-M2	1	Compressor	Copeland	400V	9686-53316001	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-R1	1	Soft start	Kimsafe AB	400V 3N	9588-56851001	/1.5C
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	400V 3N	9583-56893001	
-X2	1	Terminal block	Phoenix	Duo	9583-55837001	

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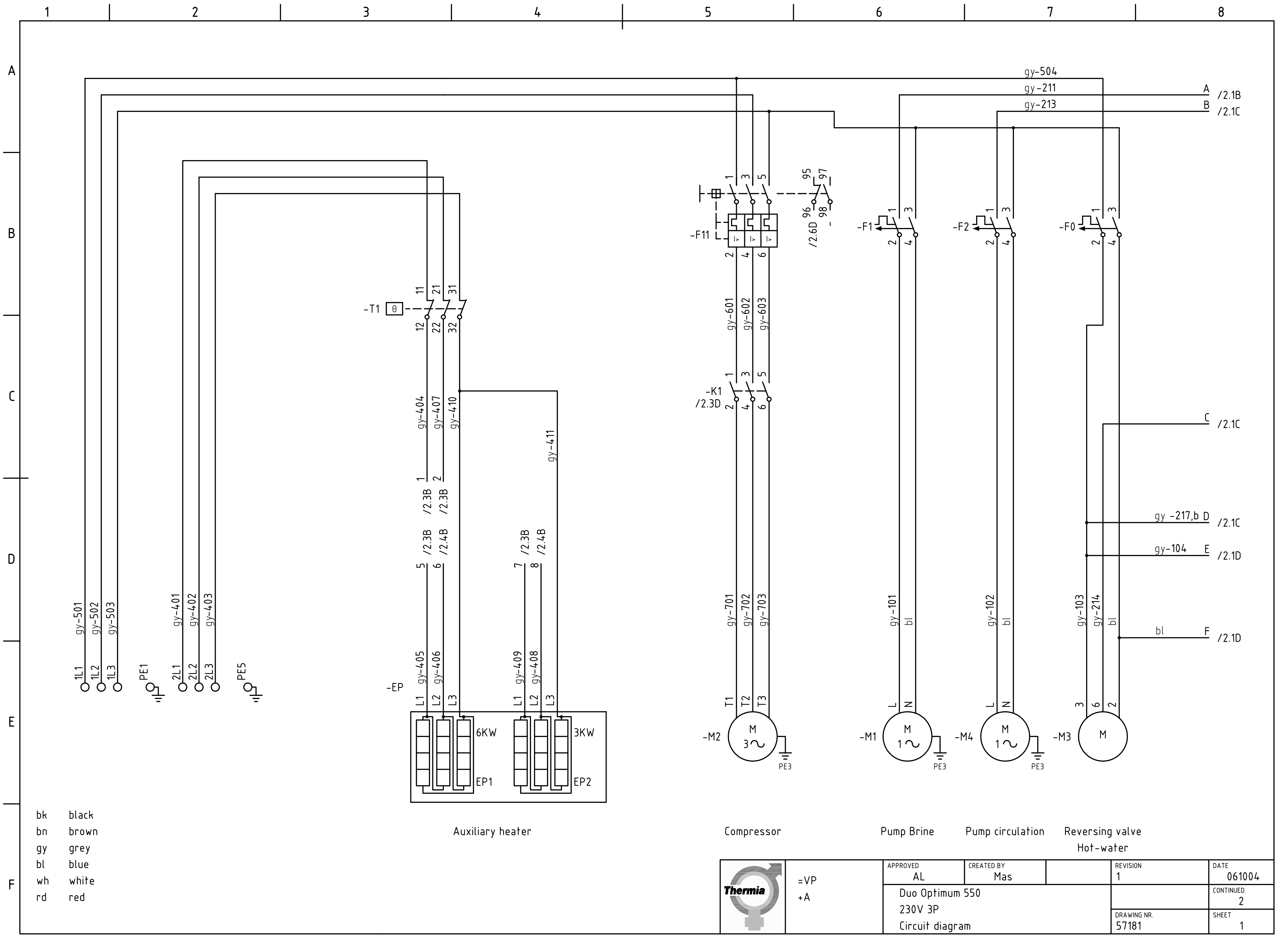
APPROVED AL
CREATED BY STP
REVISION 1
DATE 060908

Duo Optimum 16
400V 3N
Part list

CONTINUED -
DRAWING NR. 57157
SHEET 5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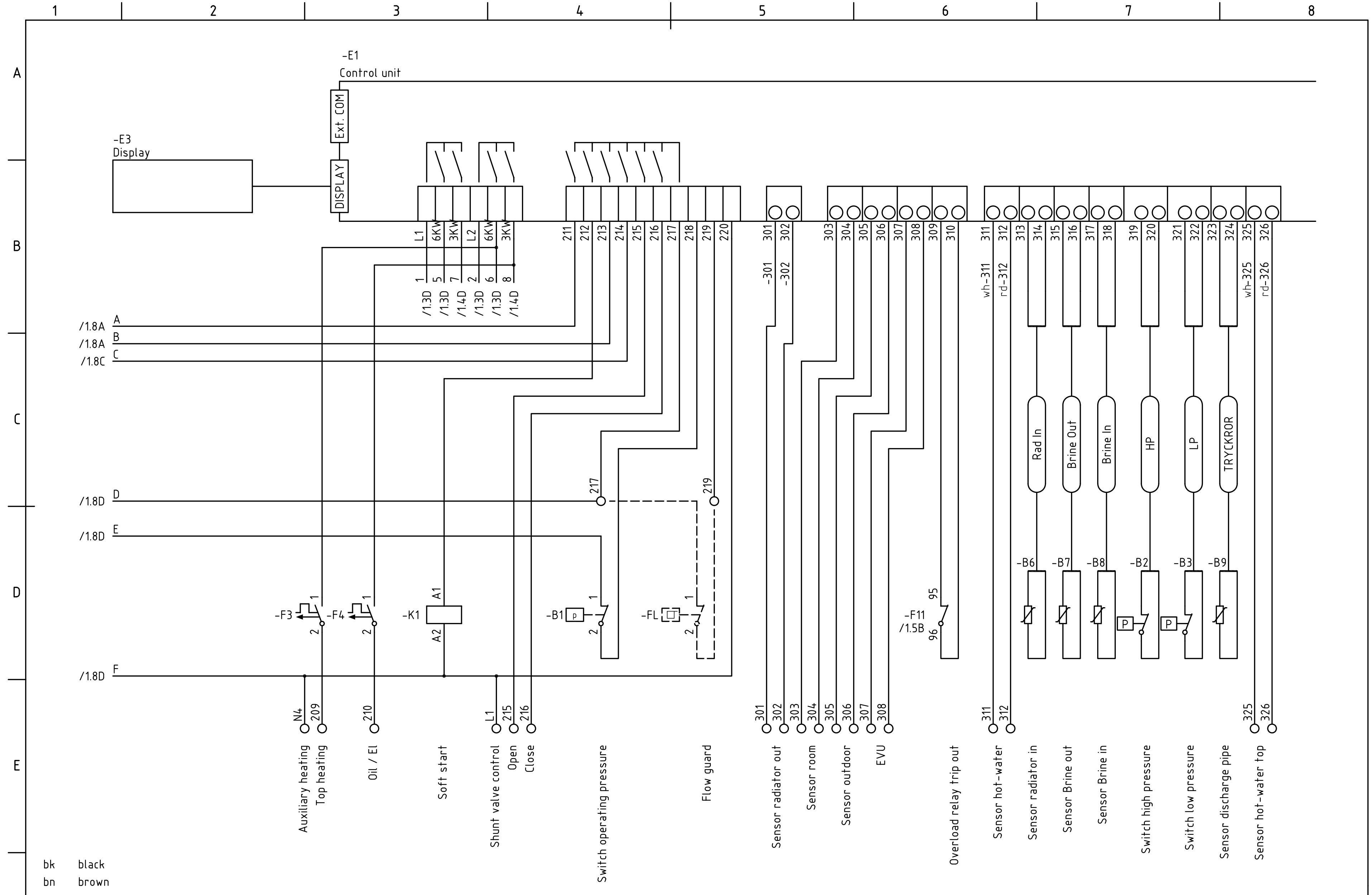
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
11
21
31
12
22
32
gy-601
gy-602
gy-603
-K1
/2.3D
gy-404
gy-407
gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-502
gy-503
gy-401
gy-402
gy-403
gy-701
gy-702
gy-703
gy-101
bl
gy-102
bl
gy-103
gy-214
bl
1L1
1L2
1L3
PE1
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1
L2
L3
6KW
EP1
3KW
EP2
T1
T2
T3
-M2
M
3~
PE3
L
N
-M1
M
1~
PE3
-M4
M
1~
PE3
3
6
2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Mas		1	061004
Duo Optimum 550				CONTINUED 2
230V 3P			DRAWING NR. 57181	SHEET 1
Circuit diagram				


1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1 /1.3D
5 /1.3D
7 /1.4D
2 /1.3D
6 /1.3D
8 /1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4
209
210
L1
215
216
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED
AL

CREATED BY
AL

REVISION
1

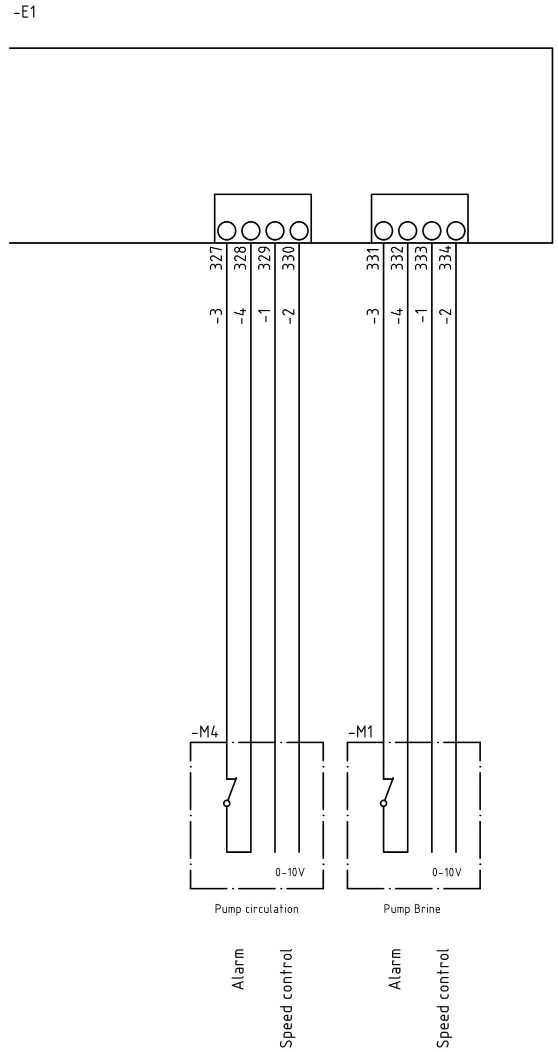
DATE
061004

Duo Optimum 550
230V 3P
Circuit diagram

CONTINUED
3

DRAWING NR.
57181

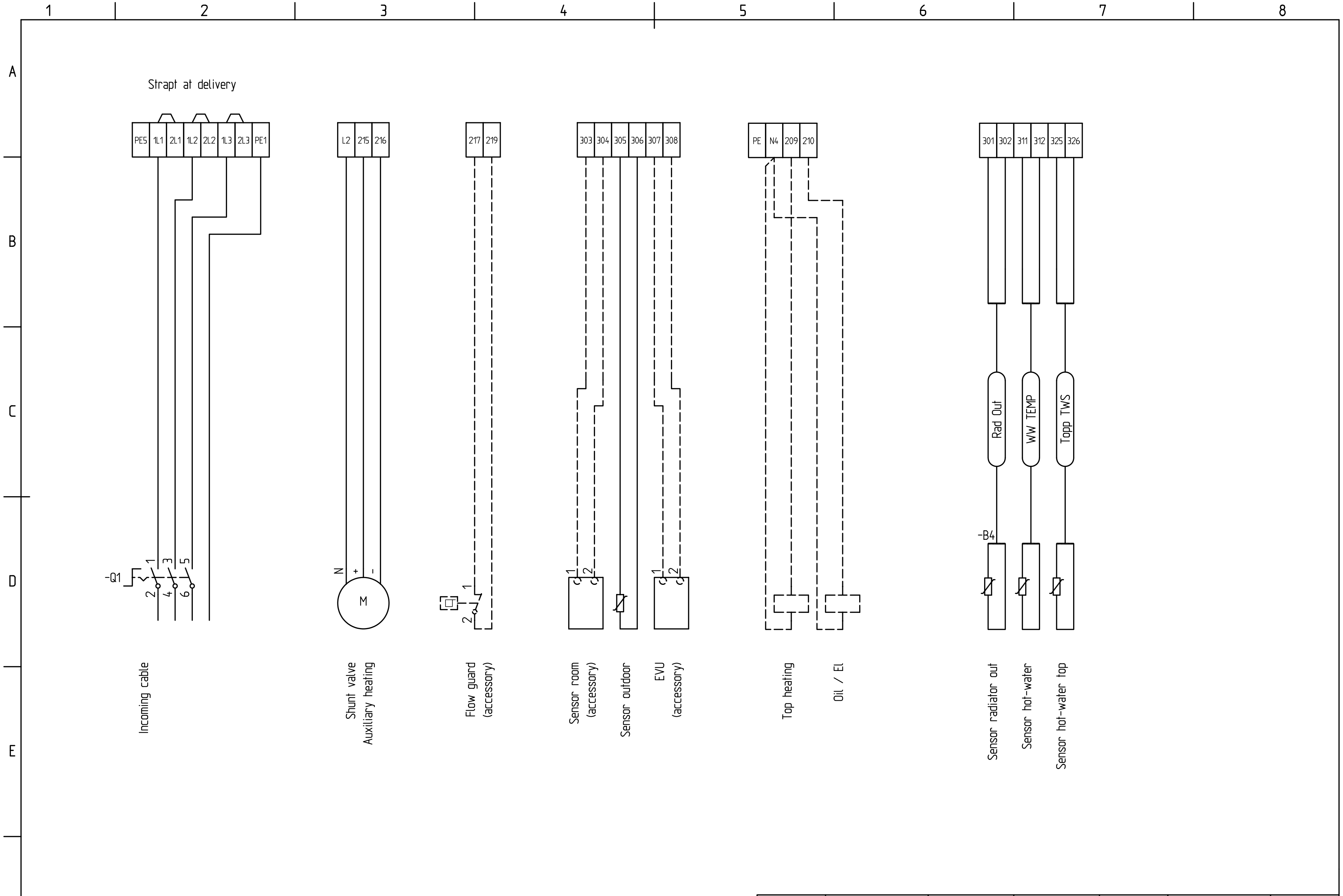
SHEET
2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

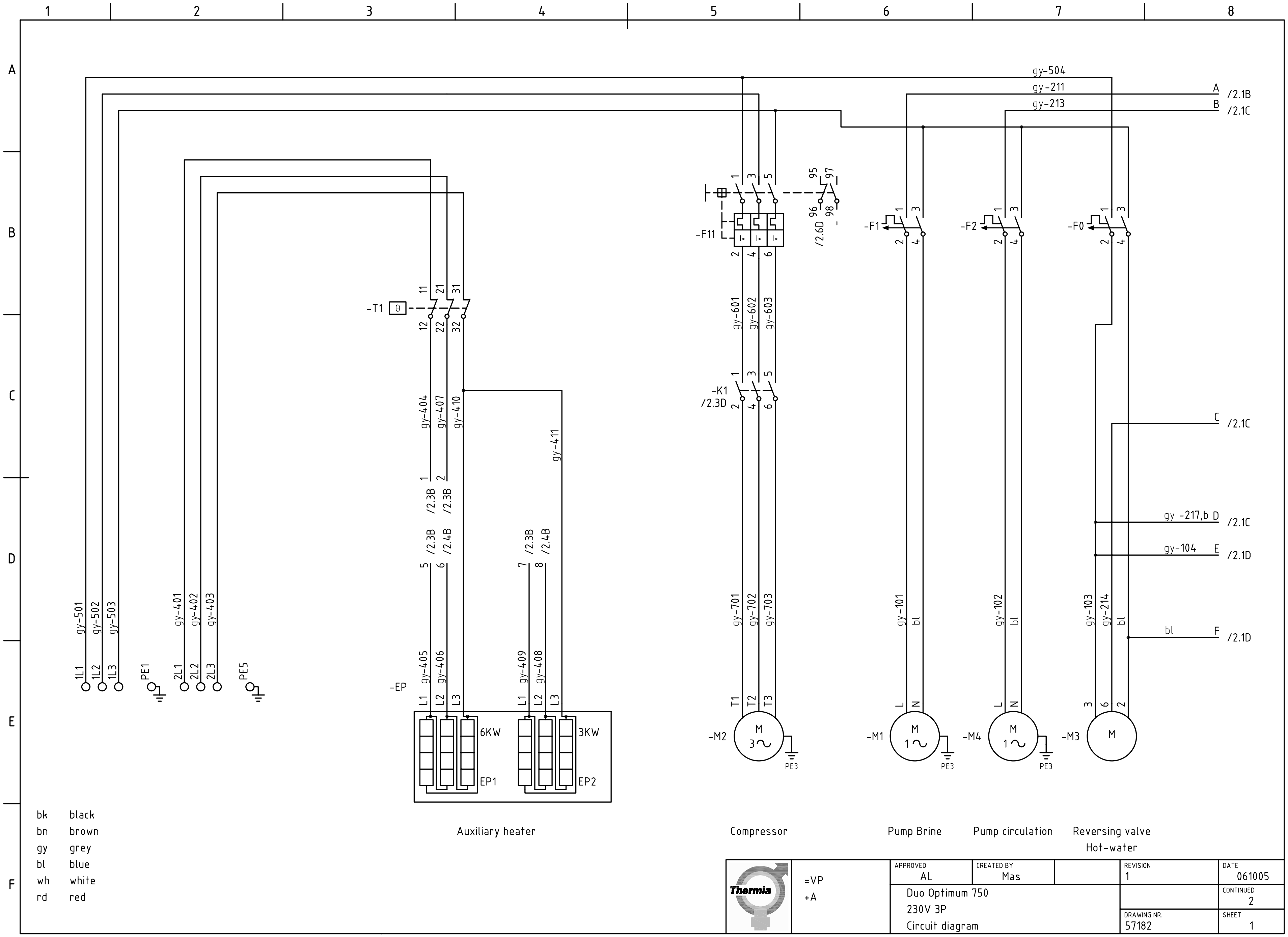
APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061004
CONTINUED 4
Duo Optimum 550
230V 3P
Circuit diagram
DRAWING NR. 57181
SHEET 3


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
1
2
1
2
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Mas	REVISION 1	DATE 061004
		Duo Optimum 550 230V 3P External connections		DRAWING NR. 57181	CONTINUED 5 SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	9-14A	9588-53794001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	12A 5,5KW 230V	9588-38378002	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51332002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061004
	Duo Optimum 550 230V 3P Part list			DRAWING NR. 57181	CONTINUED - SHEET 5


1
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3
4
5
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
/2.6D
95
97
96
98
-F1
-F2
-F0
-T1
11
21
31
12
22
32
gy-601
gy-602
gy-603
-K1
/2.3D
gy-404
gy-407
gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-502
gy-503
gy-401
gy-402
gy-403
1L1
1L2
1L3
PE1
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1
L2
L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1
T2
T3
-M2
M
3~
PE3
gy-101
bl
L
N
-M1
M
1~
PE3
gy-102
bl
-M4
M
1~
PE3
gy-103
gy-214
bl
3
6
2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

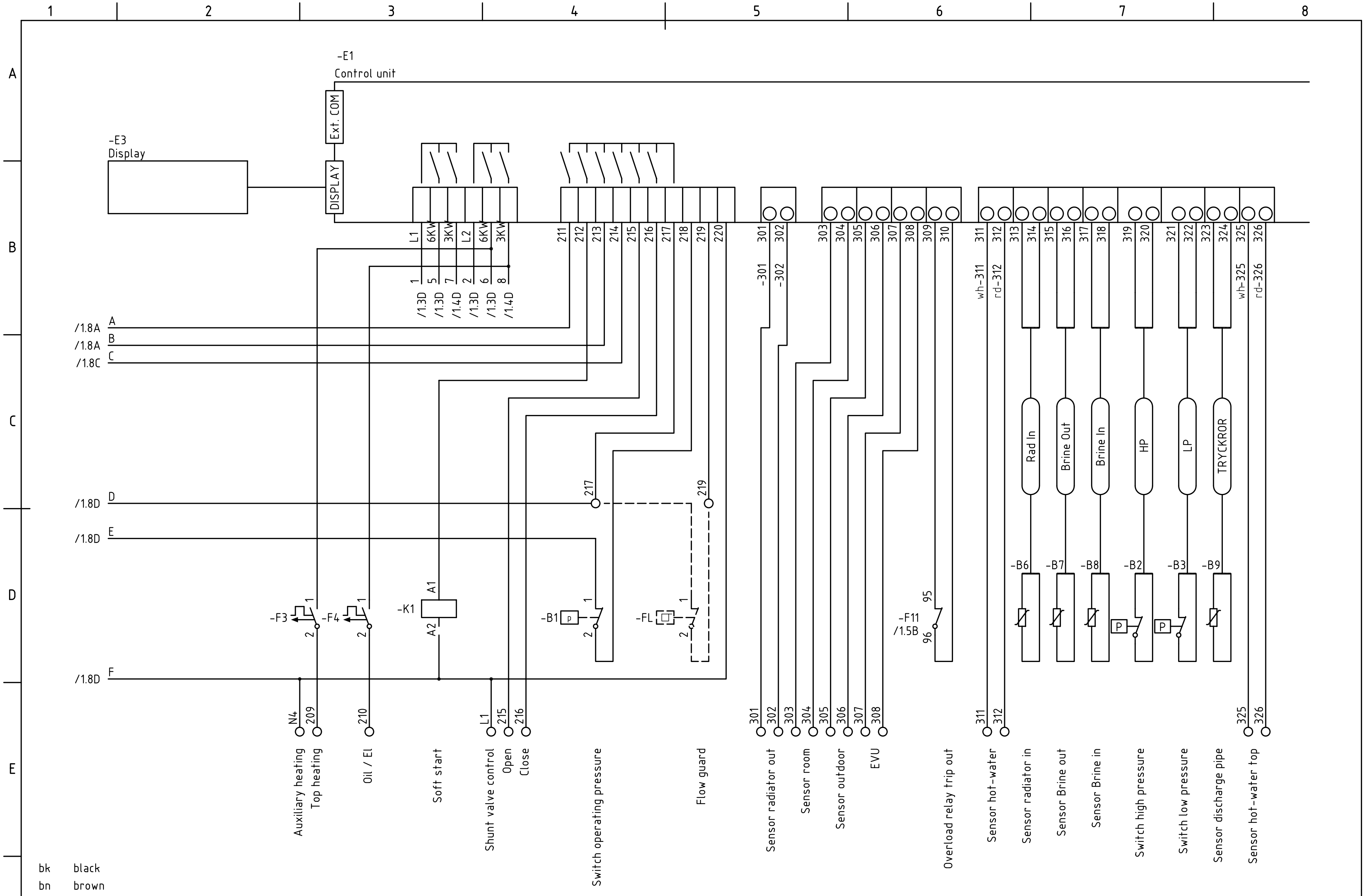
Thermia

=VP
+A

APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061005
CONTINUED 2

Duo Optimum 750
230V 3P
Circuit diagram

DRAWING NR. 57182
SHEET 1


1
2
3
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5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
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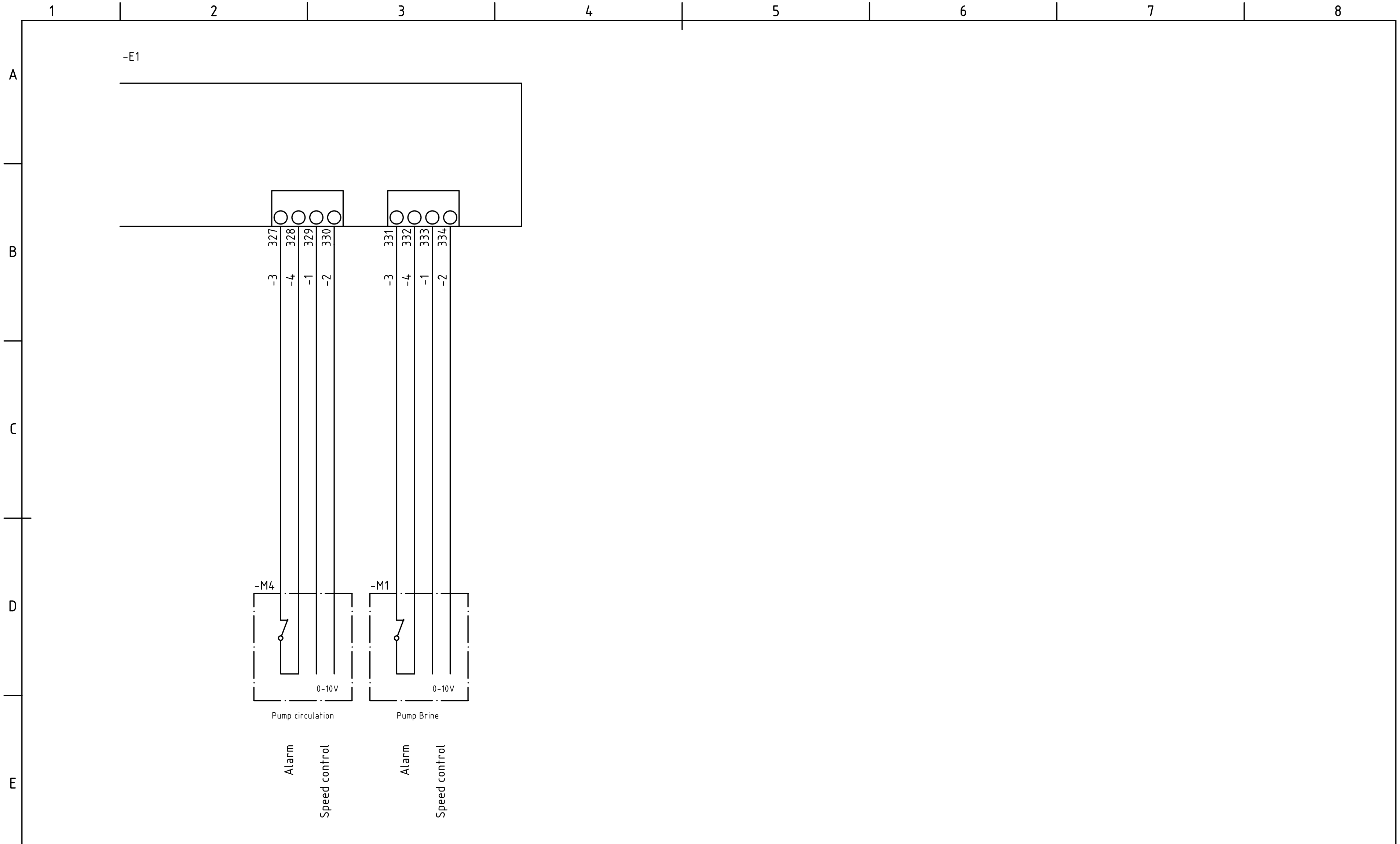
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
N4
209
210
L1
215
216
325
326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061005
Duo Optimum 750 230V 3P Circuit diagram			DRAWING NR. 57182	CONTINUED 3 SHEET 2


1
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6
7
8
A
B
C
D
E
-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

F

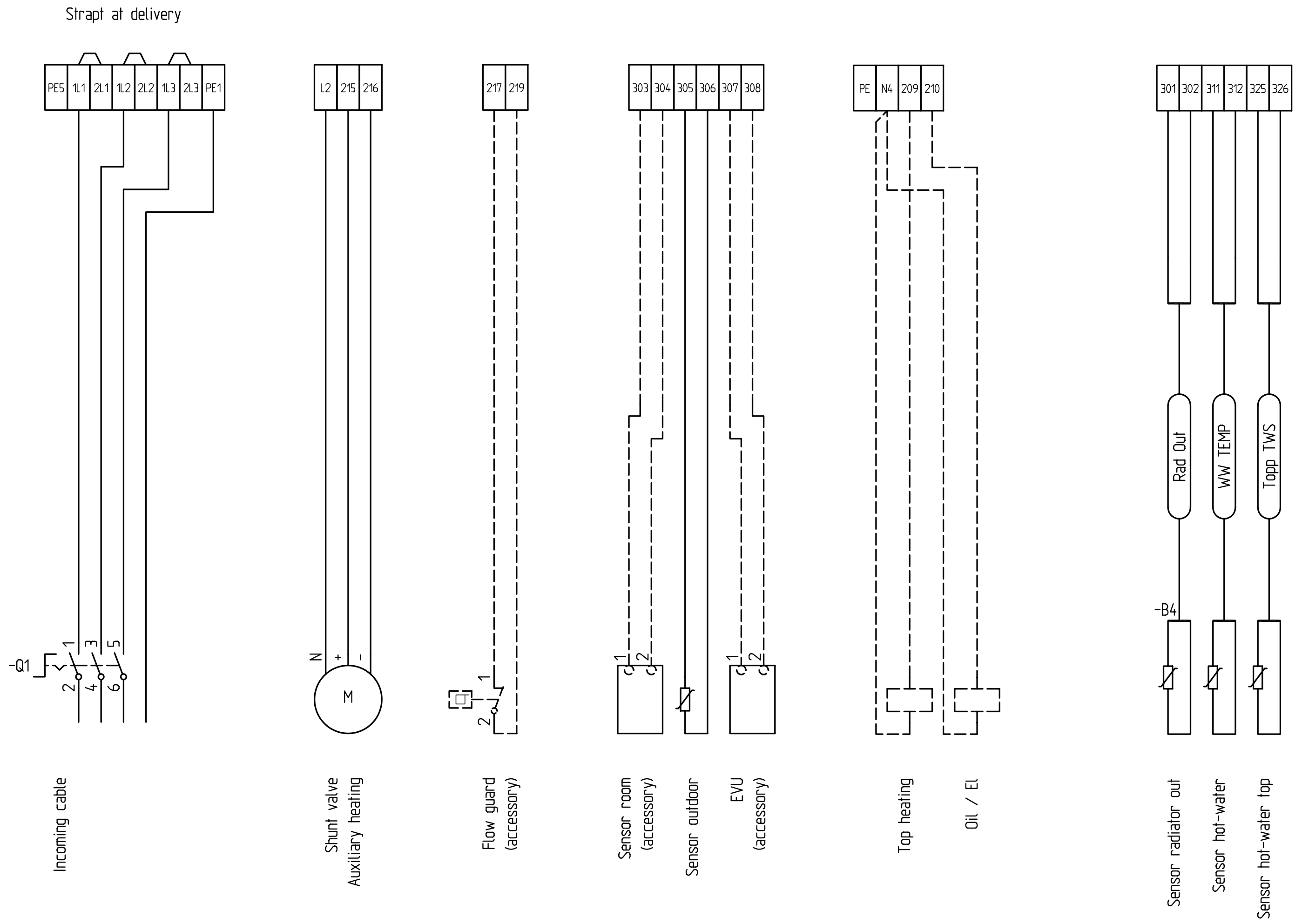
bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061005

Duo Optimum 750
230V 3P
Circuit diagram

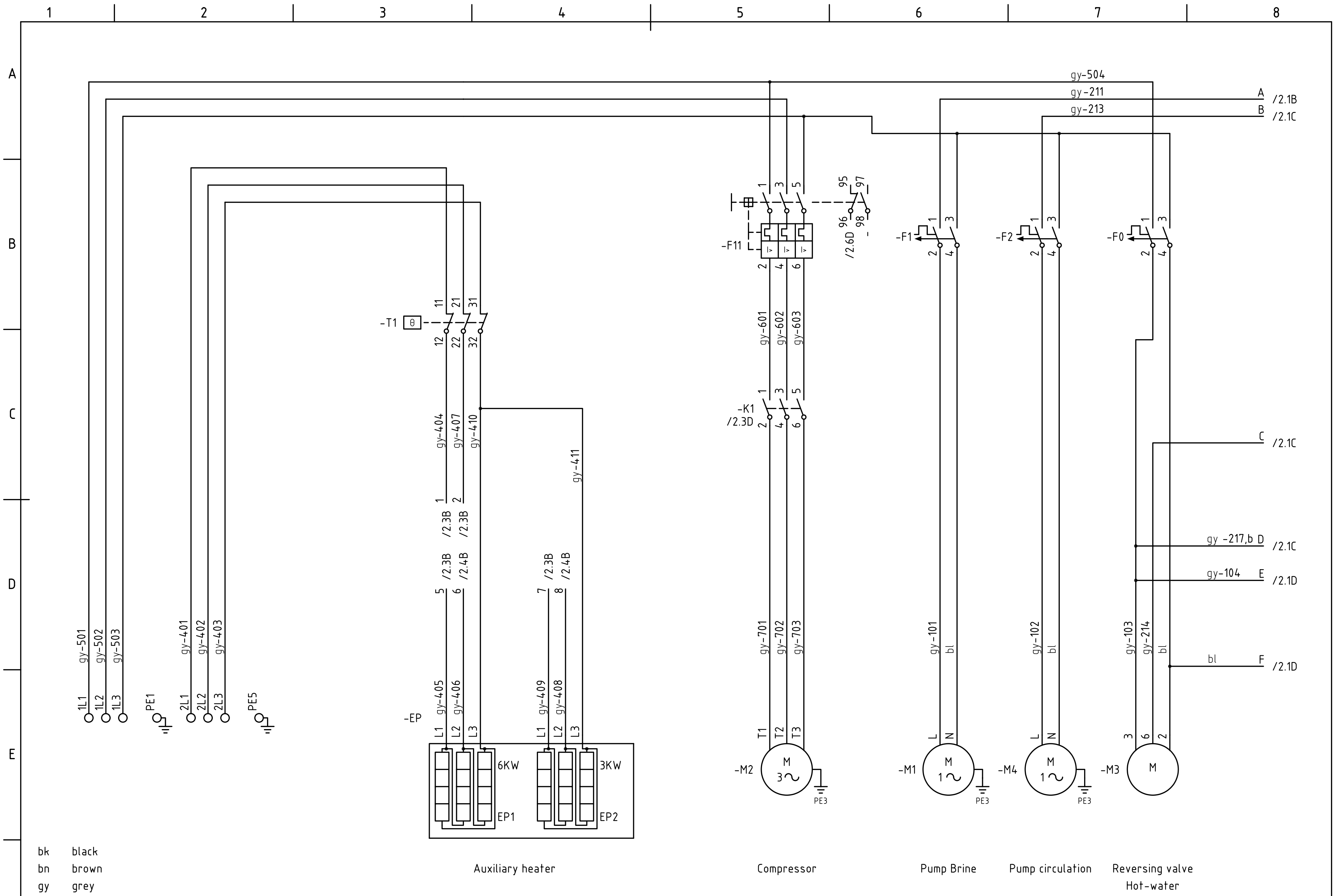
CONTINUED 4
DRAWING NR. 57182
SHEET 3


Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
-Q1
1
3
5
2
4
6
Incoming cable
L2
215
216
N
+
-
M
Shunt valve
Auxiliary heating
217
219
1
2
Flow guard
(accessory)
303
304
305
306
307
308
1
2
Sensor room
(accessory)
Sensor outdoor
1
2
EVU
(accessory)
PE
N4
209
210
Top heating
Oil / El
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
Sensor radiator out
Sensor hot-water
Sensor hot-water top

Thermia	=VP +A	APPROVED AL	CREATED BY Mas	REVISION 1	DATE 061005
		Duo Optimum 750 230V 3P External connections		DRAWING NR. 57182	CONTINUED 5 SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	9-14A	9588-53794001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51333002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061005
	Duo Optimum 750 230V 3P Part list				CONTINUED -
				DRAWING NR. 57182	SHEET 5


1
2
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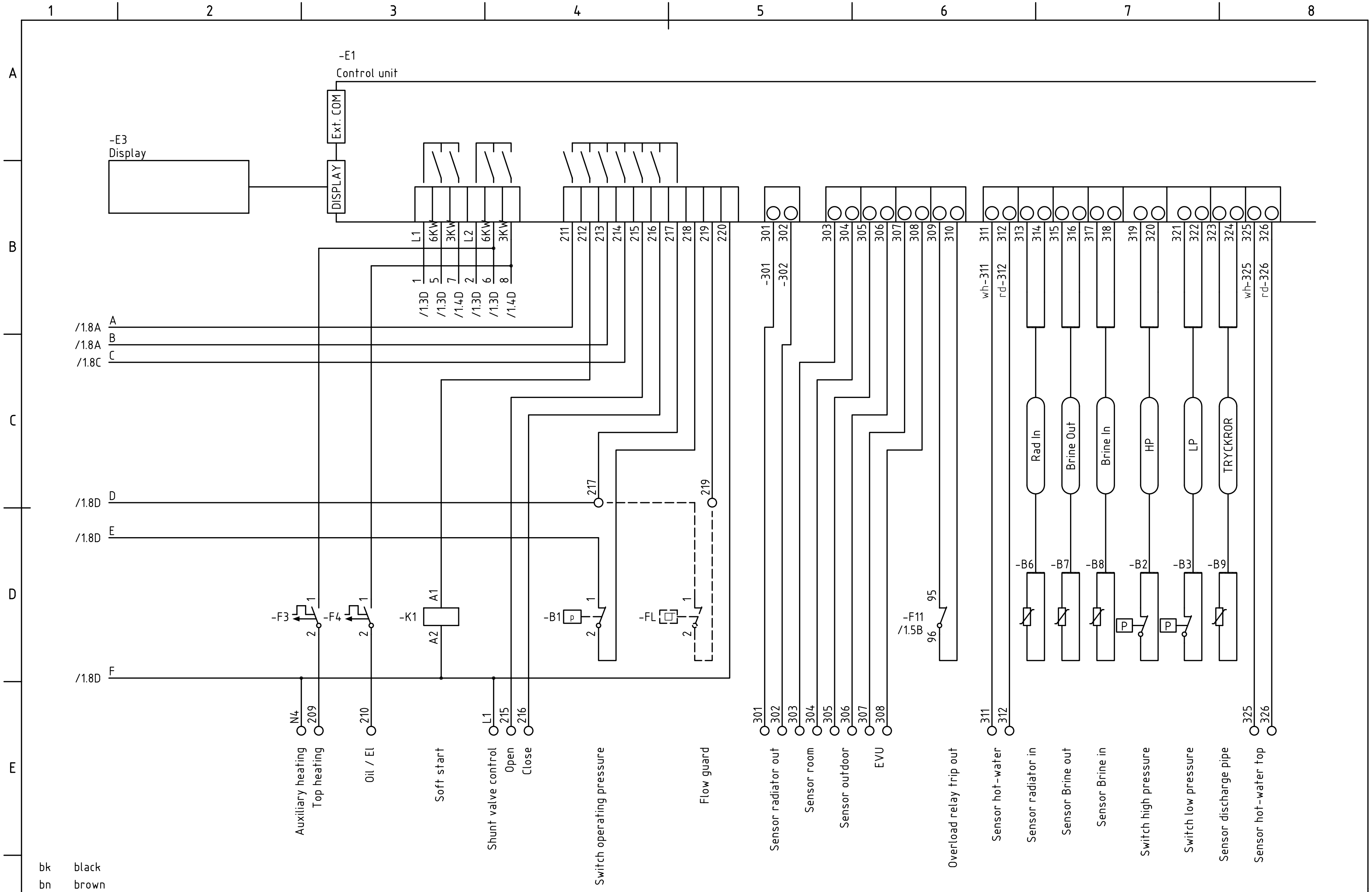
6
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
gy-601
gy-602
gy-603
-K1
/2.3D
gy-404
gy-407
gy-410
gy-411
C /2.1C
/2.3B
/2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501
gy-502
gy-503
gy-401
gy-402
gy-403
1L1
1L2
1L3
PE1
2L1
2L2
2L3
PE5
-EP
gy-405
gy-406
gy-409
gy-408
L1
L2
L3
6KW
EP1
3KW
EP2
gy-701
gy-702
gy-703
T1
T2
T3
-M2
M 3~
PE3
gy-101
bl
L
N
-M1
M 1~
gy-102
-M4
gy-103
gy-214
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY Mas		REVISION 1	DATE 061005
	Duo Optimum 1050 230V 3P Circuit diagram			DRAWING NR. 57183	CONTINUED 2 SHEET 1

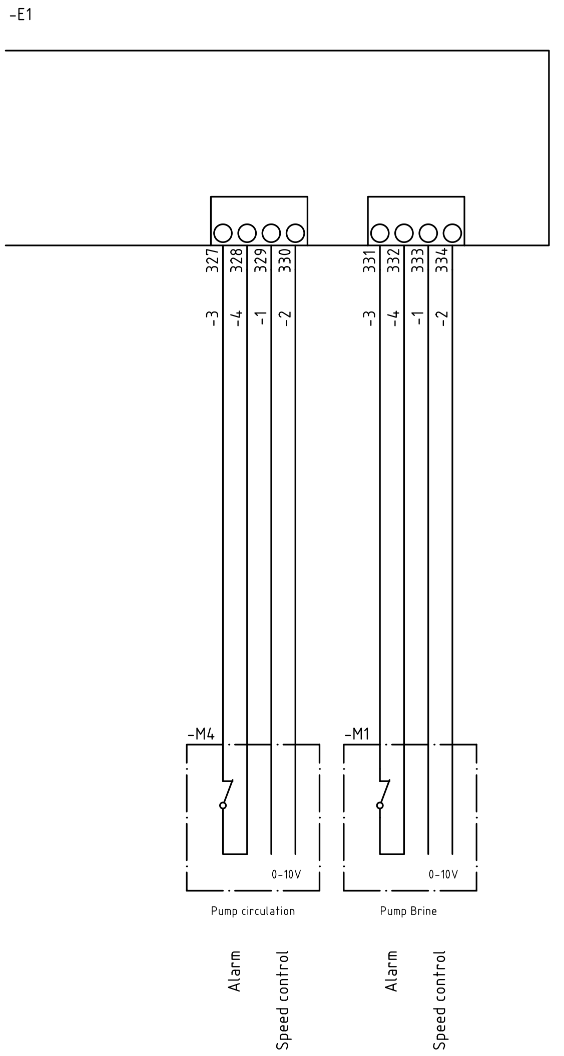

1
2
3
4
5
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
305
306
307
308
309
310
311
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6
-B7
-B8
-B2
-B3
-B9
-F3
-F4
-K1
A1
A2
-B1
p
-FL
217
219
-F11
/1.5B
95
96
P
P
N4
209
210
L1
215
216
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061005
Duo Optimum 1050 230V 3P Circuit diagram				CONTINUED 3
			DRAWING NR. 57183	SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

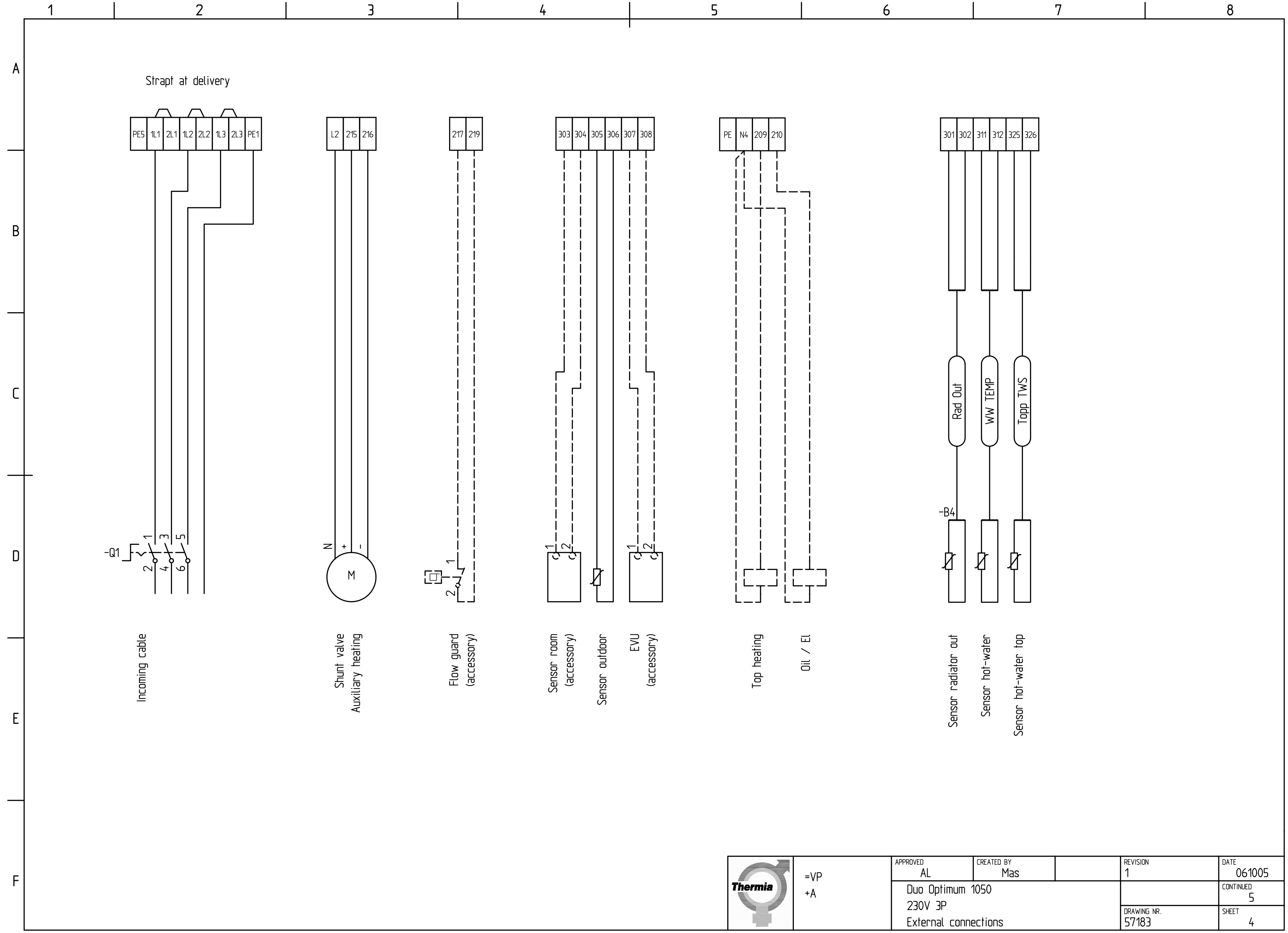
bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP
+A

APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061005

Duo Optimum 1050
230V 3P
Circuit diagram

CONTINUED 4
DRAWING NR. 57183
SHEET 3


1
2
3
4
5
6
7
8
A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
1
3
5
2
4
6
N
+
-
M
1
2
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia

=VP +A	APPROVED AL	CREATED BY Mas	REVISION 1	DATE 061005
	Duo Optimum 1050 230V 3P External connections		DRAWING NR. 57183	CONTINUED 5 SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-555051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	13-18A	9588-54715001	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	18A 7,5KW 230V	9588-38378003	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51335002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

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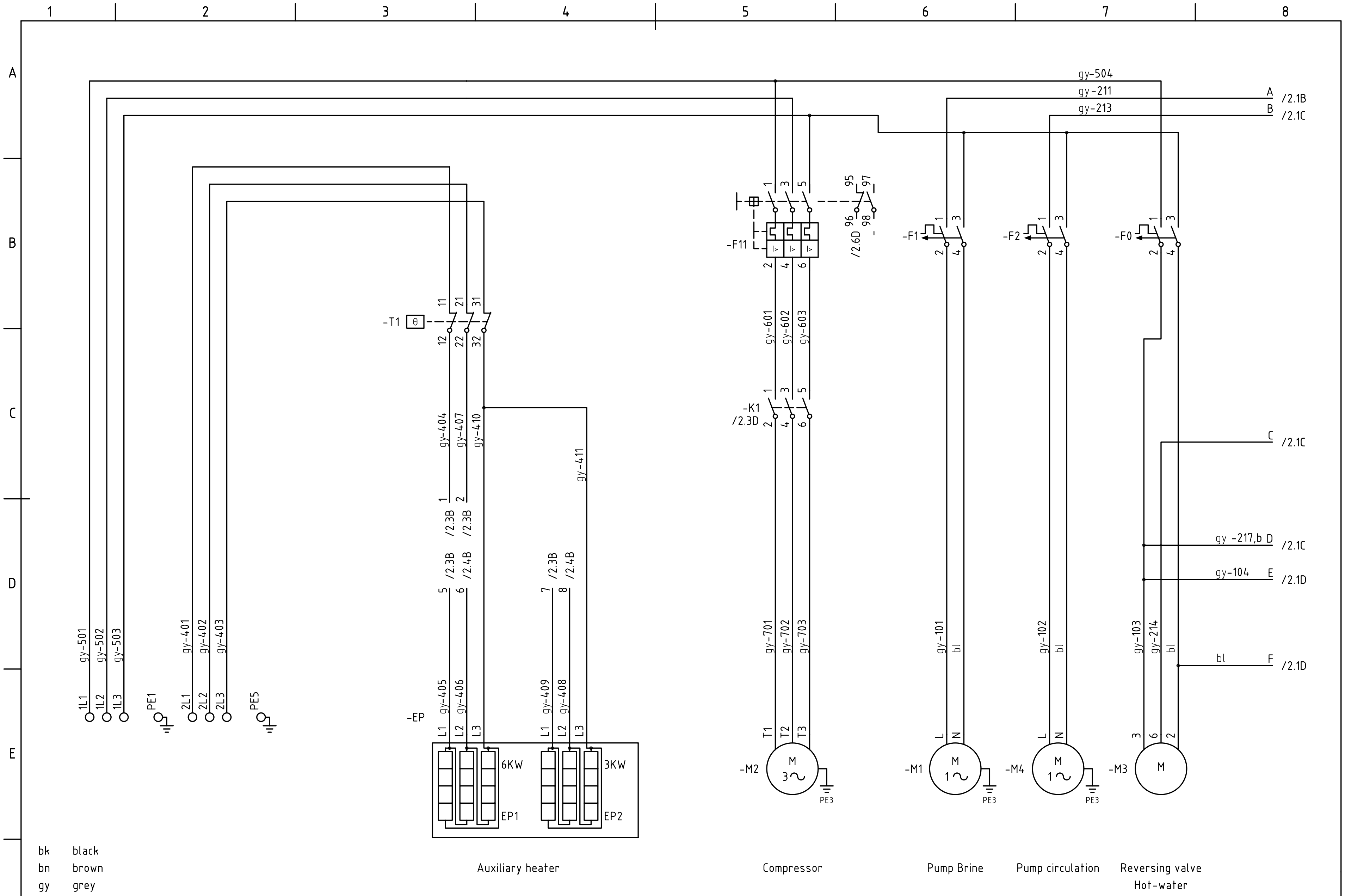
+A

APPROVED	CREATED BY		REVISION	DATE
AL	Bja		1	061005
Duo Optimum 1050 230V 3P Part list			DRAWING NR. 57183	CONTINUED - SHEET 5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7
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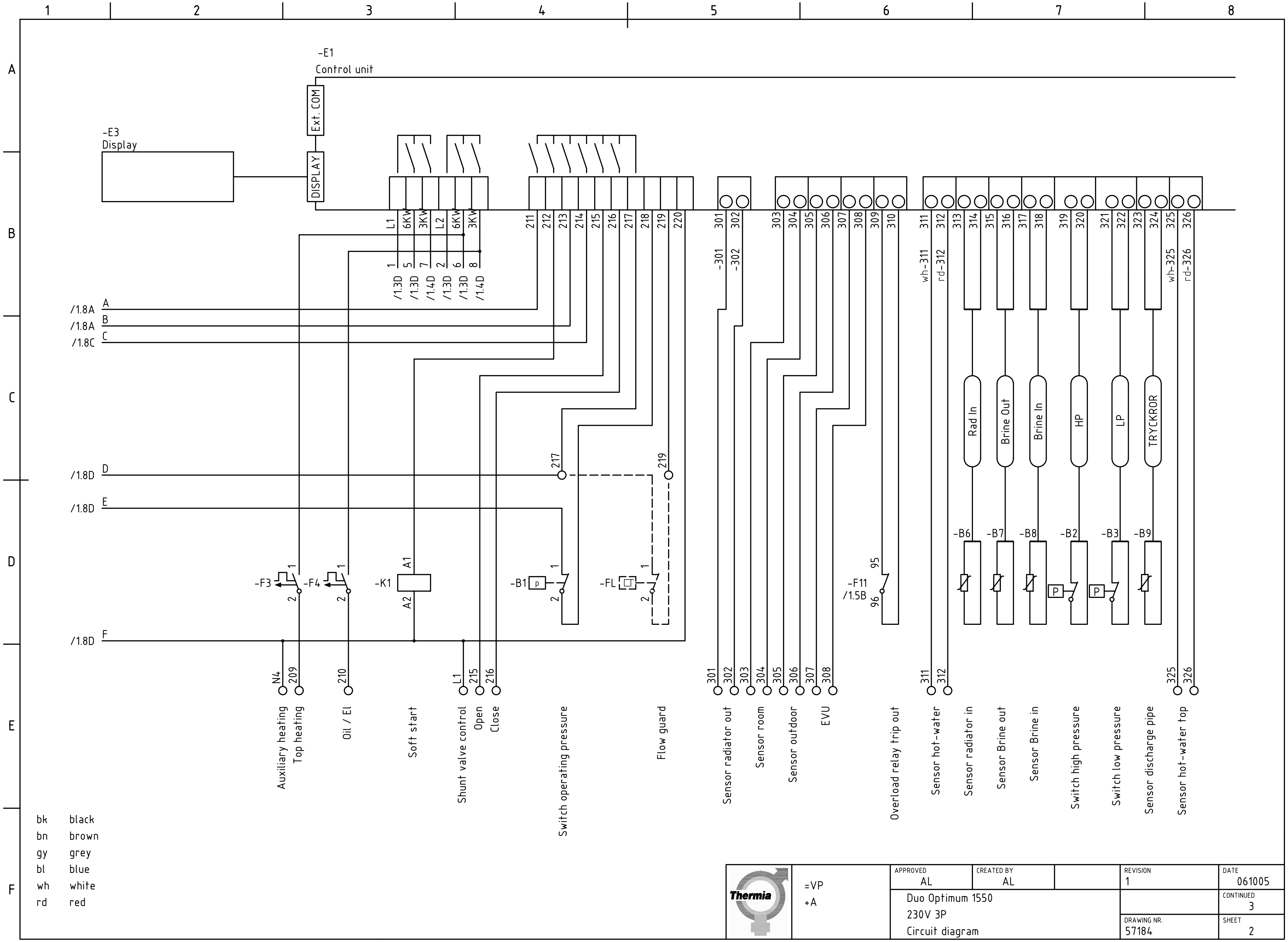
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-T1 θ
11 21 31
12 22 32
gy-404 gy-407 gy-410
gy-411
1 2
/2.3B /2.3B
5 6
/2.3B /2.4B
7 8
/2.3B /2.4B
-F11
1 3 5
2 4 6
95 97
96 98
/2.6D
gy-601 gy-602 gy-603
-K1
/2.3D
1 3 5
2 4 6
-F1
-F2
-F0
1 3
2 4
C /2.1C
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501 gy-502 gy-503
1L1 1L2 1L3
PE1
gy-401 gy-402 gy-403
2L1 2L2 2L3
PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M
3~
PE3
gy-101 bl
L N
-M1
M
1~
PE3
gy-102 bl
L N
-M4
M
1~
PE3
gy-103 gy-214 bl
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

		APPROVED	CREATED BY		REVISION	DATE
Thermia	=VP +A	AL	Mas		1	061005
		Duo Optimum 1550 230V 3P Circuit diagram			DRAWING NR. 57184	CONTINUED 2 SHEET 1


1
2
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4
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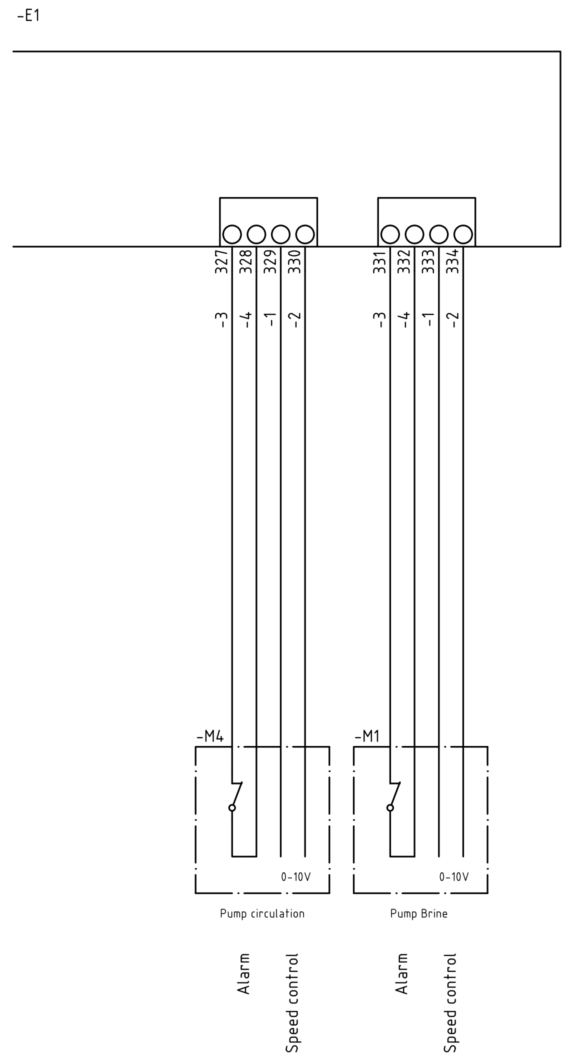
6
7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
-E3
Display
DISPLAY
L1 6KW 3KW L2 6KW 3KW
1 5 7 2 6 8
/1.3D /1.3D /1.4D /1.3D /1.3D /1.4D
211 212 213 214 215 216 217 218 219 220
301 302
-301 -302
303 304 305 306 307 308 309 310
311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326
wh-311 rd-312
wh-325 rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
217
219
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
-B6 -B7 -B8 -B2 -B3 -B9
P
P
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
1
2
N4 209 210 L1 215 216
301 302 303 304 305 306 307 308
311 312
325 326
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top
Thermia

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

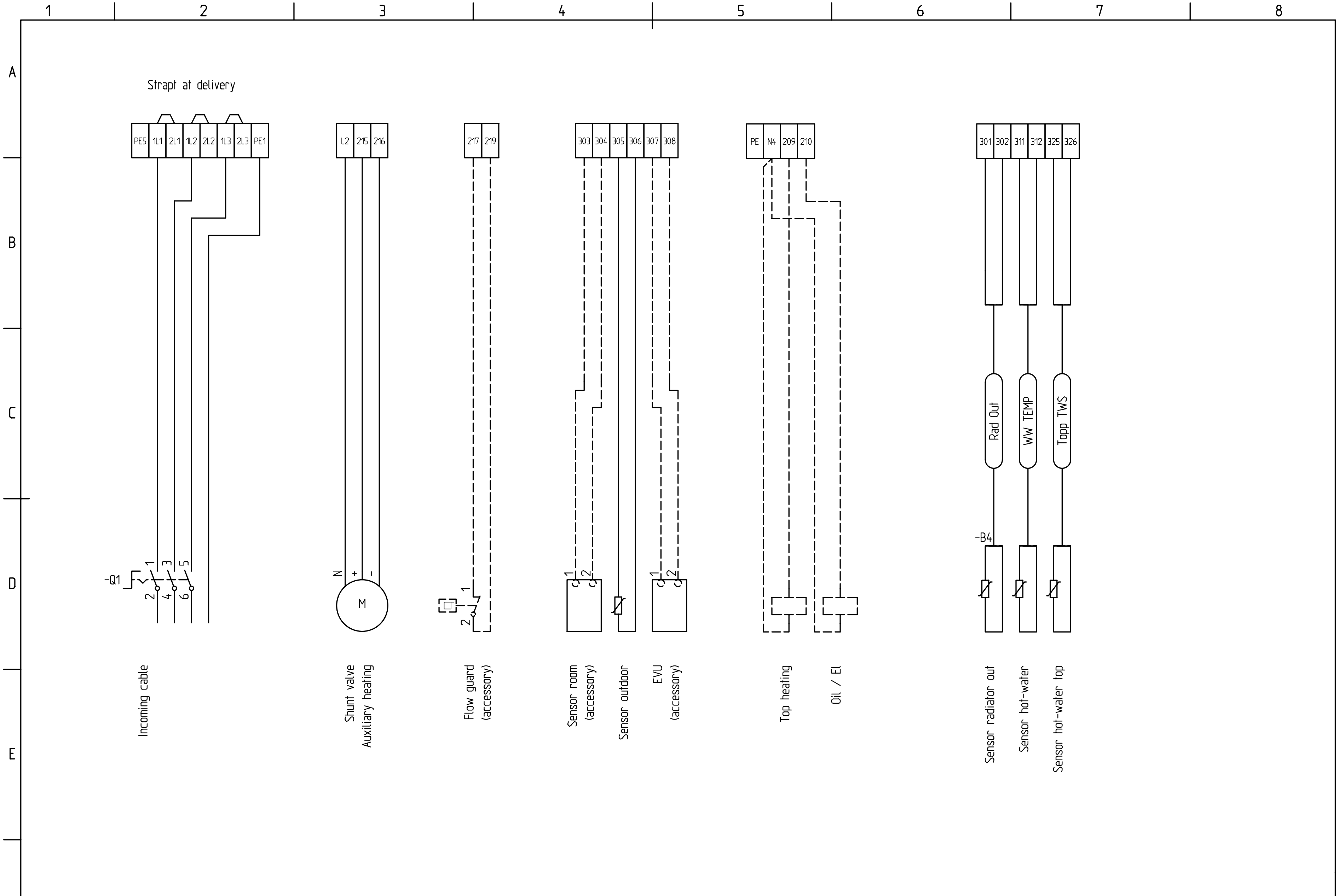
=VP
+A

APPROVED	CREATED BY		REVISION	DATE
AL	AL		1	061005
Duo Optimum 1550				CONTINUED 3
230V 3P			DRAWING NR. 57184	SHEET 2
Circuit diagram				


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY Mas	REVISION 1	DATE 061005
	Duo Optimum 1550 230V 3P Circuit diagram		DRAWING NR. 57184	CONTINUED 4 SHEET 3


1
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7
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A
B
C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
-Q1
1
3
5
2
4
6
Incoming cable
L2
215
216
N
+
-
M
Shunt valve
Auxiliary heating
217
219
1
2
Flow guard
(accessory)
303
304
305
306
307
308
1
2
1
2
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
PE
N4
209
210
Top heating
Oil / El
301
302
311
312
325
326
Rad Out
WW TEMP
Topp TWS
-B4
Sensor radiator out
Sensor hot-water
Sensor hot-water top

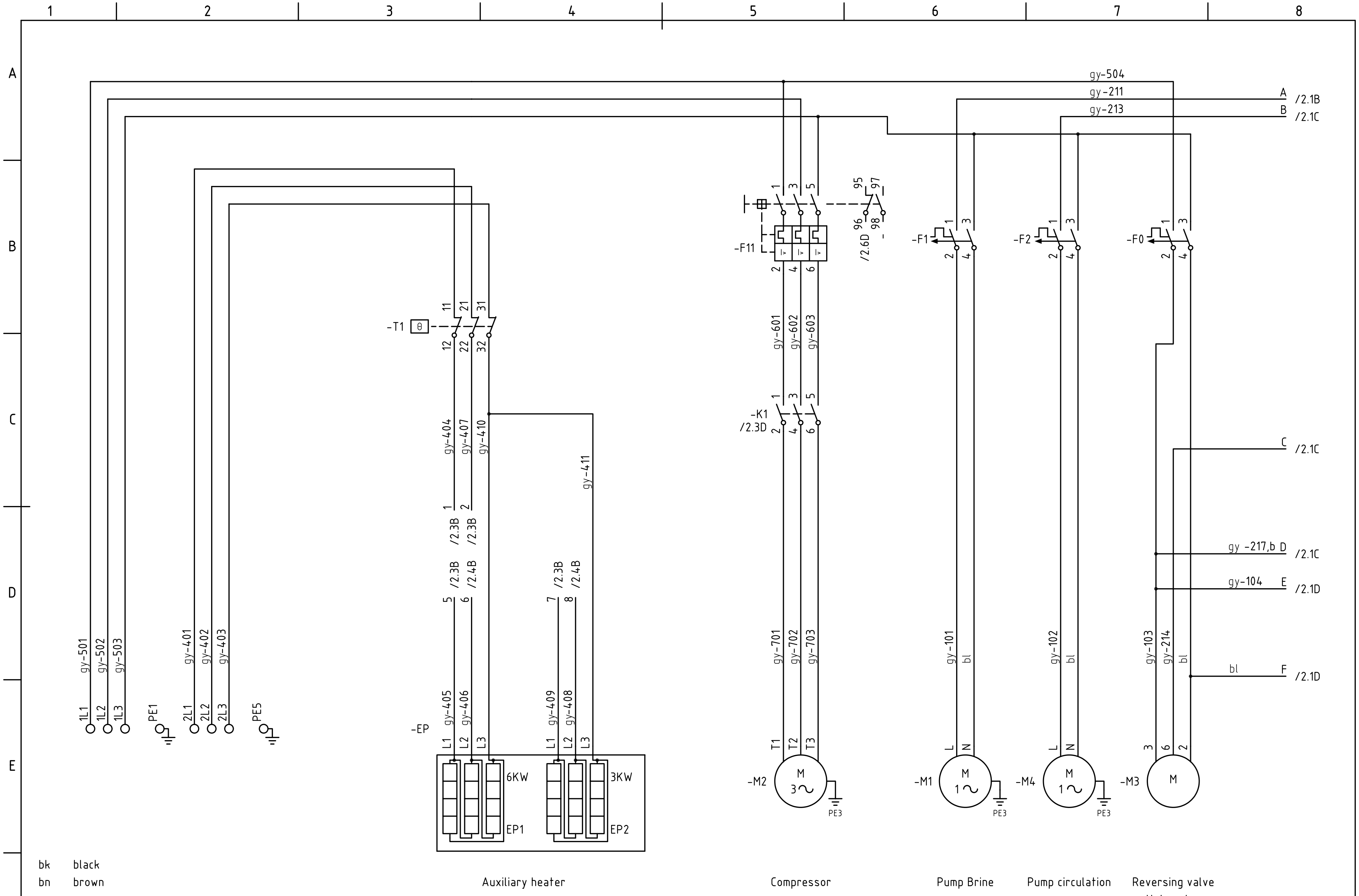
Thermia

=VP
+A

APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061005
Duo Optimum 1550
230V 3P
External connections
CONTINUED 5
DRAWING NR. 57184
SHEET 4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	24-32A	9588-52640004	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	32A 15KW 230VA	9588-38378005	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51337002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

Thermia	=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061005
		Duo Optimum 1550 230V 3P Part list			DRAWING NR. 57184	CONTINUED - SHEET 5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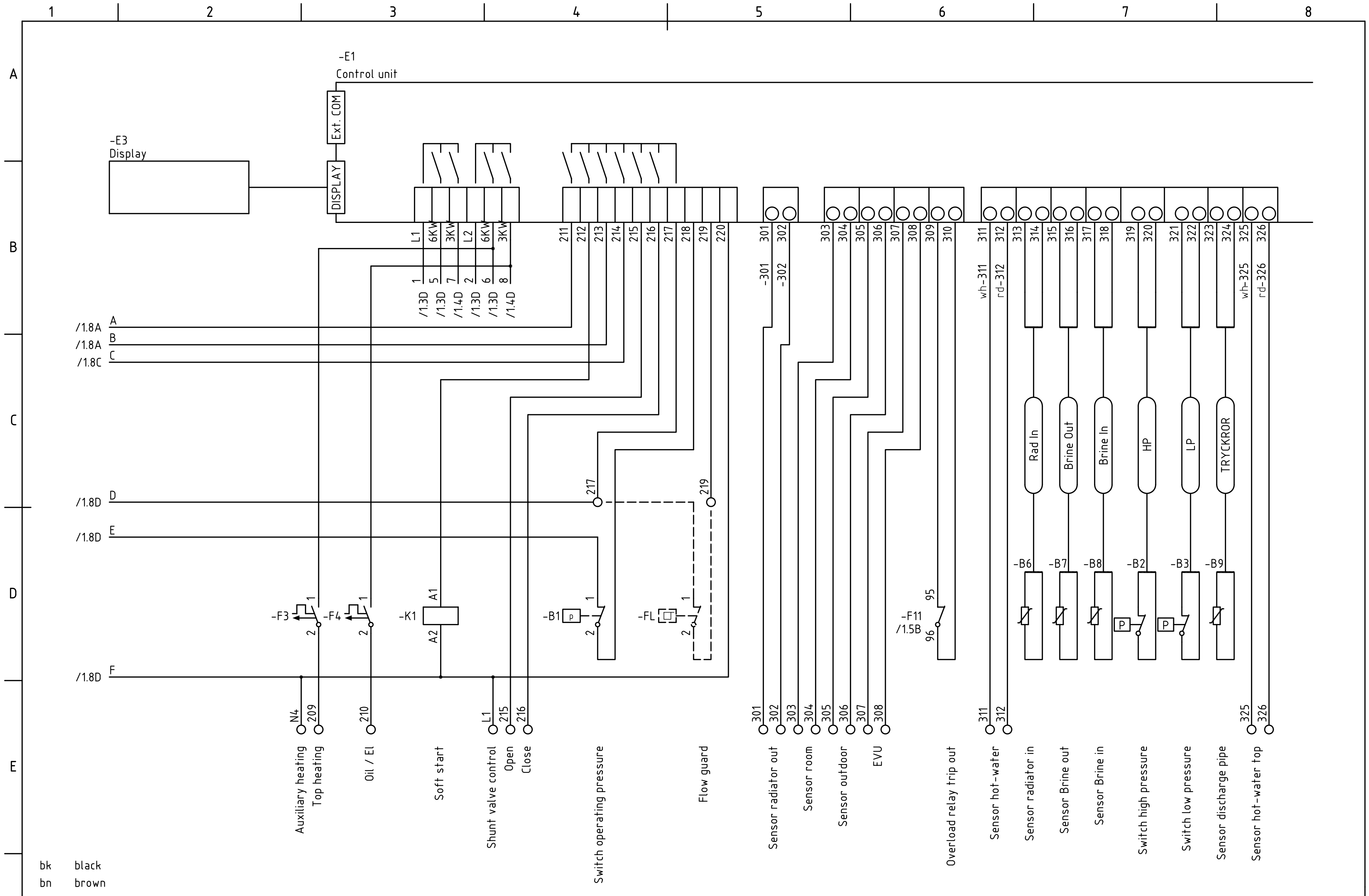
7
8
A
B
C
D
E
F
gy-504
gy-211
gy-213
A /2.1B
B /2.1C
-F11
95
97
96
98
/2.6D
-F1
-F2
-F0
-T1
θ
11 21 31
12 22 32
gy-601 gy-602 gy-603
-K1
/2.3D
gy-404 gy-407 gy-410
gy-411
C /2.1C
1 /2.3B
2 /2.3B
5 /2.3B
6 /2.4B
7 /2.3B
8 /2.4B
gy -217,b D /2.1C
gy-104 E /2.1D
bl F /2.1D
gy-501 gy-502 gy-503
gy-401 gy-402 gy-403
1L1 1L2 1L3
PE1
2L1 2L2 2L3
PE5
-EP
gy-405 gy-406
gy-409 gy-408
L1 L2 L3
6KW
EP1
3KW
EP2
gy-701 gy-702 gy-703
T1 T2 T3
-M2
M
3~
PE3
gy-101 bl
L N
-M1
M
1~
PE3
gy-102 bl
-M4
M
1~
PE3
gy-103 gy-214 bl
3 6 2
-M3
M
Auxiliary heater
Compressor
Pump Brine
Pump circulation
Reversing valve
Hot-water

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

=VP
+A

APPROVED AL
CREATED BY Mas
REVISION 1
DATE 061005
Duo Optimum 1800
230V 3P
Circuit diagram
CONTINUED 2
DRAWING NR. 57185
SHEET 1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7
8
A
B
C
D
E
F
-E1
Control unit
Ext. COM
DISPLAY
-E3
Display
L1
6KW
3KW
L2
6KW
3KW
1
5
7
2
6
8
/1.3D
/1.3D
/1.4D
/1.3D
/1.3D
/1.4D
211
212
213
214
215
216
217
218
219
220
301
302
-301
-302
303
304
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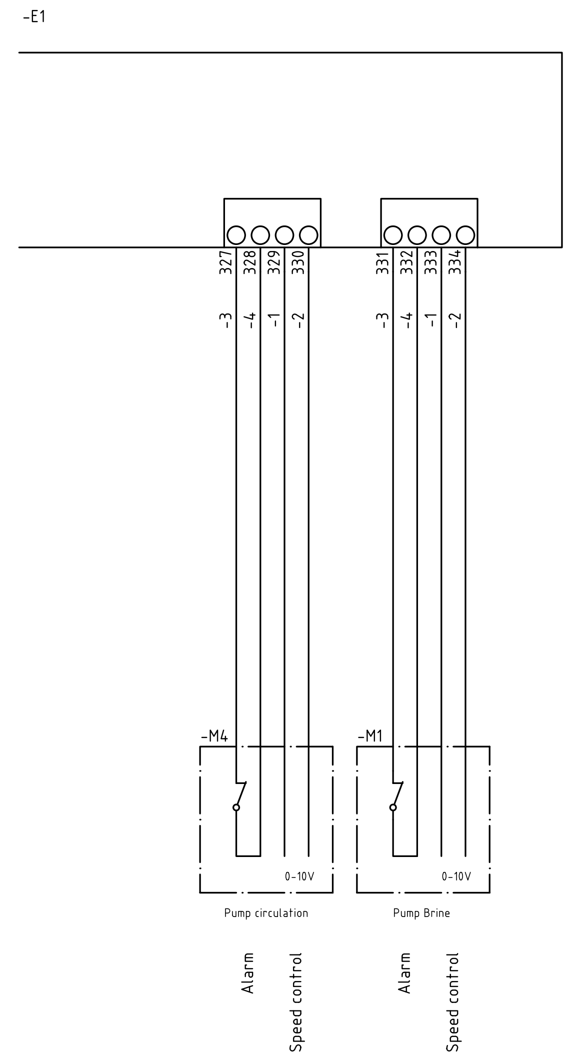
312
313
314
315
316
317
318
319
320
321
322
323
324
325
326
wh-311
rd-312
wh-325
rd-326
/1.8A A
/1.8A B
/1.8C C
/1.8D D
/1.8D E
/1.8D F
Rad In
Brine Out
Brine In
HP
LP
TRYCKROR
217
219
-B6
-B7
-B8
-B2
-B3
-B9
-F3
-F4
-K1
A1
A2
-B1
p
-FL
-F11
/1.5B
95
96
P
N4
209
210
L1
215
216
Auxiliary heating
Top heating
Oil / El
Soft start
Shunt valve control
Open
Close
Switch operating pressure
Flow guard
Sensor radiator out
Sensor room
Sensor outdoor
EVU
Overload relay trip out
Sensor hot-water
Sensor radiator in
Sensor Brine out
Sensor Brine in
Switch high pressure
Switch low pressure
Sensor discharge pipe
Sensor hot-water top

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

Thermia

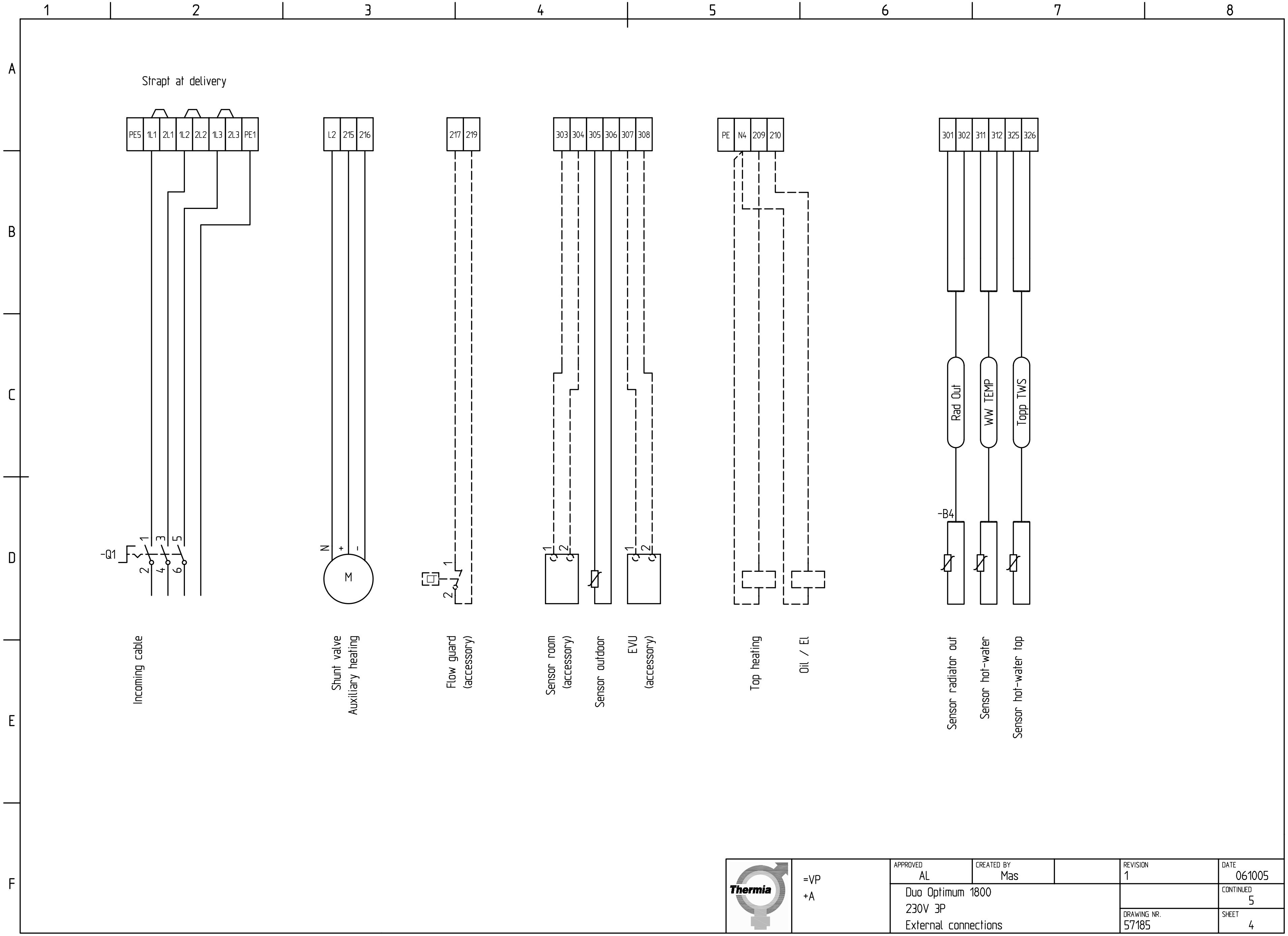
=VP
+A

APPROVED AL
CREATED BY AL
REVISION 1
DATE 061005
Duo Optimum 1800
230V 3P
Circuit diagram
CONTINUED 3
DRAWING NR. 57185
SHEET 2


-E1
327
328
329
330
331
332
333
334
-3
-4
-1
-2
-3
-4
-1
-2
-M4
-M1
0-10V
0-10V
Pump circulation
Pump Brine
Alarm
Speed control
Alarm
Speed control

bk	black
bn	brown
gy	grey
bl	blue
wh	white
rd	red

=VP +A	APPROVED AL	CREATED BY Mas		REVISION 1	DATE 061005
	Duo Optimum 1800 230V 3P Circuit diagram			DRAWING NR. 57185	CONTINUED 4 SHEET 3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A
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C
D
E
F
Strapt at delivery
PE5
1L1
2L1
1L2
2L2
1L3
2L3
PE1
L2
215
216
217
219
303
304
305
306
307
308
PE
N4
209
210
301
302
311
312
325
326
-Q1
M
N
+
-
Rad Out
WW TEMP
Topp TWS
-B4
Incoming cable
Shunt valve
Auxiliary heating
Flow guard
(accessory)
Sensor room
(accessory)
Sensor outdoor
EVU
(accessory)
Top heating
Oil / El
Sensor radiator out
Sensor hot-water
Sensor hot-water top
Thermia
=VP
+A
APPROVED
AL
CREATED BY
Mas
REVISION
1
DATE
061005
Duo Optimum 1800
230V 3P
External connections
CONTINUED
5
DRAWING NR.
57185
SHEET
4

Number	Quantity	Designation	Manufacture	Electrical data	Part number	Sheet
-B1	1	Switch operating pressure	Saginomiya	26,5 bar	9578-37142002	/2.4D
-B1	1	Switch operating pressure	Saginomiya	28,5 bar	9578-55974001	/2.4D
-B2	1	Switch high pressure	Saginomiya	31 bar	9578-55973001	/2.7B
-B3	1	Switch low pressure	Saginomiya	0,8 bar	9578-55212001	/2.7B
-B6	1	Sensor radiator in	Betatherm	NTC	9588-53809001	/2.6B
-B7	1	Sensor Brine out	Betatherm	NTC	9588-53808001	/2.7B
-B8	1	Sensor Brine in	Betatherm	NTC	9588-53807001	/2.7B
-B9	1	Sensor discharge pipe	Betatherm	NTC	9588-55322001	/2.8B
-E1	1	Control unit	Kimsafe AB		9588-56464001	/2.3B
-E3	1	Display	Kimsafe AB		9588-56562001	/2.1B
-EP	1	Auxiliary heater	Backer	3kW/6kW	9571-55051A00	/1.3E
-F0	1	Circuit breaker	Merlin Gerin	2P C10A	9588-52666001	/1.7B
-F1	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.6B
-F2	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/1.7B
-F3	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F4	1	Circuit breaker	Merlin Gerin	2P C3A	9585-54631001	/2.3D
-F11	1	Motor circuit breaker	Telemecanique	24-32A	9588-52640004	/1.5B
-F11	1	Auxiliary switch block	Telemecanique		9575-53795001	/1.5B
-K1	1	Contactor	Telemecanique	32A 15KW 230VA	9588-38378005	/2.3D
-M1	1	Pump circulation	Wilo		9675-57045002	/1.6E
-M2	1	Compressor	Copeland	230V 3N	9686-51338002	/1.5E
-M3	1	Reversing valve	Honeywell		9360-37547001	/1.7E
-M4	1	Pump circulation	Wilo		9675-57045002	/1.7E
-T1	1	Temperature guard	Ego	30A 230V	9579-54513001	/1.3C
-X1	1	Terminal block	Phoenix	Diplomat 230V	9583-54439001	
-X2	1	Terminal block	Phoenix	Duo	9583-54533001	

=VP +A	APPROVED AL	CREATED BY Bja		REVISION 1	DATE 061005
	Duo Optimum 1800 230V 3P Part list			DRAWING NR. 57185	CONTINUED - SHEET 5

If these instructions are not followed during installation, operation and maintenance, the obligations of Thermia Värme AB under the warranty regulations are no longer binding.

Thermia Värme AB • Box 950 • 671 29 Arvika • Sweden
www.thermia.com

www.setcommunication.se 2004